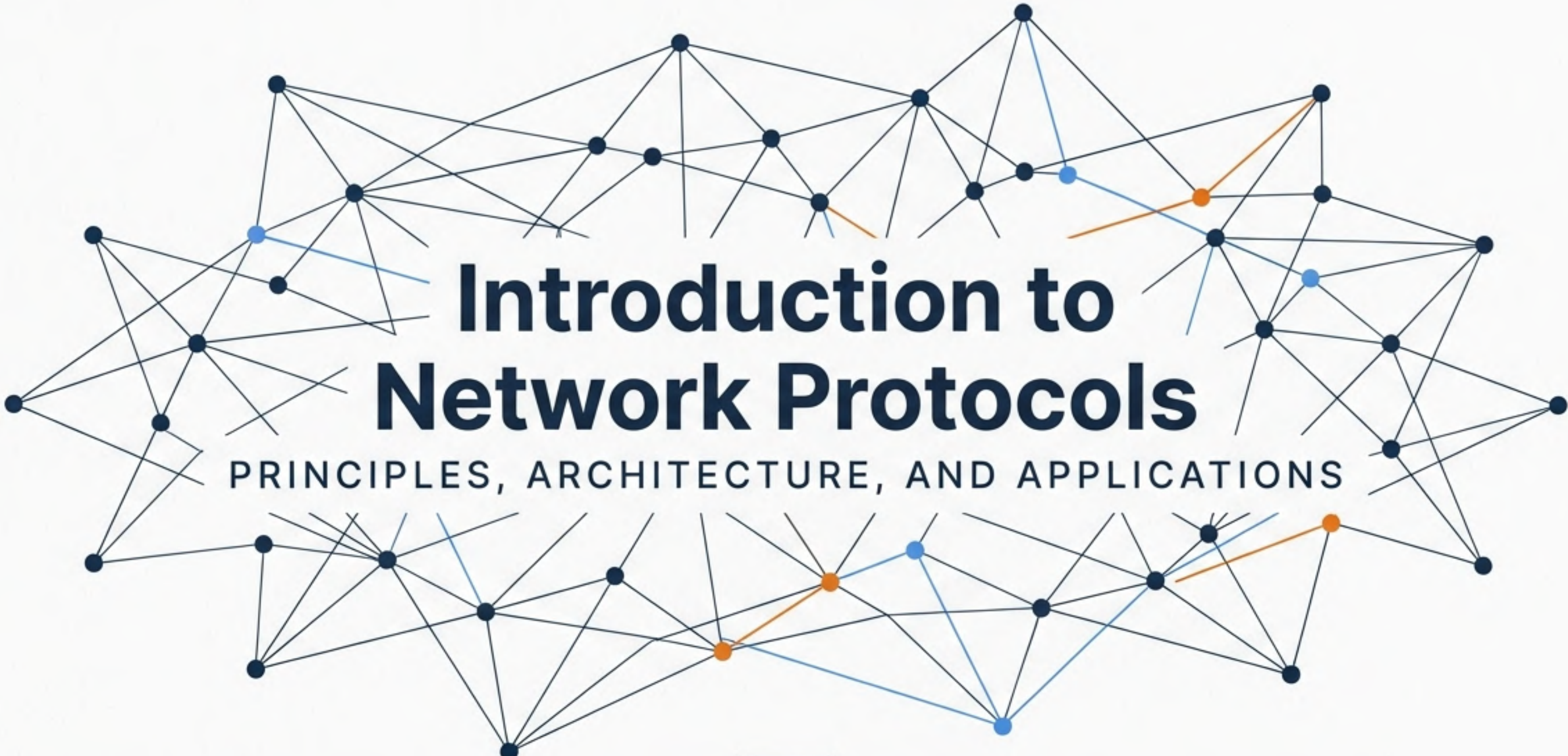




Introduction to Network Protocols

PRINCIPLES, ARCHITECTURE, AND APPLICATIONS

By: Dr. Richa Kumari

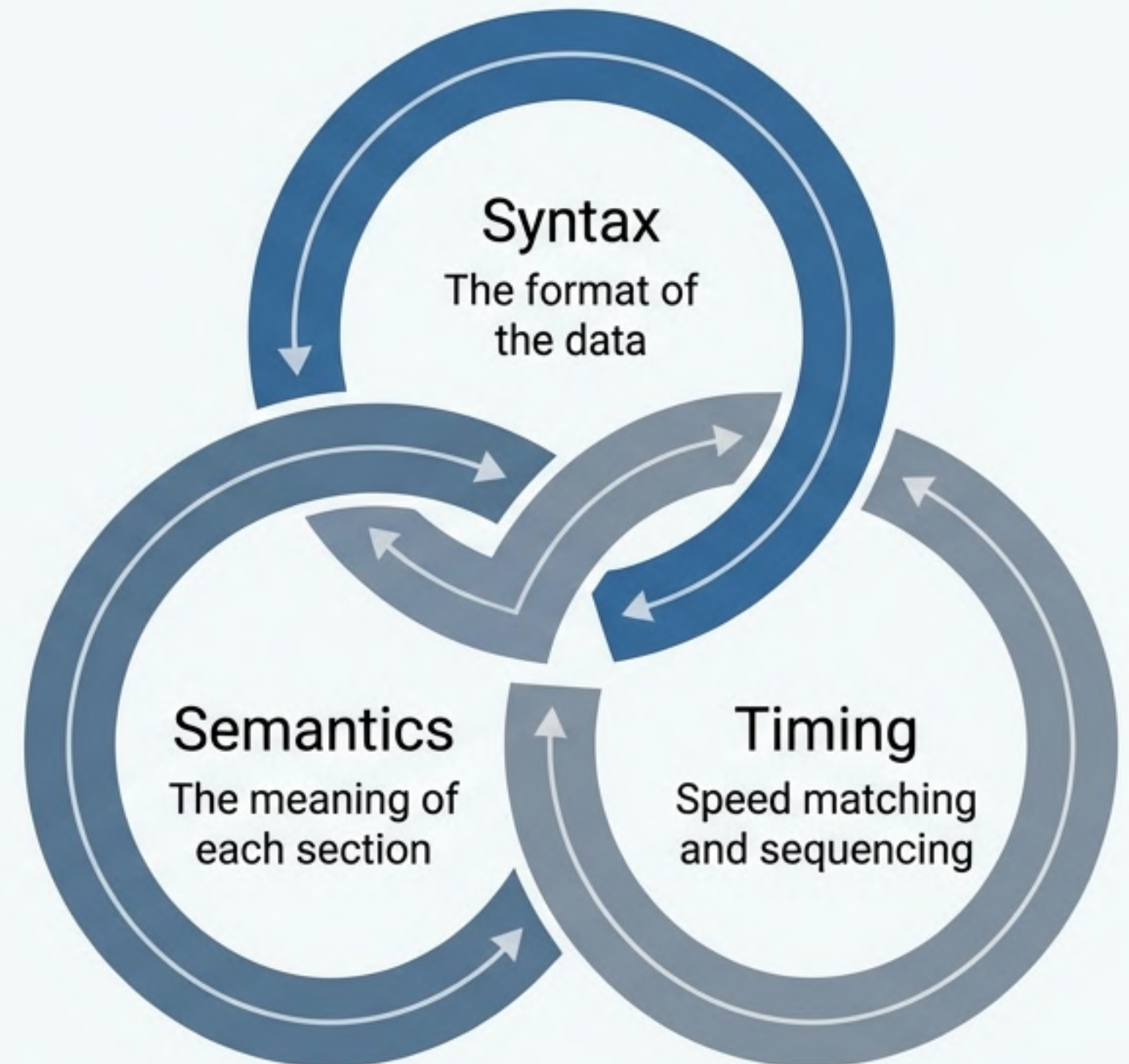
Defining Network Protocols

A protocol is a set of rules that governs data communications. It determines what is communicated, how it is communicated, and when it is communicated. Without these standards, devices would lack a common language.

Key Functions

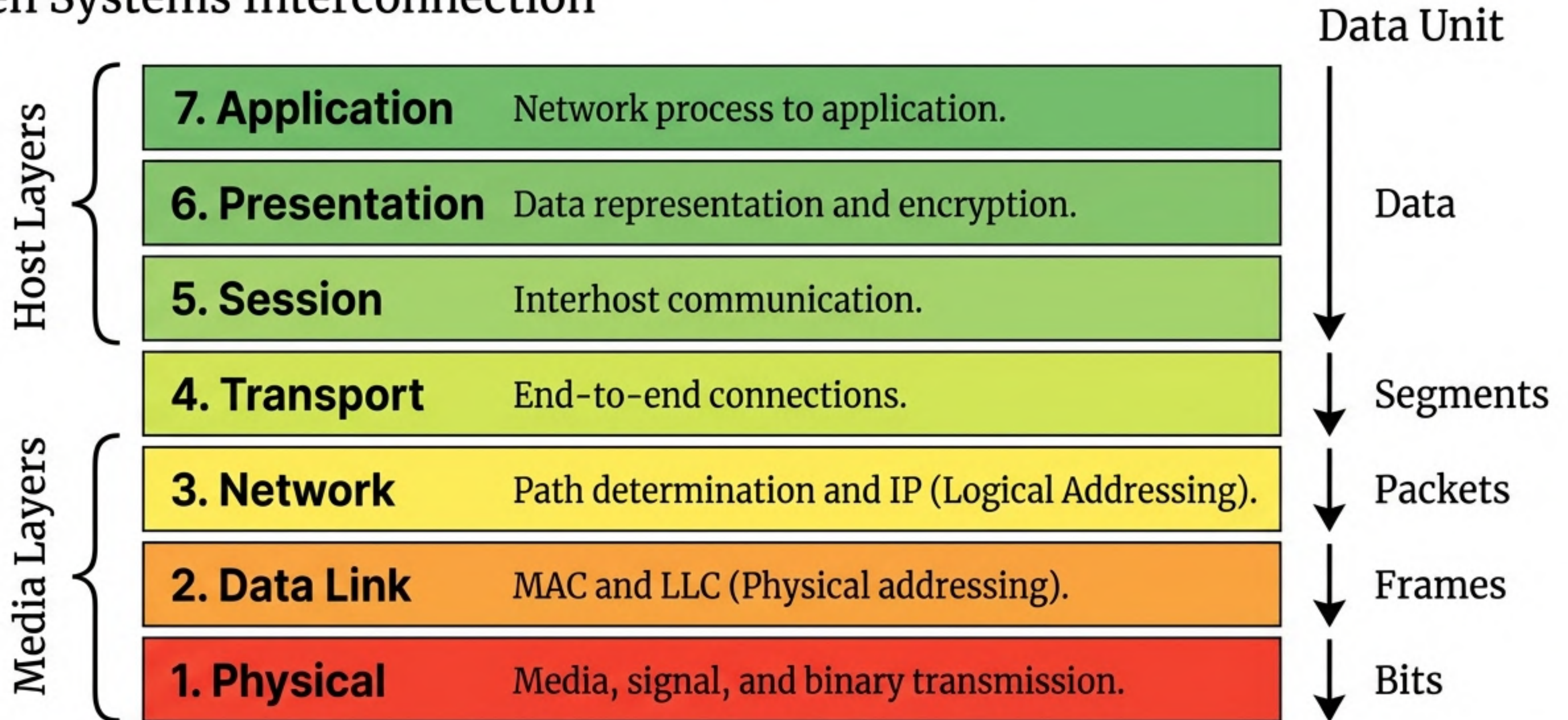
- **Formatting:** Defining the structure of data.
- **Processing:** Handling requests and responses.
- **Transmission:** Ensuring messages reach the destination.

Three Key Elements of a Protocol



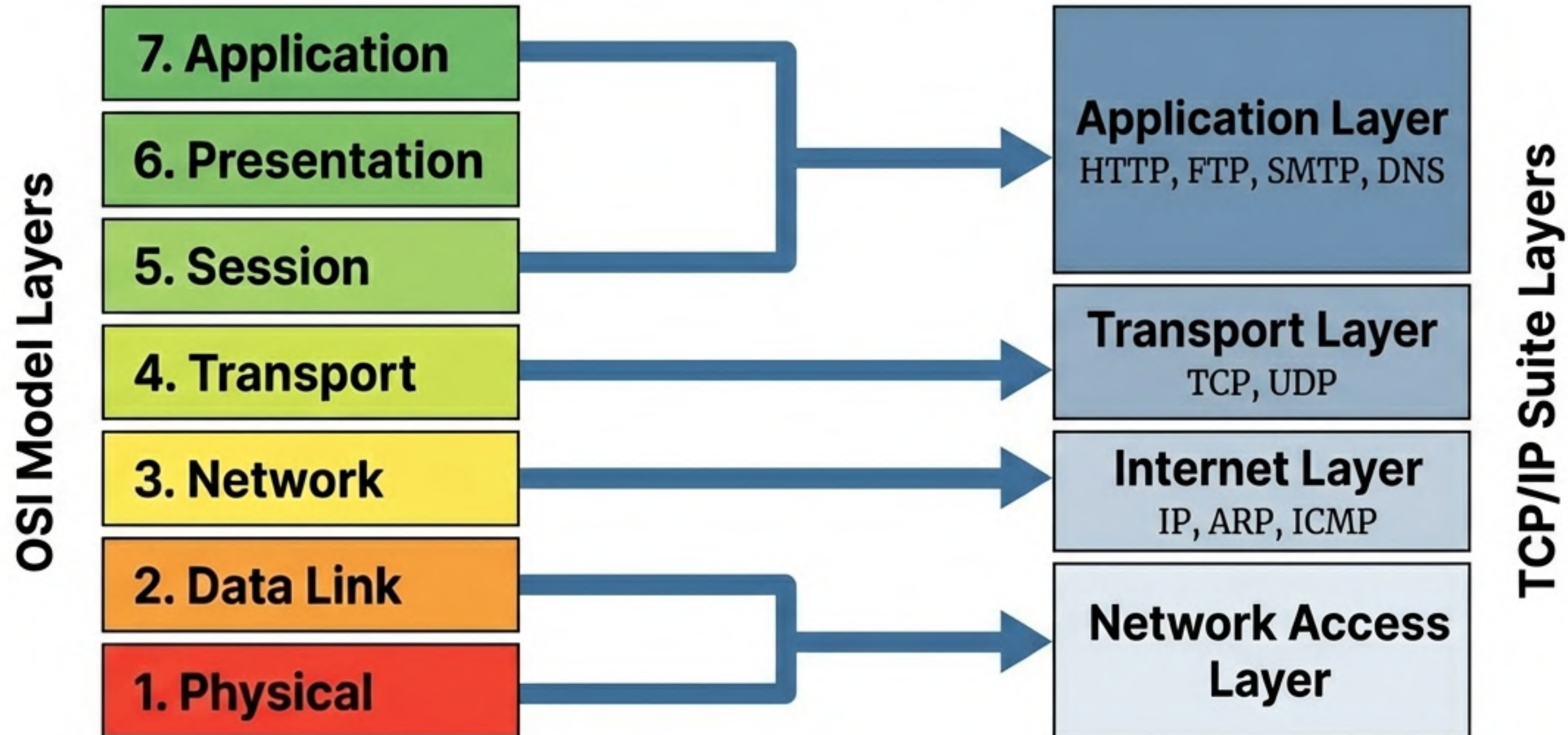
The OSI Reference Model

Open Systems Interconnection



The TCP/IP Protocol Suite

Practical Implementation vs. Theoretical Model



Internet Protocol (IP): The Network Layer

IP is the principal communications protocol in the Internet Protocol Suite for relaying datagrams across network boundaries. It handles addressing and routing.

Key Characteristics



Unreliable / Best Effort

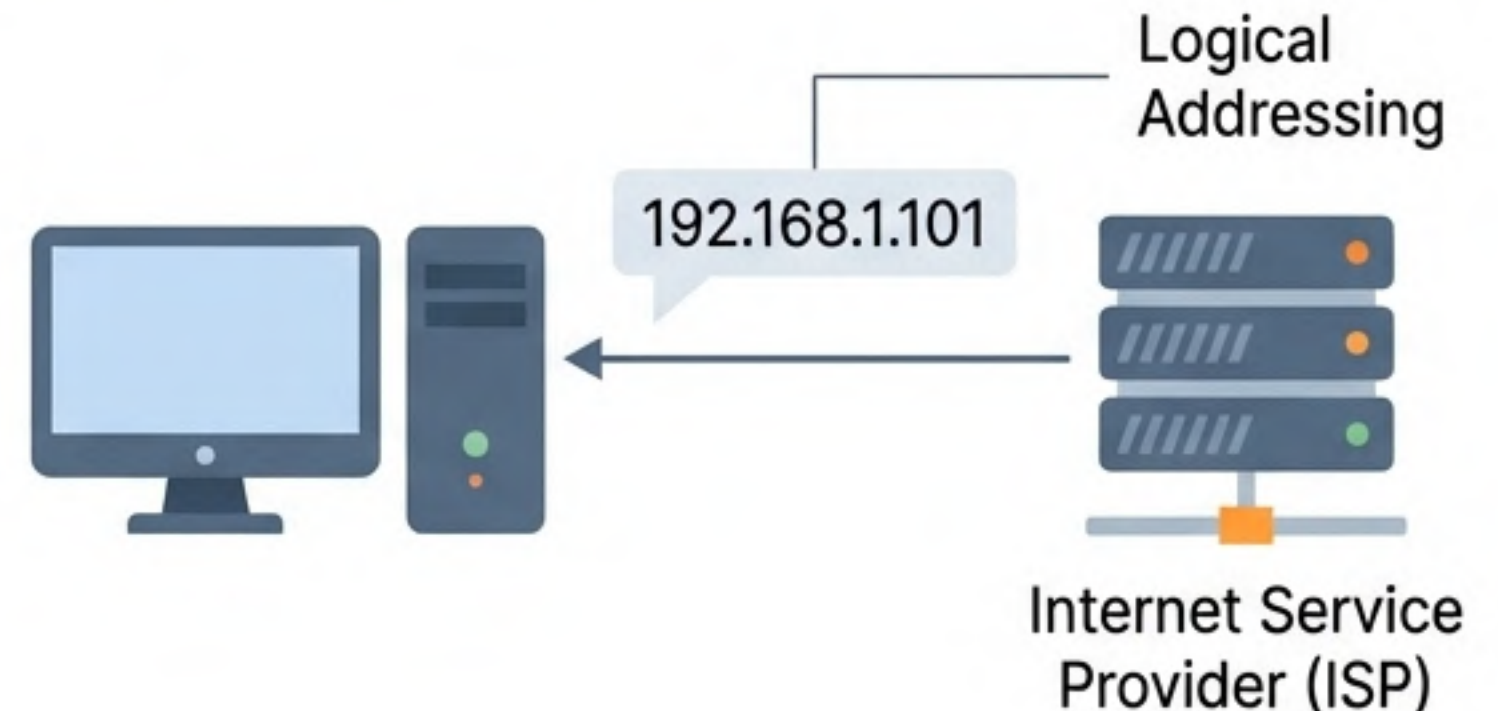
No guarantee of delivery. Packets may be lost, duplicated, or corrupted.



Connectionless

Each packet treated independently. Packets may take different routes to the same destination.

Addressing



Transmission Control Protocol (TCP)

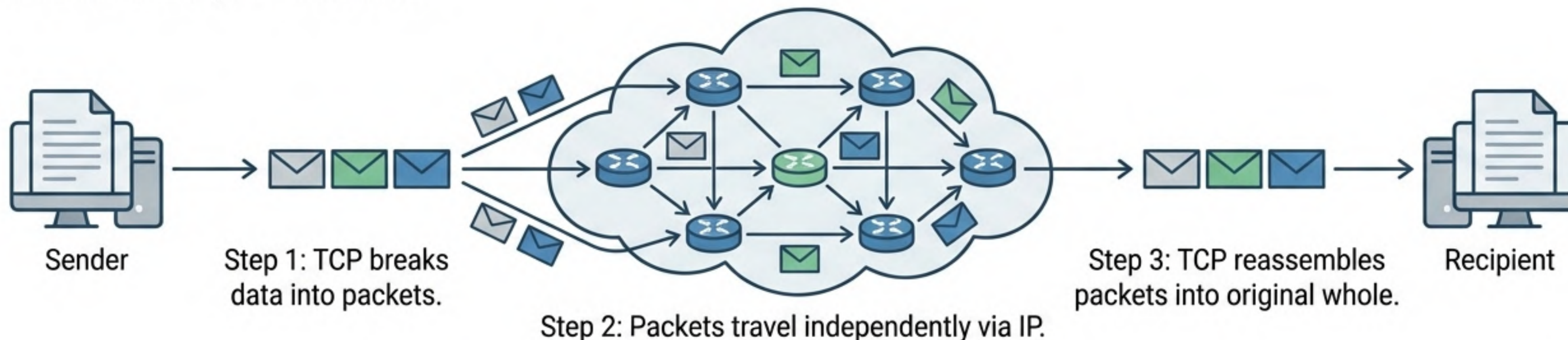
Reliability at the Transport Layer

TCP provides reliable, ordered, and error-checked delivery of a stream of octets. It manages the fragmentation of data into packets and their reassembly.

Key Mechanisms

- Virtual Circuits (Connection-Oriented)
- Error Detection & ARQ
- Packet Reassembly

How TCP/IP Works



TCP Ports and Multiplexing

Ports act as specific doors on a single IP address, allowing multiple applications to run simultaneously. A Socket = IP Address + Port Number.

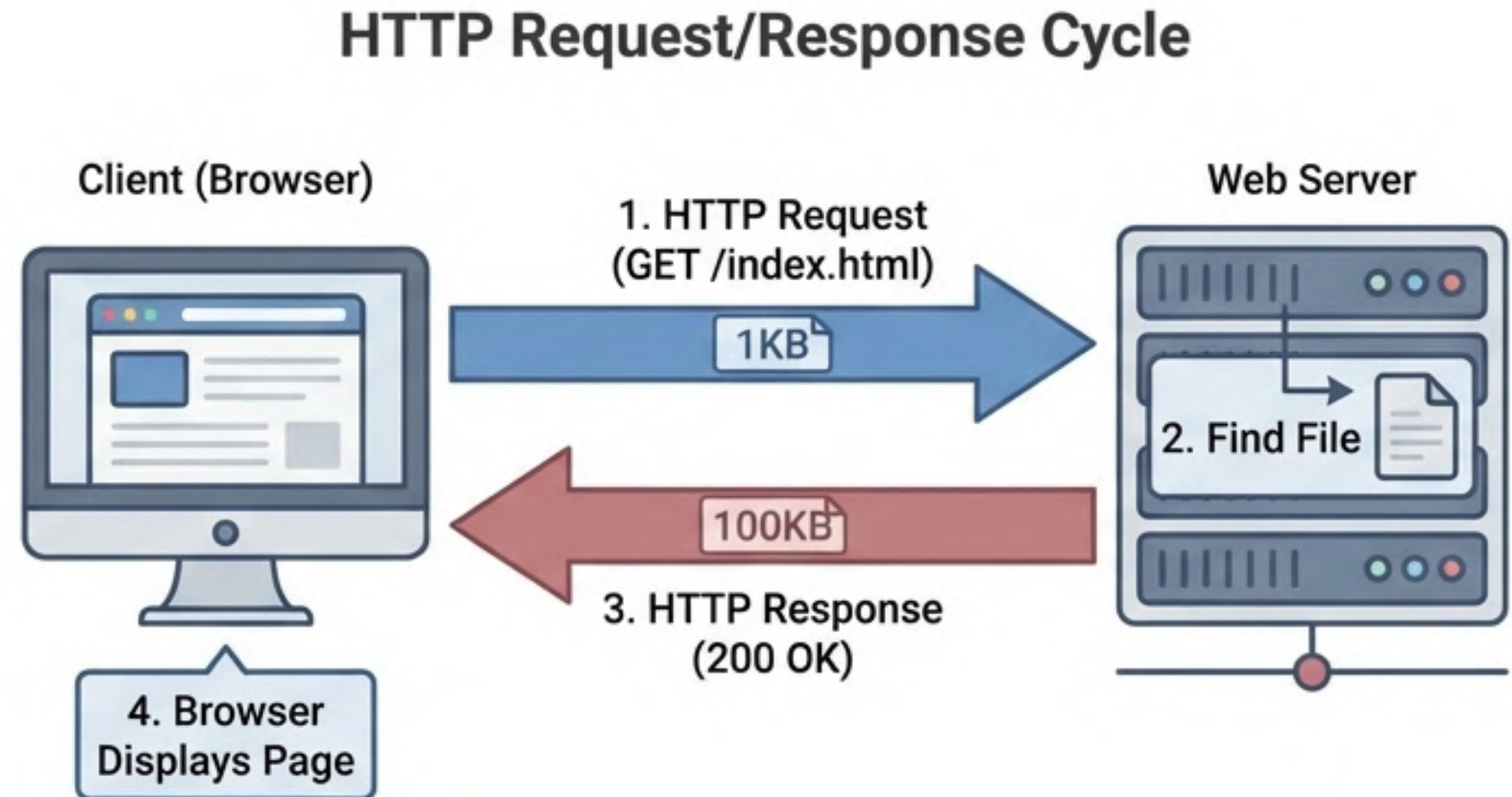
Well-Known Ports (0-1023)		
Protocol	Port Number	Service Type
FTP Data	20	File Transfer
FTP Control	21	File Transfer
SMTP	25	Email Transmission
DNS	53	Name Resolution
HTTP	80	Web Browsing
POP3	110	Email Retrieval

Dynamic/Private Ports (1024-65535) are generated dynamically for client-side connections.

Hypertext Transfer Protocol (HTTP)

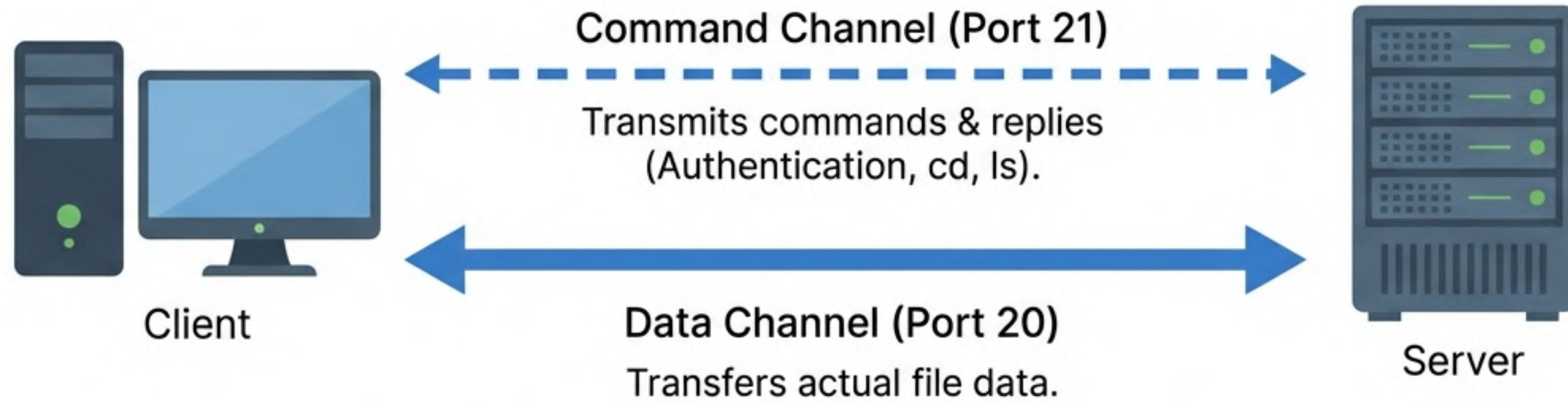
The Foundation of the World Wide Web

- Application protocol for distributed hypermedia systems.
- Stateless: Each request is independent.
- Hypertext: Structured text using logical links.



File Transfer Protocol (FTP)

Dual-Channel Architecture

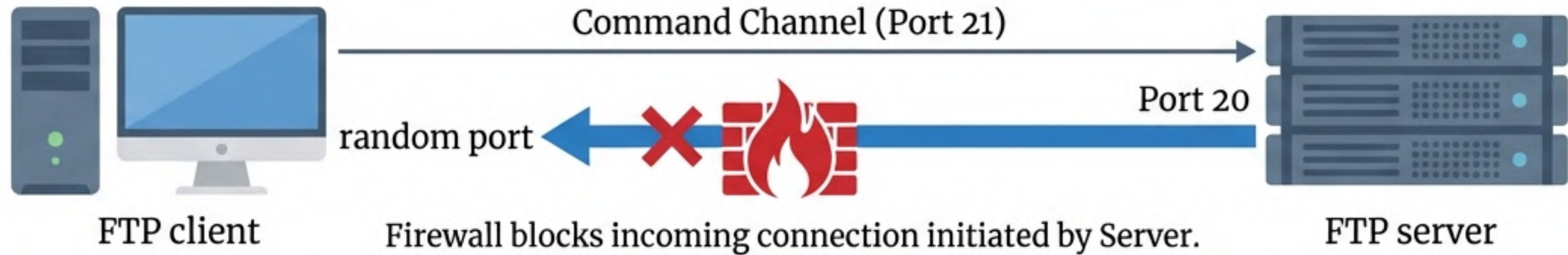


FTP is a standard network protocol used to transfer files between hosts. Unlike HTTP, it maintains a stateful control connection separate from the data stream. This conceptual separation of ports 20 and 21 ensures efficient command processing while allowing for data to be transferred on a dedicated channel.

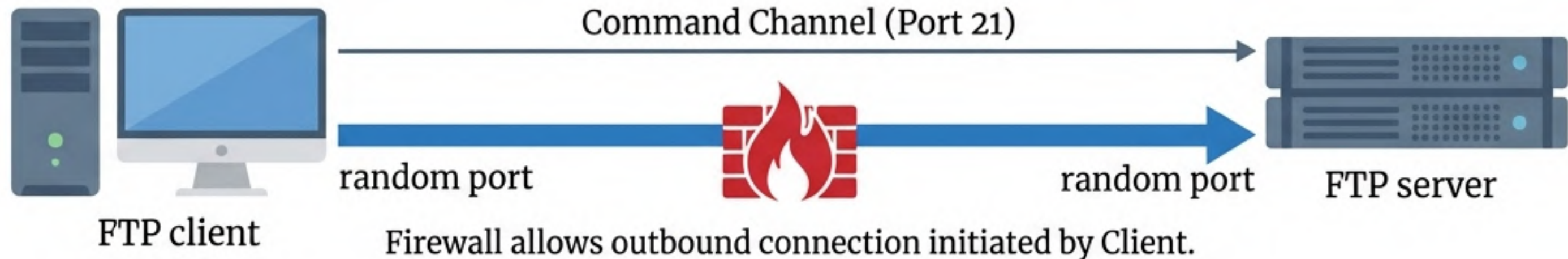
FTP Modes: Active vs. Passive

Firewalls and Connection Initiation

Active Mode (The Problem)

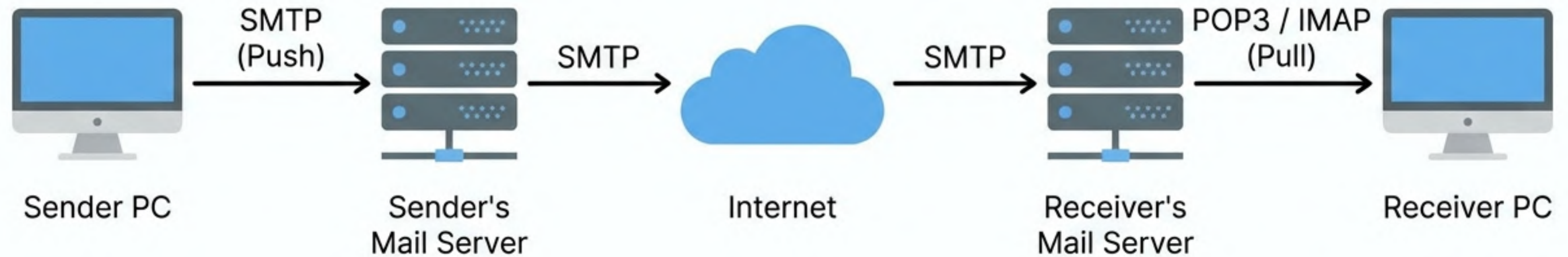


Passive Mode (The Solution)



Electronic Mail: SMTP

Simple Mail Transfer Protocol



- **SMTP** (Port 25): Used for sending mail (Push protocol).
- **POP3/IMAP**: Used for retrieving mail (Pull protocol).
- **MTA**: Mail Transfer Agents handle the server-to-server relay.

Domain Name System (DNS)

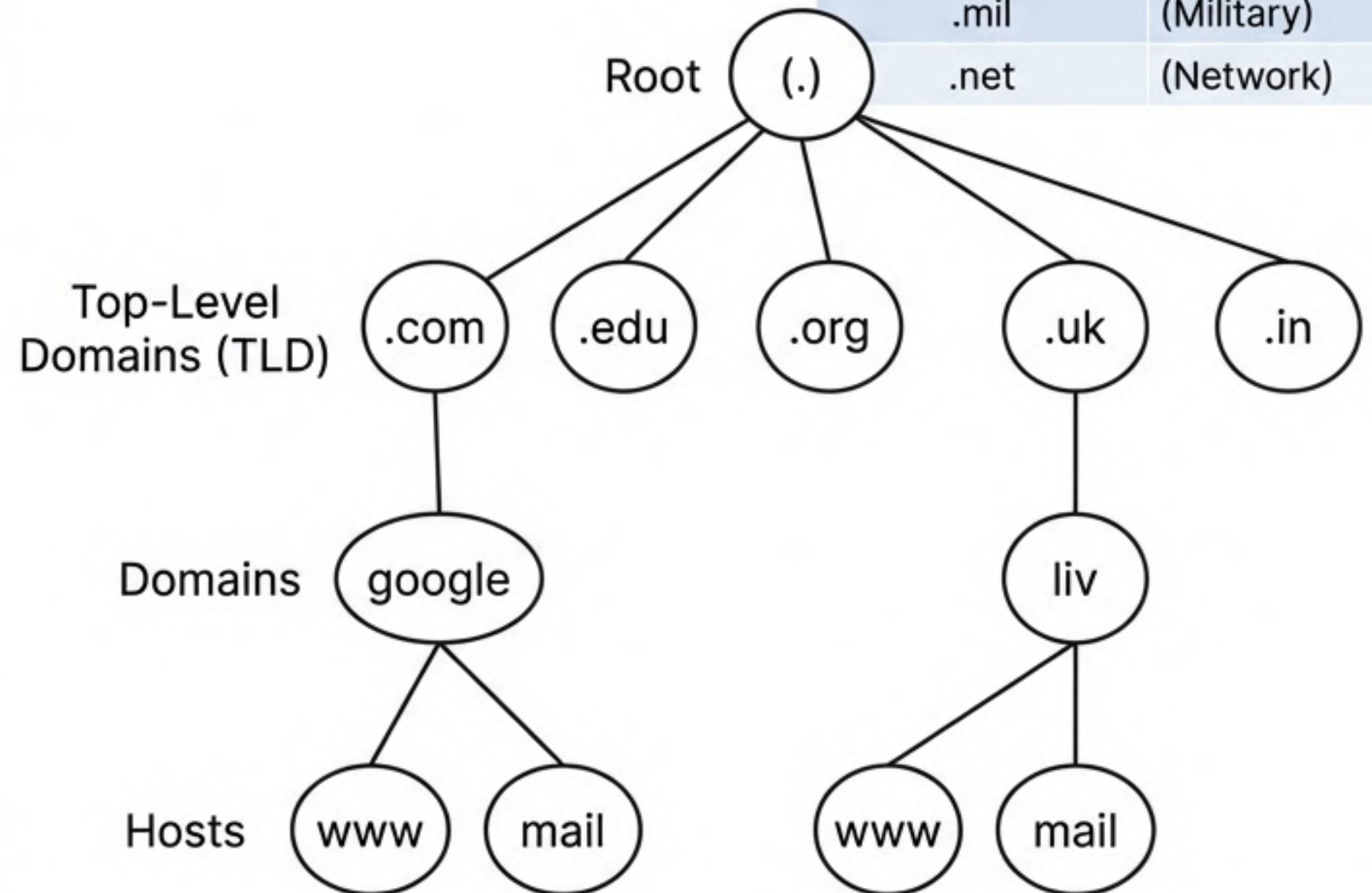
The Phonebook of the Internet

DNS maps human-readable domain names to numerical IP addresses. It uses a hierarchical distributed naming system.

Example: csc.liv.ac.uk

- csc: Host Number
- liv: Domain Name
- ac.uk: Top-Level Domain

Top-Level Domain	General Purpose
.com	(Commercial)
.edu	(Educational)
.org	(Nonprofit)
.gov	(Government)
.mil	(Military)
.net	(Network)



Web Browsers: The User Agent

A software application for retrieving, presenting, and traversing information resources on the World Wide Web.

Google Chrome



- Safe Browsing API
- V8 JavaScript Engine
- High speed

Mozilla Firefox



- Open-source
- Real-time add-on facility
- High extensibility

Apple Safari



- Native to Apple ecosystem
- Optimized for efficiency

Information Retrieval & Aggregation

Search Engines

Programs that search documents for specific keywords and return results based on relevance.

Types: Crawler-based (Google), Directories (Human-edited), Hybrid.

Web Portals

A unified access point bringing information from diverse sources into a uniform way.



Summary & References

Key Takeaways

- **Protocols** are the essential rules enabling digital communication (Syntax, Semantics, Timing).
- The **TCP/IP Suite** is the practical implementation of the network stack, powering the Internet.
- **Layered Architecture** separates concerns: IP handles routing, TCP ensures reliability, and Applications (HTTP, FTP, SMTP) interface with users.

References

1. Stallings, W. (2011). *Cryptography and Network Security: Principles and Practice* (5th Edition). Prentice Hall.
2. Stallings, W. (2007). *Data and Computer Communications* (8th Edition). Pearson Education.
3. Stallings, W. *Network Security Essentials: Applications and Standards* (Global Edition). Pearson.

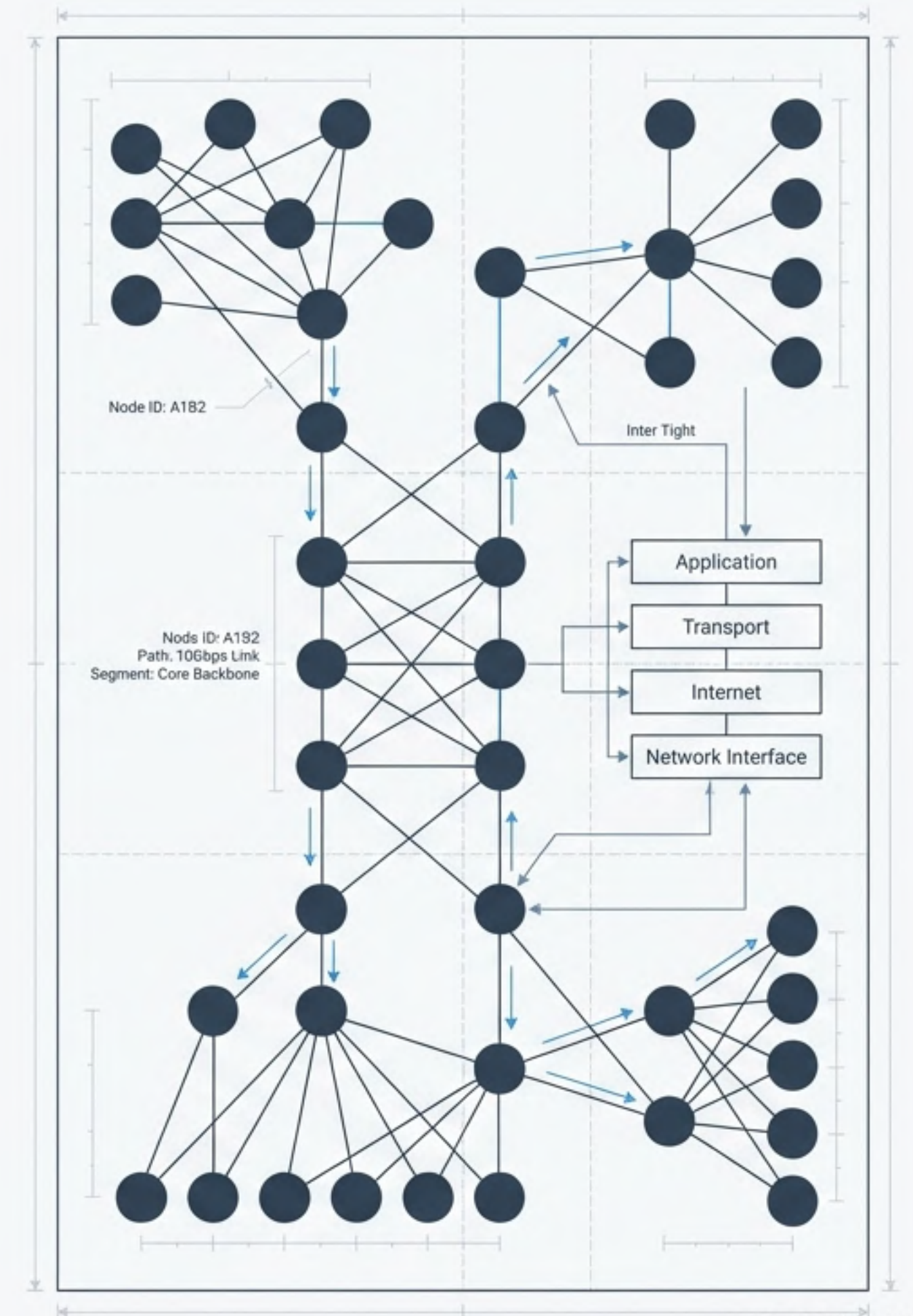
The TCP/IP Protocol Suite

Architecture, Operations, and Layered Analysis

Inter Tight Swiss Modernism

A Comprehensive Overview for Network Engineering | Third-Year Undergraduates

Presented By: Dr. Richa Kumari



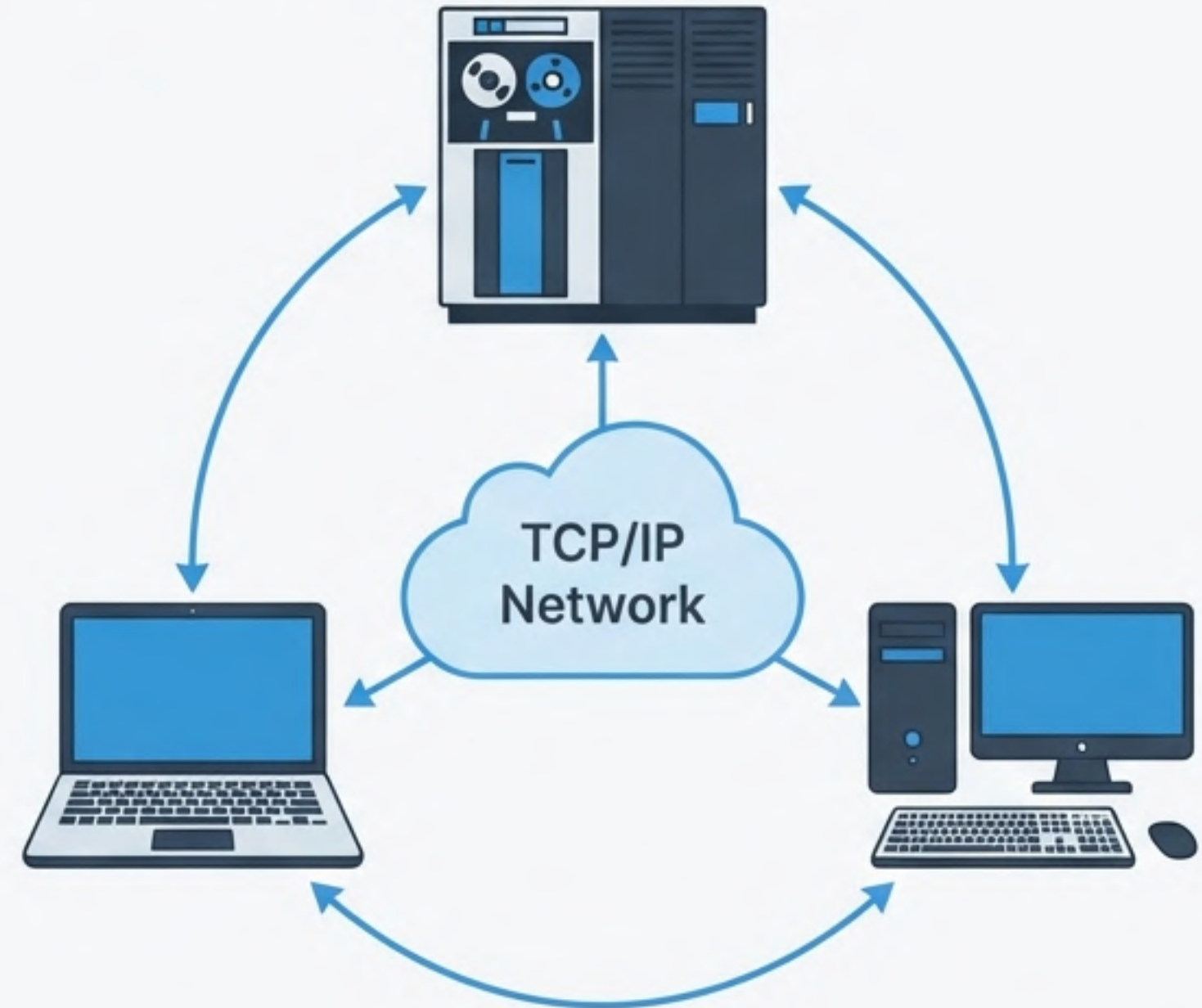
Introduction to the Internet Protocol Suite

Definition:

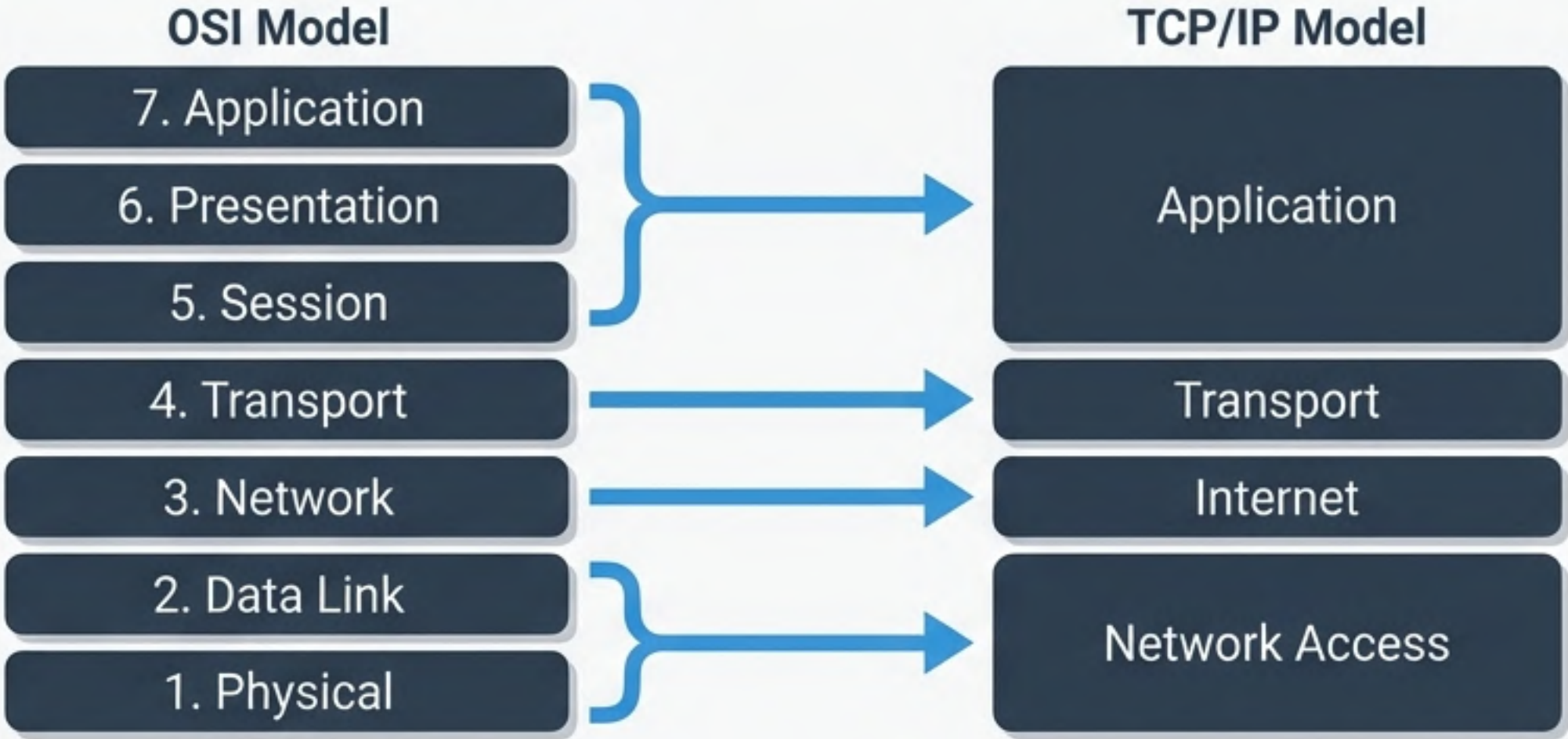
TCP/IP (Transmission Control Protocol/Internet Protocol) is the de facto standard suite of communications protocols used to interconnect network devices on the Internet.

- ✓ **Scalability:** Connects small LANs to the global Internet seamlessly.
- ✓ **Hardware Independence:** Agnostic to the underlying physical network hardware.
- ✓ **Client/Server Architecture:** Supports the request/response model essential for modern computing.
- ✓ **Routing Support:** Compatible with standard routing protocols to establish connections between disparate computers.

Heterogeneous Networking



Architectural Comparison: TCP/IP vs. OSI

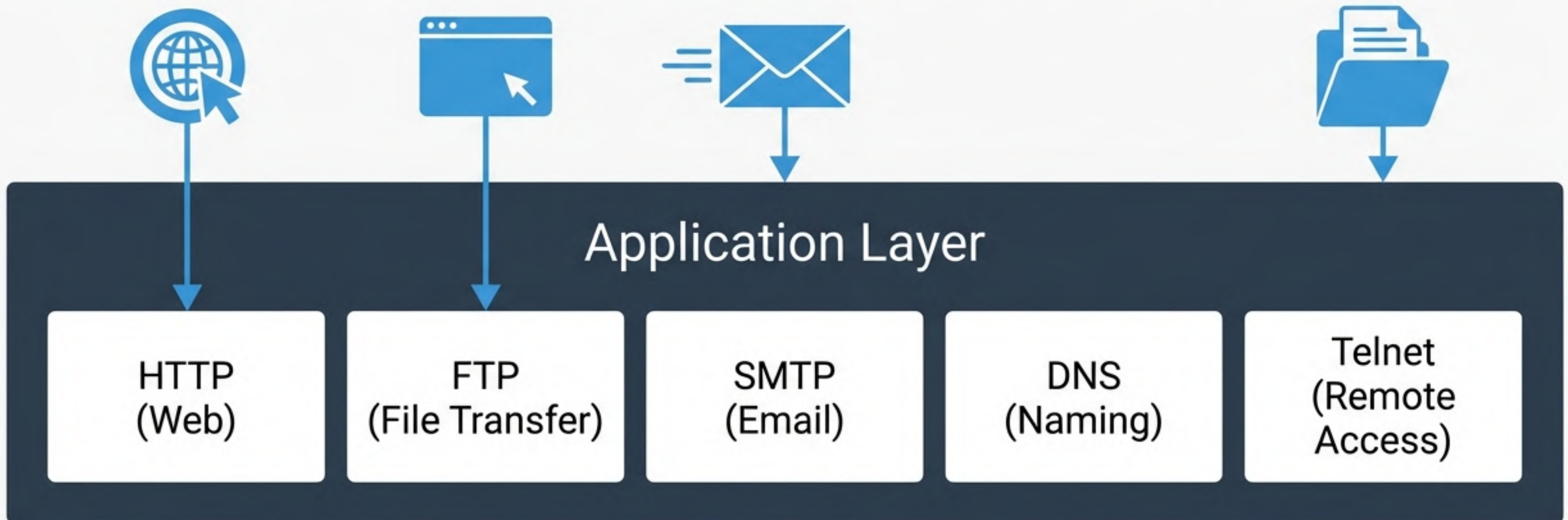


Key Distinctions

Feature	OSI Model	TCP/IP Model
Timing	Defined before the internet	Defined after the internet
Layering	Strict layering	Loosely layered
Protocol Nature	Protocol independent	Protocol dependent
Credibility	Theoretical reference	Practical standard (More credible)

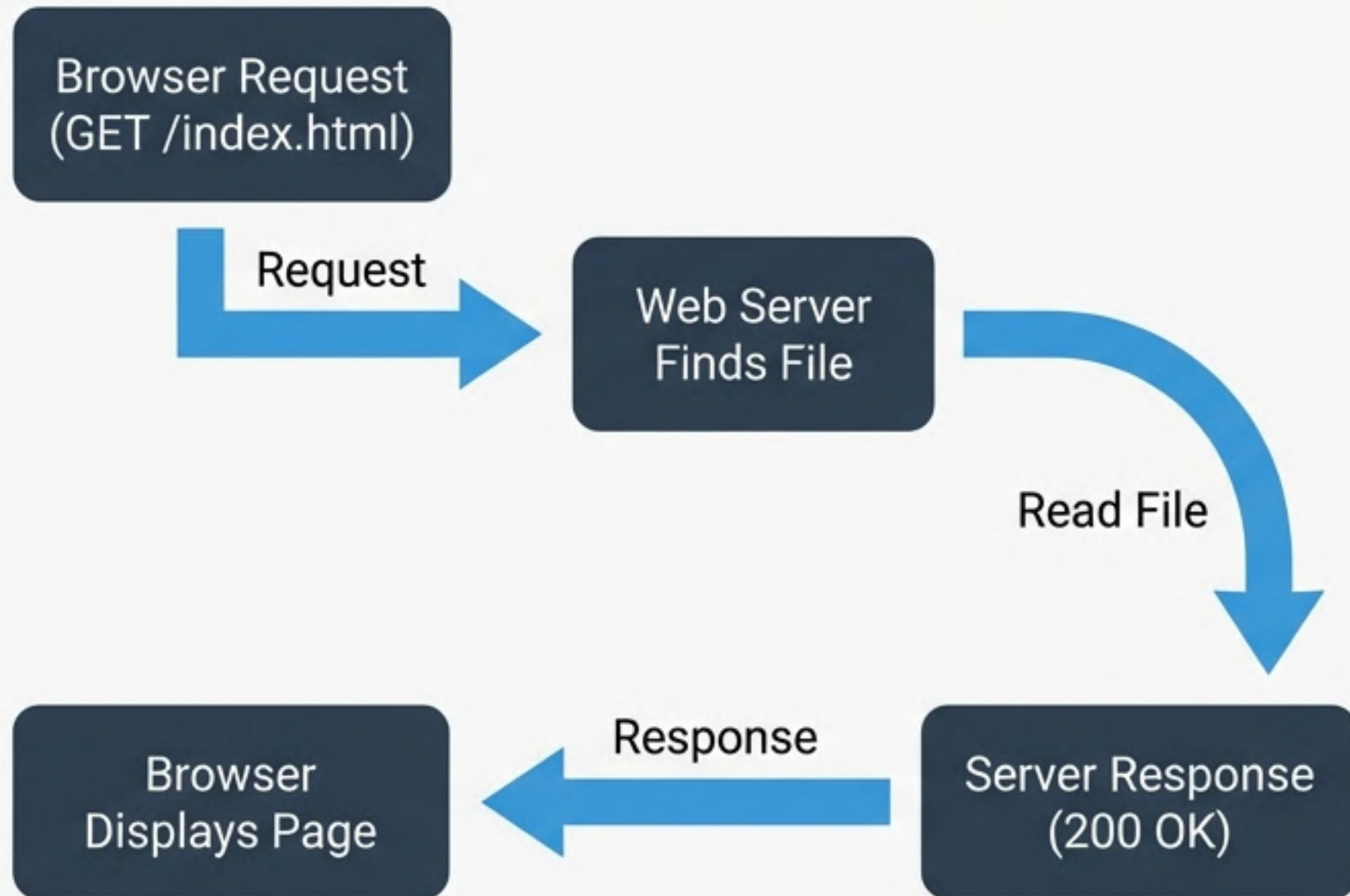
The Application Layer

The interface between software applications and network services. This layer defines data formatting, encryption, and user interfaces.



The Web: HTTP & Browsers

The Process



The Tool

The Web Browser

Definition: Software application for retrieving, presenting, and traversing information resources on the World Wide Web.

User Agents Examples:

-  Google Chrome (Uses V8 JavaScript Engine)
-  Mozilla Firefox (Open Source)
-  Apple Safari (Native to iOS)
-  Internet Explorer (Legacy Windows Default)

Information Retrieval: Search Engines & Portals

Search Engines



Function: Filter billions of web pages to transform data into relevant results.

Types:

- **Crawler-based:** Automated indexing (e.g., Google, Yahoo).
- **Directories:** Human-edited lists (e.g., dmoz.org).
- **Hybrid:** Combination of both.

Web Portals



Definition: A uniform interface bringing diverse sources into a dedicated display area.

Features: Email, forums, search, and shopping in one configured dashboard.

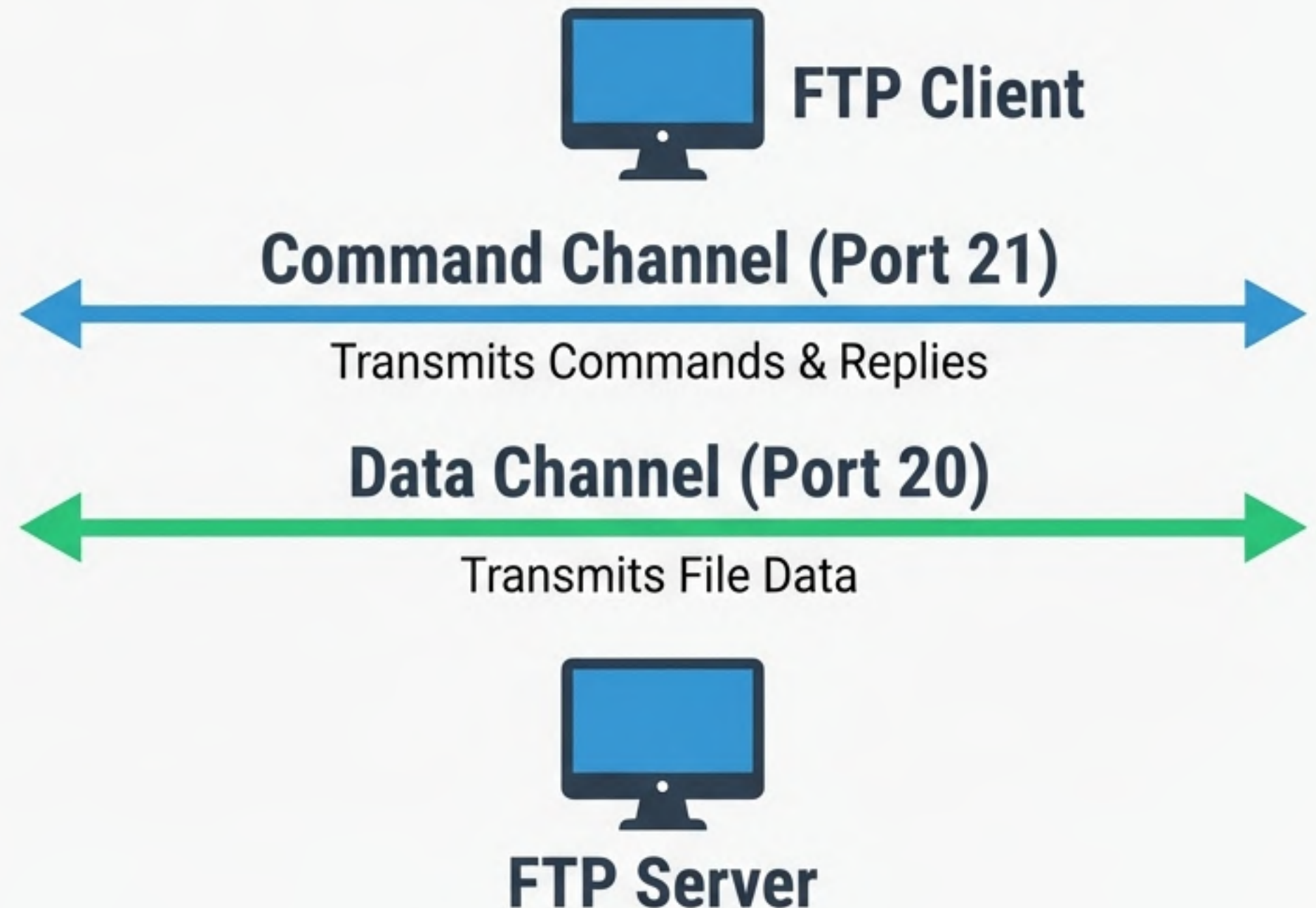
File Transfer Protocol (FTP)

Definition:

Standard network protocol used to transfer files from one host to another over a TCP-based network.

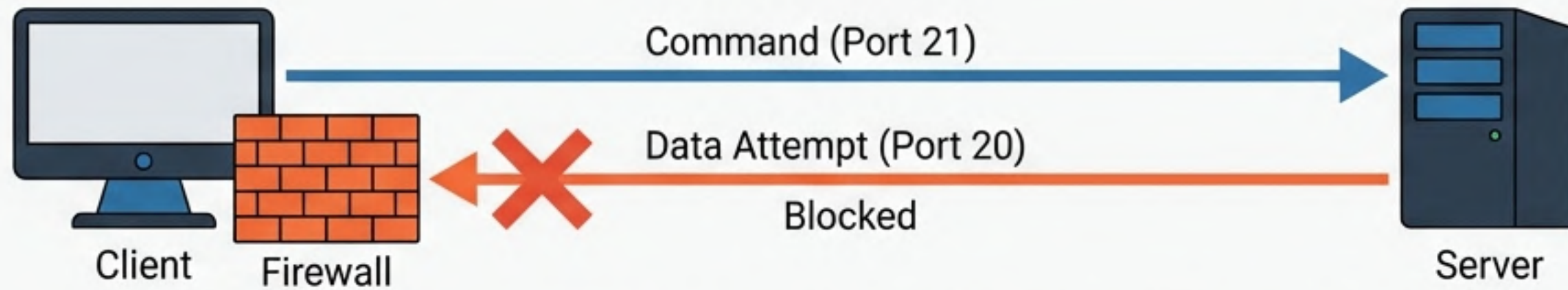
Key Architecture:

FTP is unique because it uses a Dual-Channel Architecture.



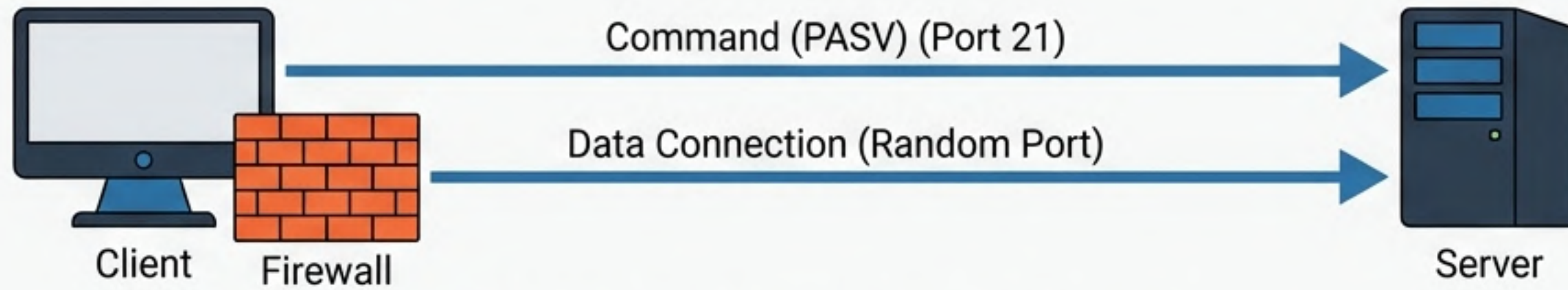
Deep Dive: Active vs. Passive FTP Modes

Active Mode (The Problem)



Roboto. Server attempts to connect back to Client. Often blocked by client-side firewall.

Passive Mode (The Solution)



Client initiates both connections. Firewall friendly.

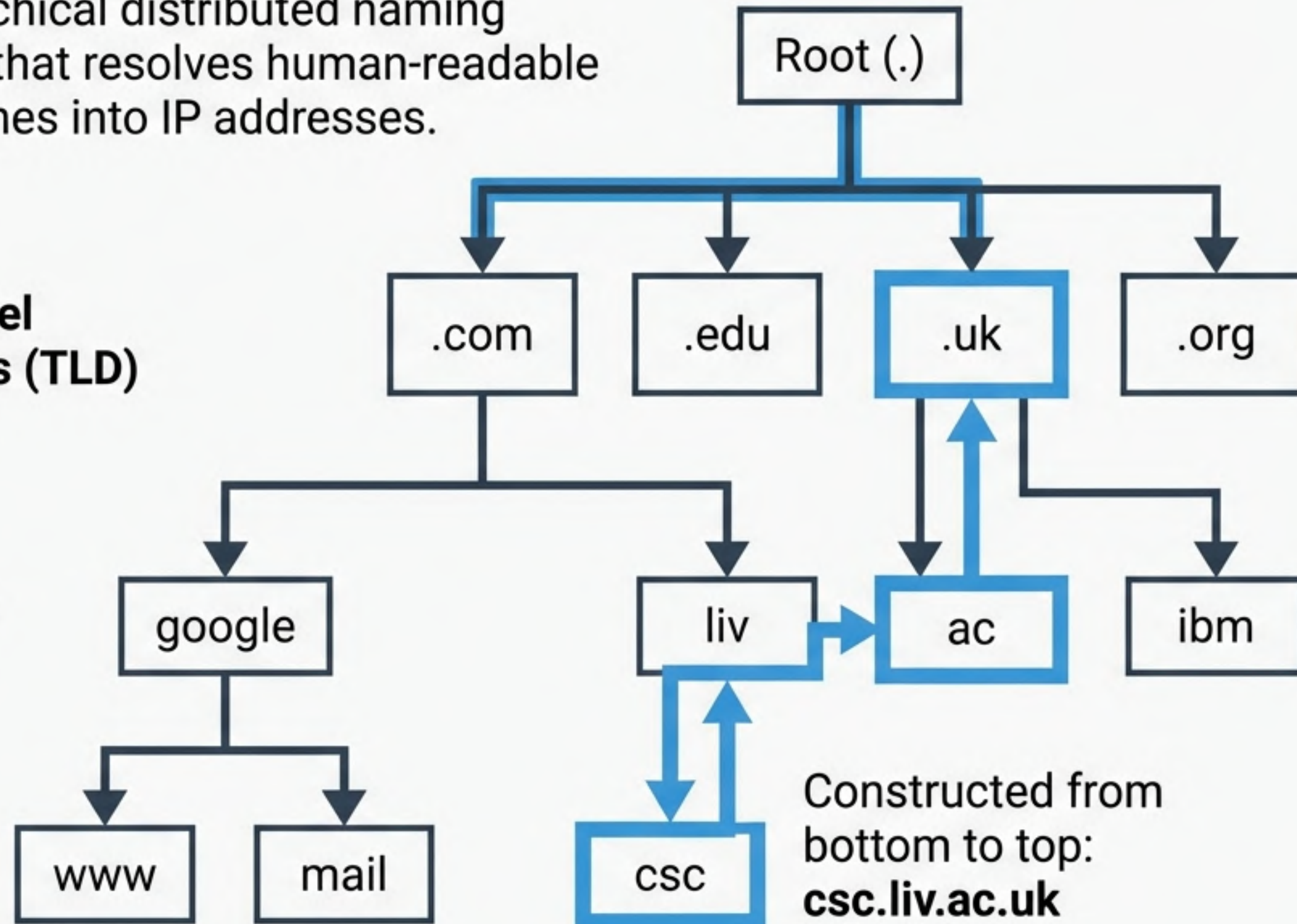
Domain Name System (DNS)

A hierarchical distributed naming system that resolves human-readable hostnames into IP addresses.

Top-Level Domains (TLD)

Domain Name

Host



TLD Categories

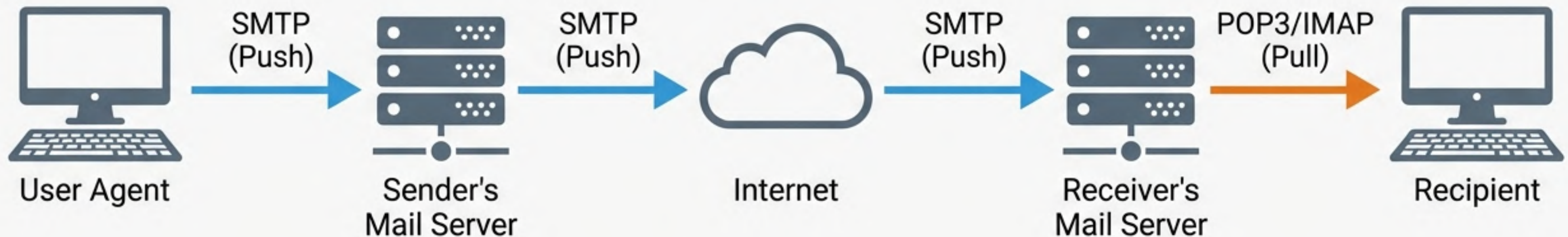
- Generic: .com, .edu, .gov, .mil, .org
- Country Code: .au (Australia), .in (India), .uk (United Kingdom)

Simple Mail Transfer Protocol (SMTP)

Role: The Internet standard for electronic mail transmission using TCP Port 25.

Mechanism: A "Push" protocol used to send mail from sender to server, and server to server.

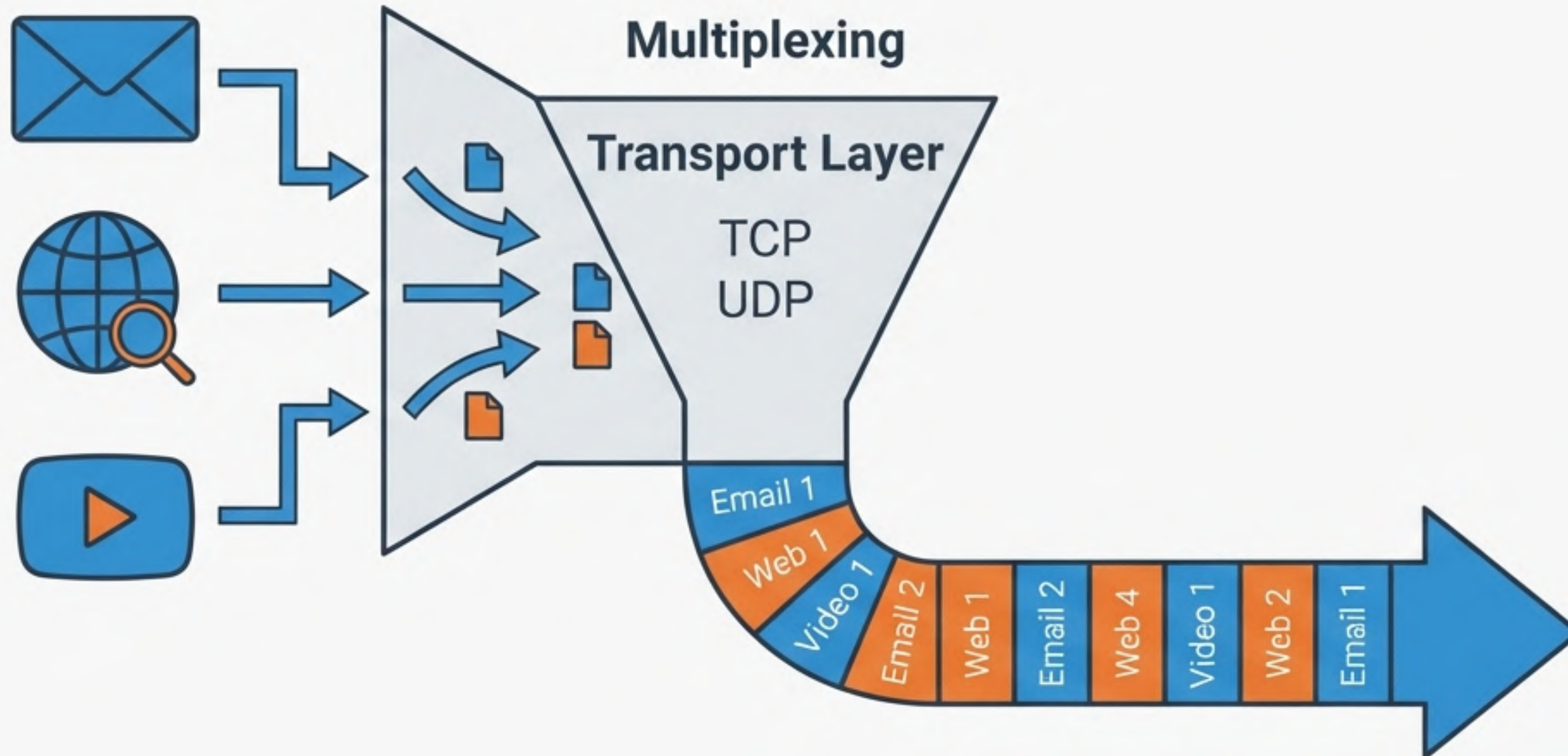
The Email Journey



Note: Emphasize that SMTP handles the sending/relay steps, while POP/IMAP handles the final retrieval.

The Transport Layer

Core Function: Provides data transfer services to the Application layer, handling multiplexing, segmentation, and error control.



Responsibilities:

- Multiplexing: Managing multiple parallel streams.
- Segmentation: Splitting data into manageable packets.
- Traffic Control: Flow control and error checking.

Transport Protocols: TCP vs. UDP

TCP (Transmission Control Protocol)

- **Type:** Connection-Oriented (Reliable)
- **Key Features:** Error detection, retransmission of lost packets, in-order delivery. Provides for virtual circuits.
- **Analogy:** A Certified Letter (Receipt guaranteed).
- **Use Cases:** Web (HTTP), Email (SMTP), File Transfer (FTP).



UDP (User Datagram Protocol)

- **Type:** Connectionless (Best-Effort)
- **Key Features:** No error correction, no retransmission, low overhead/latency.
- **Analogy:** Sending a Postcard (No guarantee).
- **Use Cases:** Streaming Video, VoIP, Gaming.

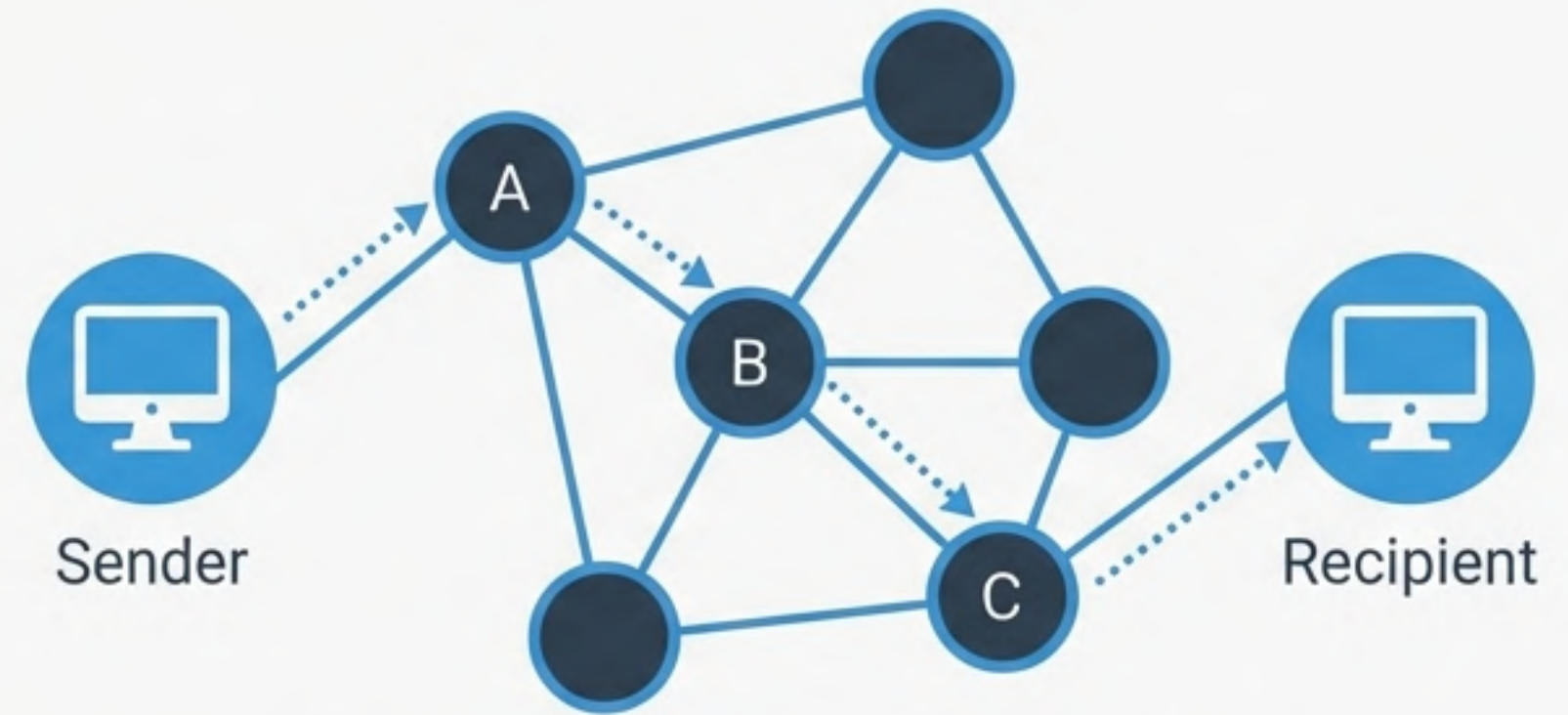


The Internet Layer (IP)

The glue that holds the internet together. IP is connectionless and provides “best-effort” delivery.

Core Functions:

- **Addressing:** Logical IP addresses for global identification.
- **Routing:** Path selection across network boundaries.
- **Fragmentation:** Breaking packets to fit network limits (MTU).



Step 2: Packets travel from router to router according to the IP protocol.

Addressing and Port Numbers

The Socket Concept

Communication Endpoint = IP Address (Host) + Port Number (Process/Application)

Protocol	Port Number	Service
Echo	7	Echo Service
FTP Data	20	File Transfer Data
FTP Control	21	File Transfer Command
Telnet	23	Remote Terminal
SMTP	25	Email Sending
DNS	53	Domain Name Service
HTTP	80	World Wide Web
POP3	110	Email Retrieval

Summary & References

Key Takeaways

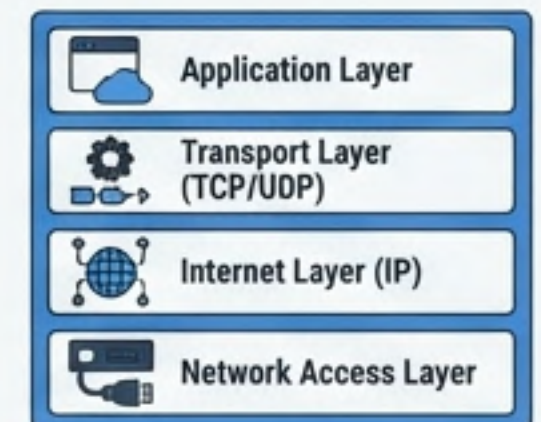
- TCP/IP is a 4-layer, protocol-dependent suite prioritizing connectivity and scalability.
- The Application layer handles user interfaces, while the Transport layer (TCP/UDP) manages reliability and speed.
- Operational modes, such as Active vs. Passive FTP, are critical for navigating firewalls and network security.
- IP provides the addressing and routing logic that allows global internetworking.

Academic References

Tanenbaum, A. S., & Wetherall, D. J. (2011). *Computer Networks* (5th ed.). Pearson.

Stallings, W. (2017). *Cryptography and Network Security: Principles and Practice* (7th ed.). Pearson.

Forouzan, B. A. (2012). *Data Communications and Networking* (5th ed.). McGraw-Hill.



TCP/IP Layers and Their Functions

Architectural Decomposition &
Security Integration

Third-Year Undergraduate Network
Security Module

By: Dr. Richa Kumari



The TCP/IP Protocol Suite: The Language of the Internet

The **TCP/IP** (Transmission Control Protocol/Internet Protocol) suite is the practical implementation of the internet, distinct from the theoretical **OSI model**. It enables open system interconnection across diverse hardware.

Key Characteristics

-  **Scalable:** Supports client/server architecture.
-  **Interoperable:** Connects diverse hardware and operating systems.
-  **Robust:** Automatically recovers from failure (rerouting).

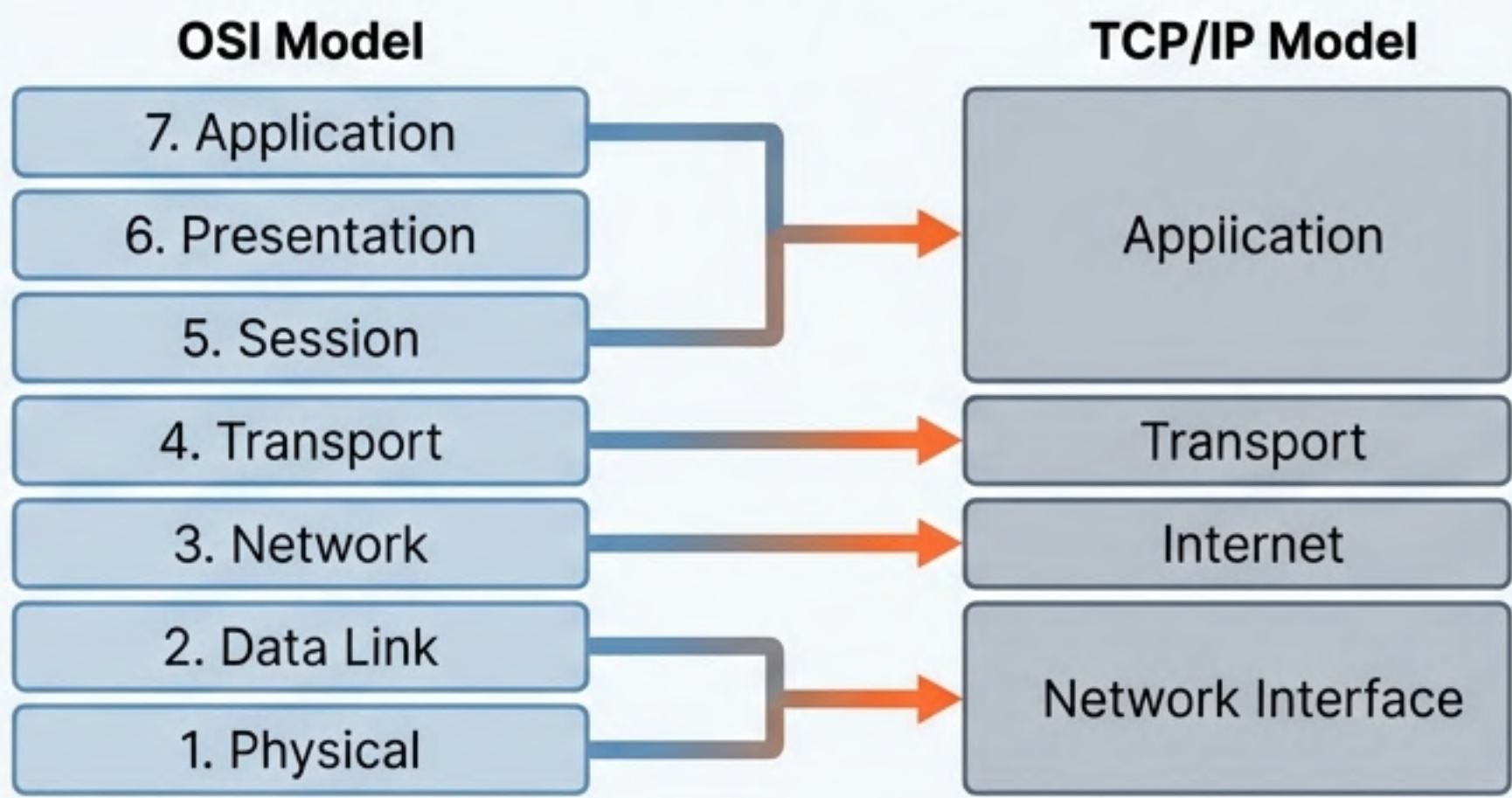
Layer 4: Application

Layer 3: Transport

Layer 2: Internet

**Layer 1: Network Interface
(Host-to-Network)**

Comparative Architecture: OSI Reference vs. TCP/IP Implementation



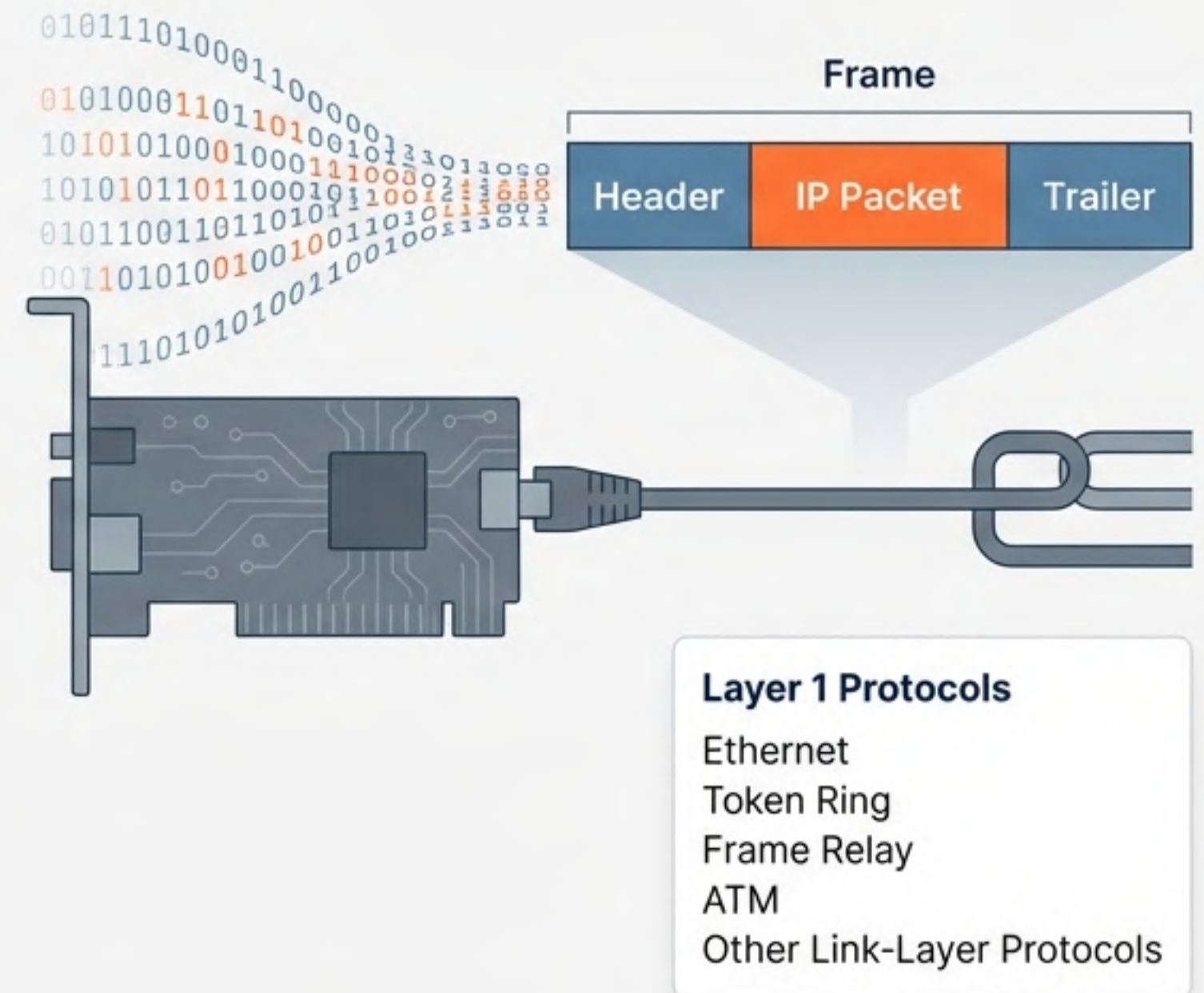
	Feature	OSI Model	TCP/IP Model
1	Nature	Reference Model (Generic)	Implementation (Protocol-Dependent)
2	Layering	Strict Layering	Loosely Layered
3	Development	Defined before protocols	Defined after protocols
4	Reliability	All layers guarantee delivery	Transport layer (TCP) responsibility

Layer 1: The Network Interface Layer

Also known as: Host-to-Network Layer

This is the foundation of the stack, responsible for placing TCP/IP packets on the network medium and receiving them. It handles the physical transmission of data bits.

- **Functions:** Encapsulates IP packets into frames; handles hardware addressing.
- **Key Protocols:** Ethernet, Token Ring, Frame Relay, ATM.



Layer 2: The Internet Layer

Connectionless Internetworking & Path Determination

Core Function



Determines the best path through the network (**Routing**).
Handles packet switching.

Primary Protocol: IP (Internet Protocol)



A routable protocol responsible for IP addressing, fragmentation, and reassembly. It operates on a "Best Effort" delivery model (unreliable, no guarantees).

Addressing



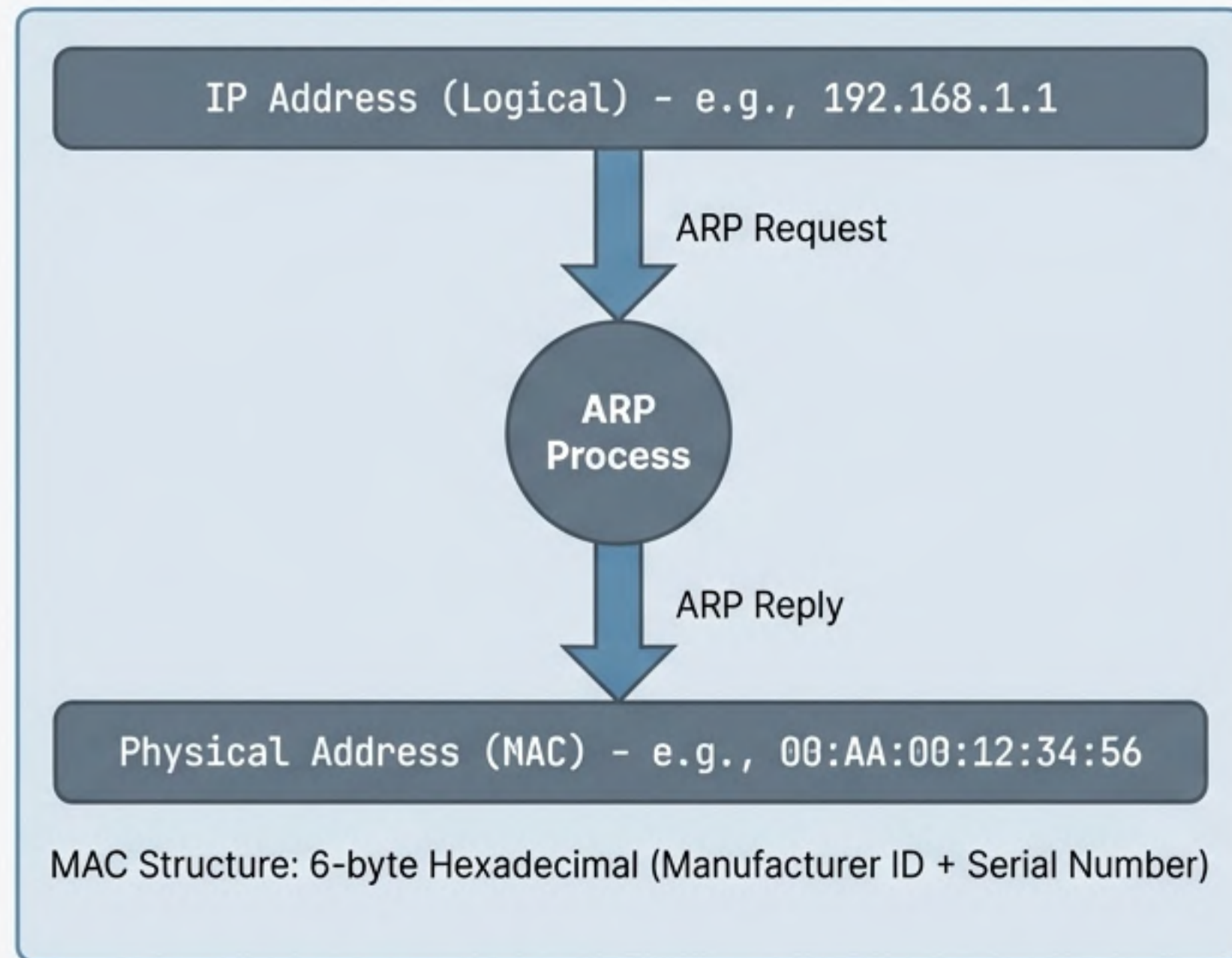
Logical addressing (IP Addresses) identifies devices across different networks.

Core Protocols: IP, ARP, ICMP, IGMP

Internet Layer Utilities: ARP and ICMP

ARP (Address Resolution Protocol)

The Bridge between Layer 3 (IP) and Layer 1 (Physical).

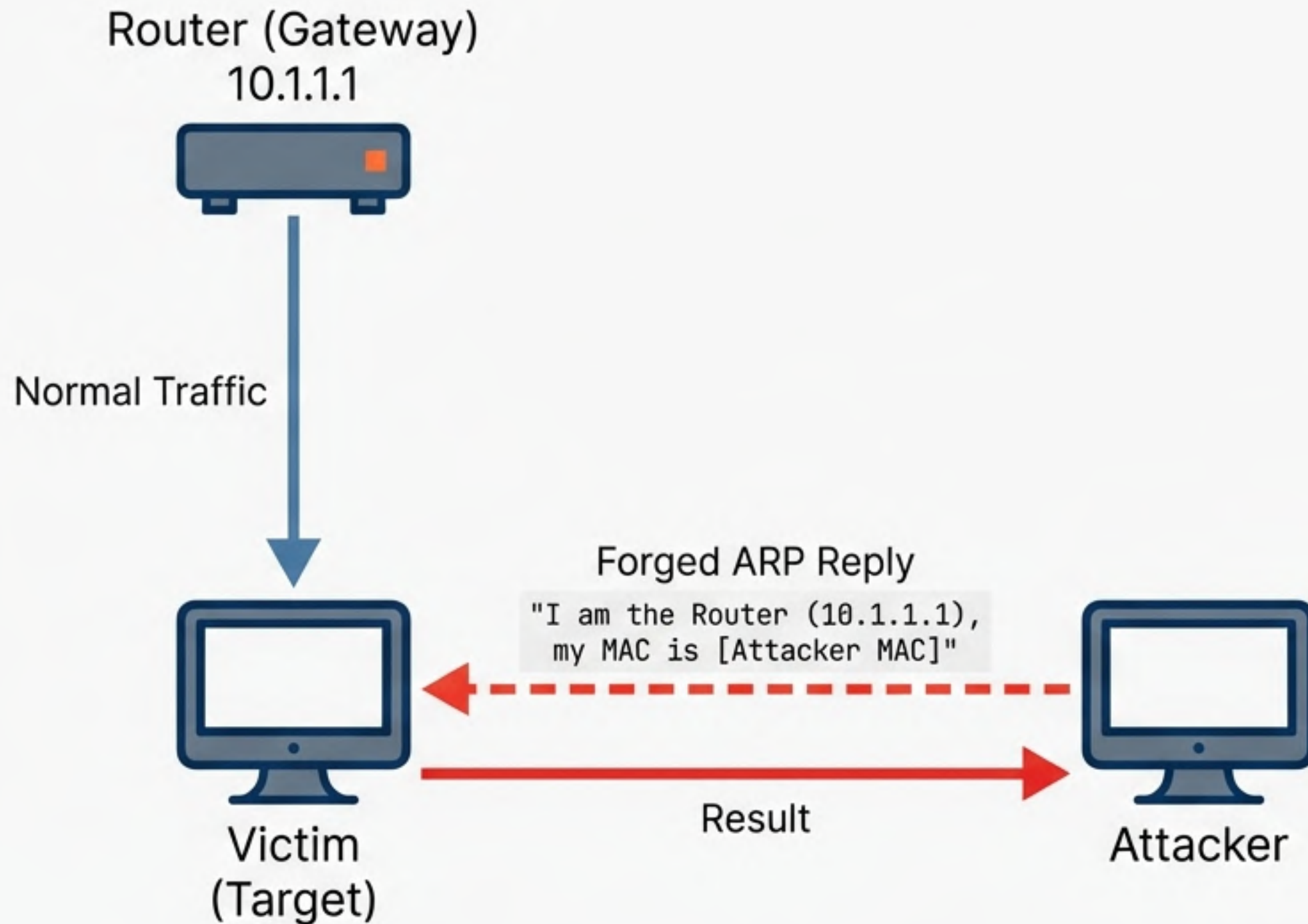


ICMP (Internet Control Message Protocol)

Network Diagnostics & Error Reporting.

- Function: Identifies problems with packet delivery.
- Utility: Used by the 'PING' command to verify physical connectivity and routing.

Layer 2 Vulnerability: ARP Poisoning



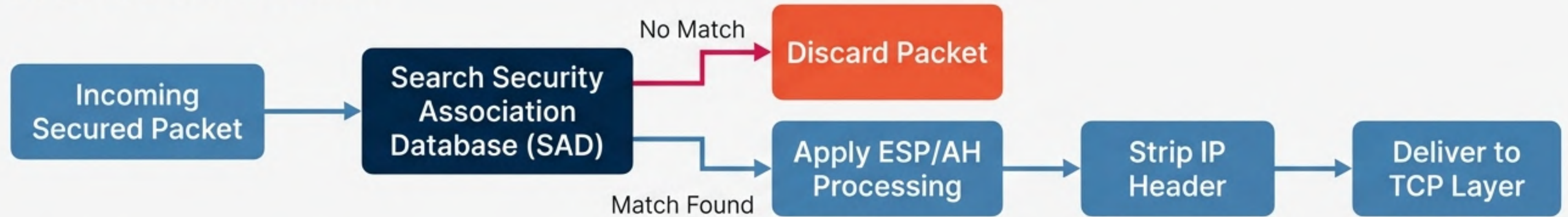
ARP Spoofing/Poisoning is a trust-based access attack.

Because **ARP** is stateless, a device will accept an ARP reply even without a request.

An attacker floods the target with forged entries, associating the router's IP with the attacker's **MAC address**. This facilitates **Man-in-the-Middle (MitM)** attacks.

Securing the Internet Layer: IPsec & ESP

The IPsec Process



ESP (Encapsulating Security Payload)

- Confidentiality (Encryption)
- Data Origin Authentication
- Connectionless Integrity
- Anti-Replay Service

Transport Mode



Note: In Transport Mode, only the payload (TCP Segment) is encrypted.

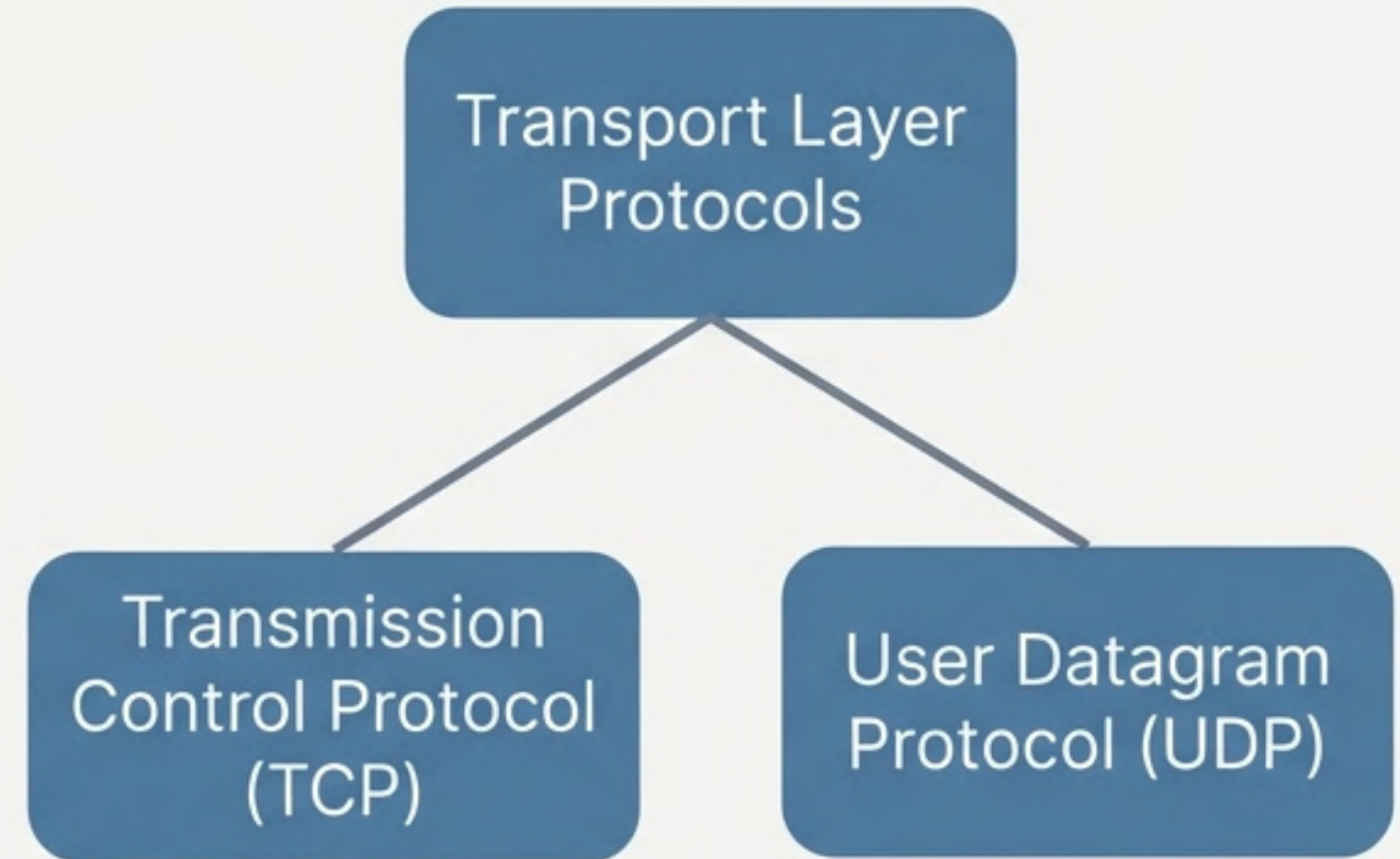
Layer 3: The Transport Layer

End-to-End Delivery & Reliability

The Transport layer sits above the Internet layer, insulating applications from the complexities of the network. It ensures data moves reliably from source to destination.

Key Responsibilities:

- **Segmentation:** Breaking message data into efficient units.
- **Sequencing:** Reordering packets upon arrival.
- **Session Control:** Maintaining conversation integrity.

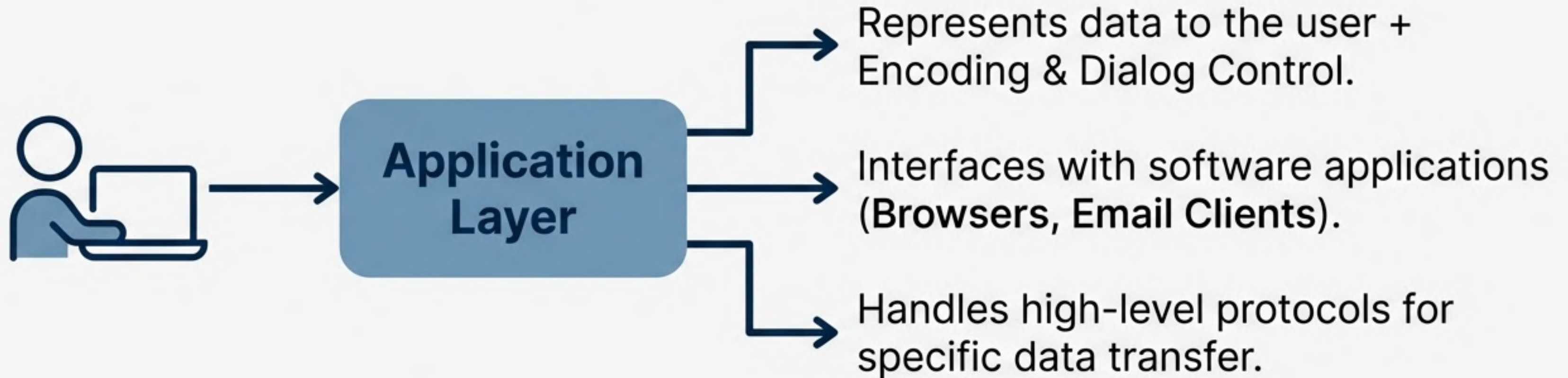


Transport Protocols: Fataals: TCP vs. UDP

TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
• Type: Connection-Oriented (The "Handshake").	• Type: Connectionless (Fire and Forget).
• Reliability: High. Guarantees delivery, error correction, and flow control.	• Reliability: "Best Effort" (Unreliable). No error correction.
• Data Transfer: Sent as a stream with no particular structure.	• Data Transfer: Sent in discrete packages.
• Overhead: High (due to acknowledgements).	• Overhead: Very Low.
• Use Case: Web browsing, Email, File Transfer (Precision is critical).	• Use Case: Streaming Video/Audio, VoIP (Speed is critical).

Layer 4: The Application Layer

User Interface & Data Representation



Data Representation:

- MIME types, JPEG, ASCII, Encryption/Decryption.

Core Application Layer Protocols

HTTP

Hypertext Transfer Protocol.
Foundation of the World
Wide Web.

FTP / TFTP

File Transfer Protocol.
Authenticated (FTP) or
Trivial (TFTP) file transfer.

SMTP

Simple Mail Transport
Protocol. Standard for
sending electronic mail.

DNS

Domain Name Service.
Resolves Host Names
(google.com) to IP
Addresses.

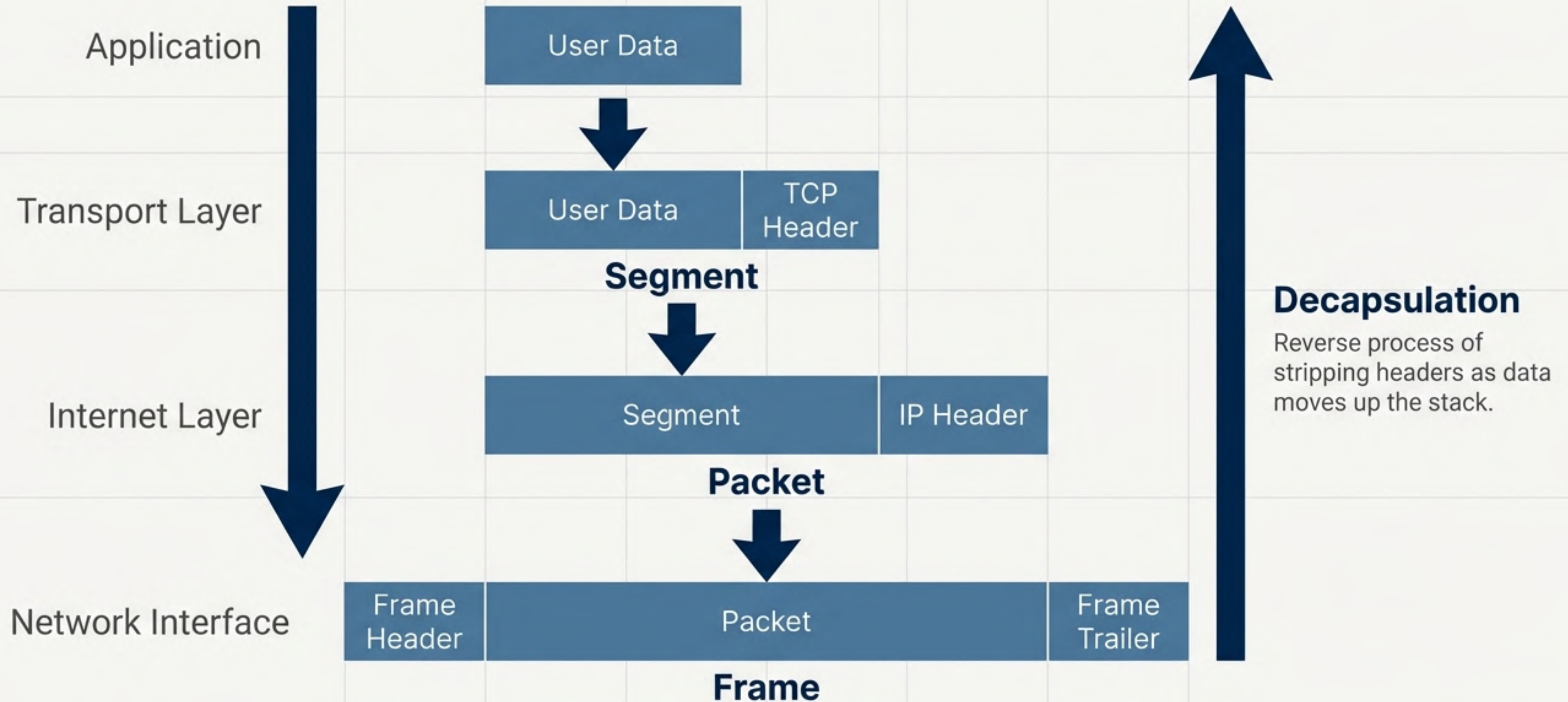
Telnet

Remote command-line
connection protocol.

SNMP

Simple Network
Management Protocol.

The Mechanism of Data Encapsulation



Network Security Begins at the Host



Host Security is the comprehensive approach to hardening each individual device on the network.

References & Further Reading

- Stallings, W. (2017). Cryptography and Network Security: Principles and Practice. Pearson.
- Stallings, W. Network Security Essentials: Applications and Standards. Pearson.
- Forouzan, B. A. TCP/IP Protocol Suite. McGraw-Hill Education.
- RFC 791 (Internet Protocol) & RFC 793 (Transmission Control Protocol).

Network Architecture Models

The OSI Reference Model & Security Architecture

Third-Year Undergraduate Lecture Series

By: Dr. Richa Kumari

Introduction to Open Systems Interconnection (OSI)

Developed by the International Organization for Standardization (ISO) circa 1984. The OSI Model is a framework for defining standards for connecting computers and other devices.

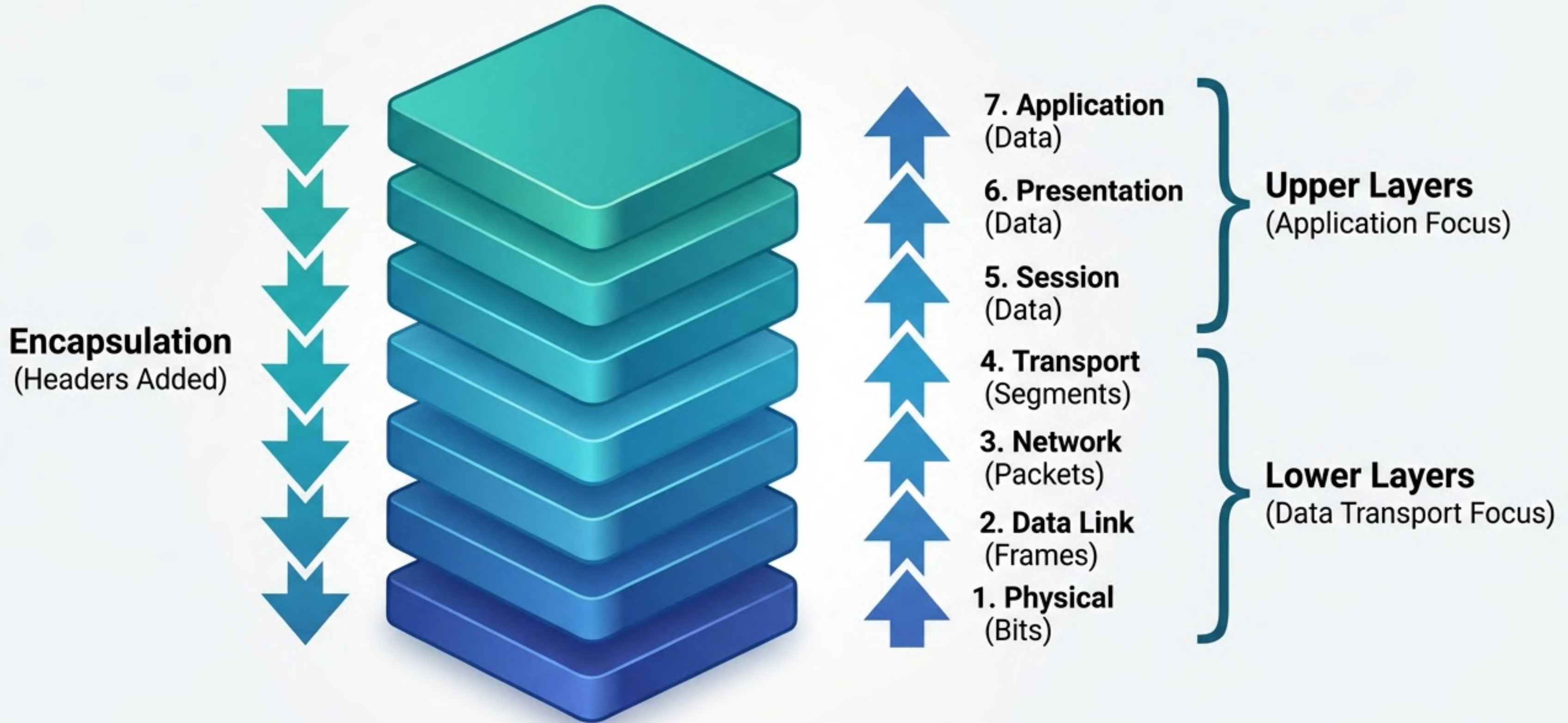
What is an Open System?

A system built on standardized protocols and interfaces, allowing it to communicate effectively with other systems regardless of the vendor.



Core Purpose: Interoperability (Removing proprietary barriers) & Standardization (Organizing functions into logical layers).

The OSI Reference Model: 7-Layer Architecture



Layer 1: The Physical Layer & Physical Security

Technical Overview

Function: Transmission of raw **Bits** over a physical medium.

Defines: Mechanical, electrical, and procedural interfaces (Voltage, pin counts, cable types).

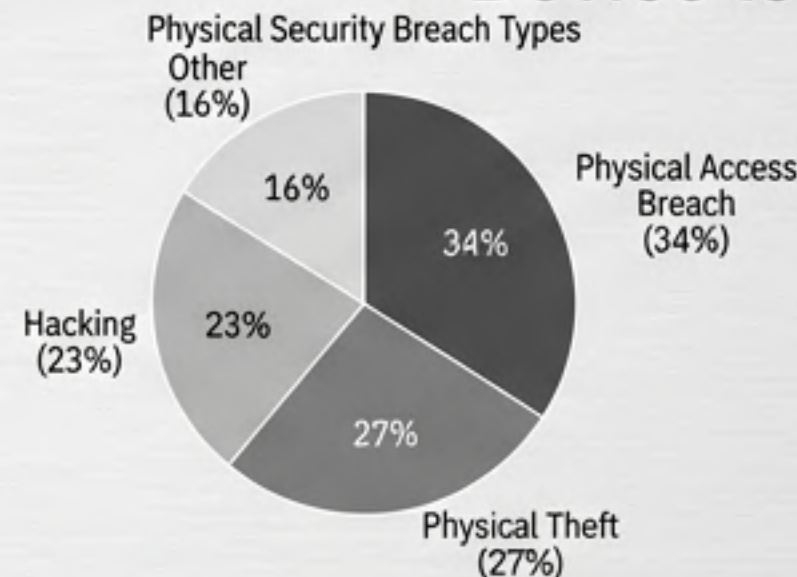
Protocols/Standards: IEEE 802.3 (Ethernet), 802.11 (Wireless), RS-232.

Physical Security

“First Rule of Network Security: Physically Safeguard Your Systems and Networks”.

Natural Threats: Floods, Fire, Earthquakes, Temperature/Humidity.

Man-Made Threats: Theft, Vandalism, Device loss, Social Engineering.



Layer 2: The Data Link Layer

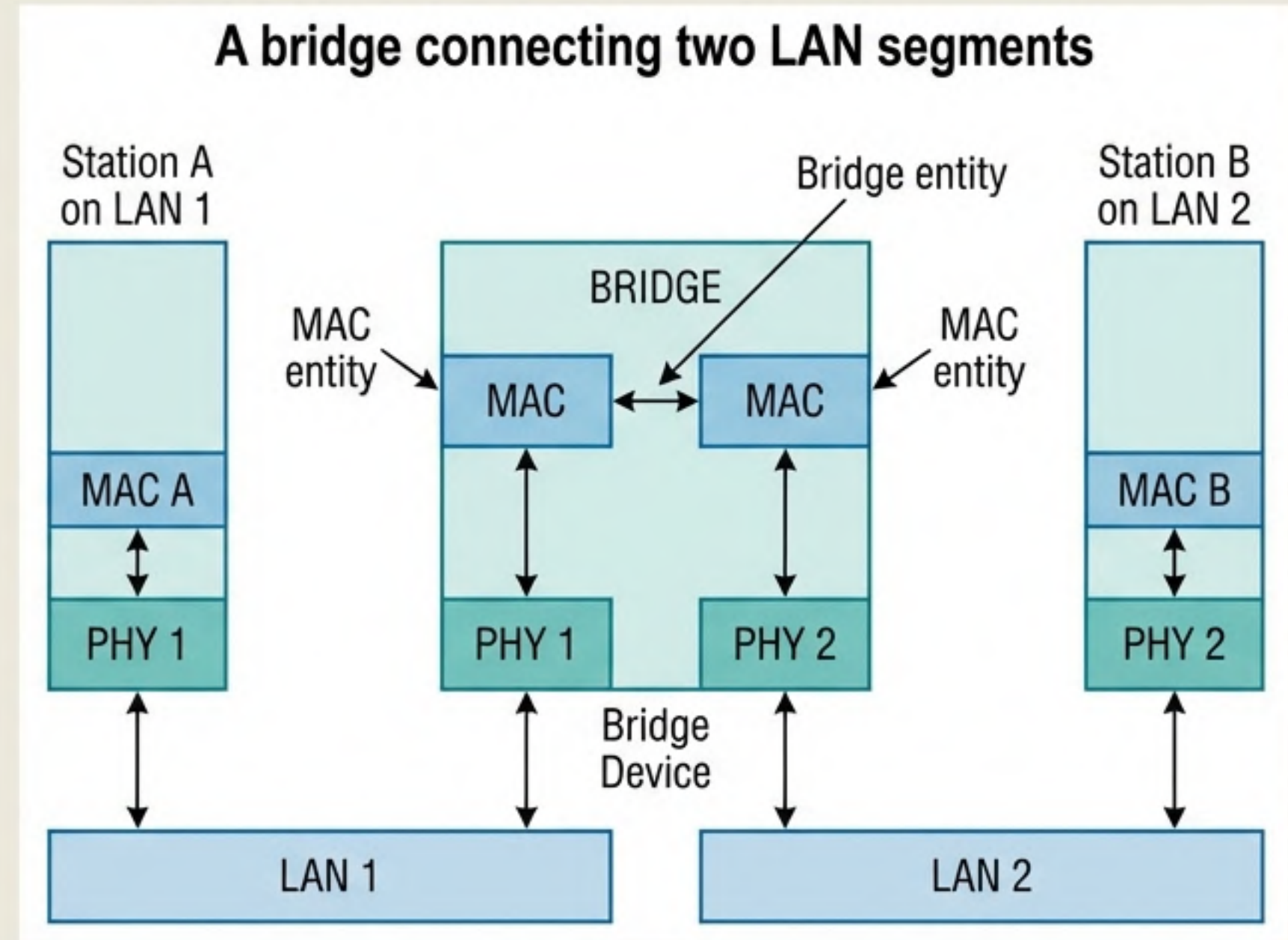
Function: Provides reliable Node-to-Node delivery. Transforms the raw transmission into a link free of errors.

Data Unit: Frame.

Key Devices: Switches, Bridges, NICs.

Sublayers

1. **MAC** (Media Access Control): Controls media access; defines physical addressing (MAC Address).
2. **LLC** (Logical Link Control): Interface between network layer and hardware.



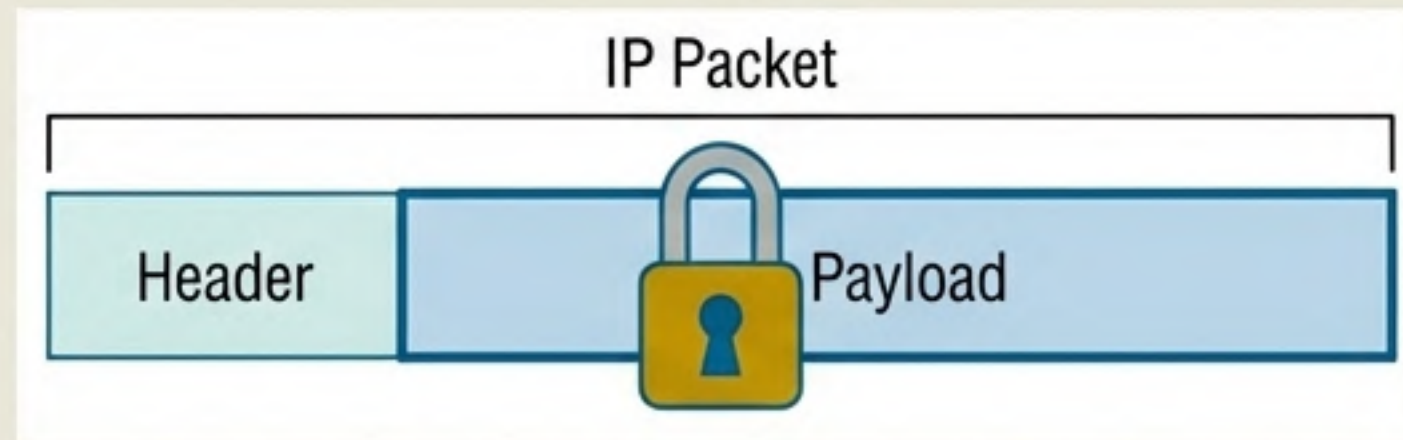
Layer 3: The Network Layer

Function: Source-to-Destination delivery across multiple networks.

- **Logical Addressing:** Translates logical addresses (IP) to physical addresses (MAC).
- **Routing:** Path selection and packet switching (e.g., using RIP).
- **Data Unit:** Packet.

Security Context: IPSec

- **IPSec** operates at Layer 3 to secure IP communications.
- **Encapsulating Security Payload (ESP):** Provides confidentiality, data origin authentication, and anti-replay services.
- **Processing:** Inbound packets are filtered; if they match a security policy, they are decrypted before passing to Layer 4.



Layer 4: The Transport Layer

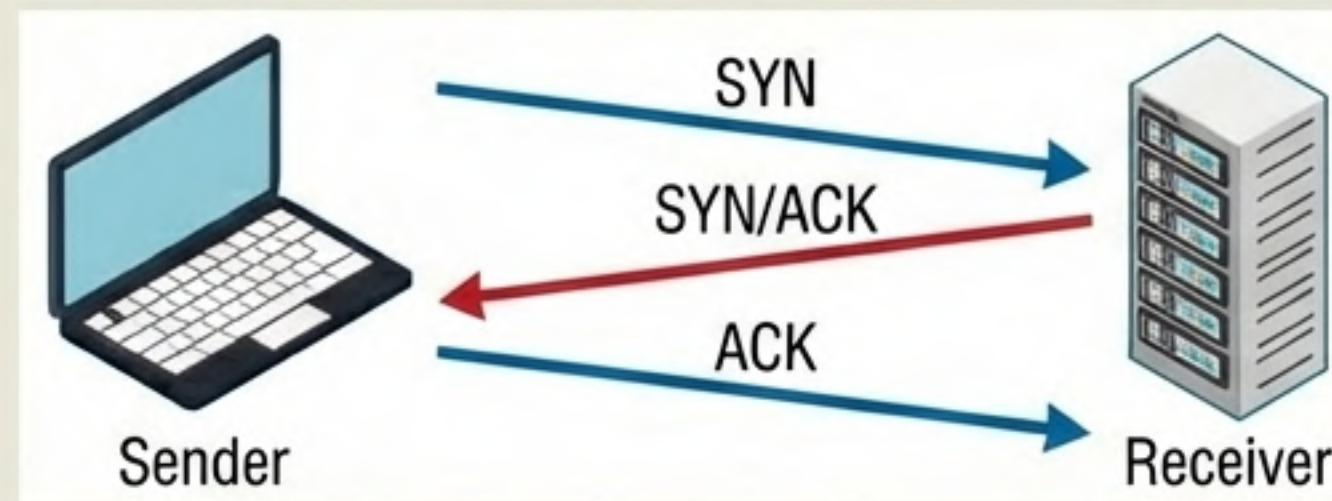
Process-to-Process Delivery | Data Unit: Segment

TCP (Transmission Control Protocol)

- Connection-oriented & Reliable.
- Heavyweight (20 byte header).
- Features: Flow control, Sequencing, Error recovery.
- Uses 3-Way Handshake.

UDP (User Datagram Protocol)

- Connectionless & Unreliable.
- Lightweight (8 byte header).
- "Fire and Forget" speed.
- Used for streaming/VoIP.



The Upper Layers: Session & Presentation

Layer 5: Session Layer

Role: Establishes, manages, and terminates sessions between hosts.

Functions:

- Dialog Control (Who speaks when).
- Synchronization (Checkpoints).
- Logical port transfers.

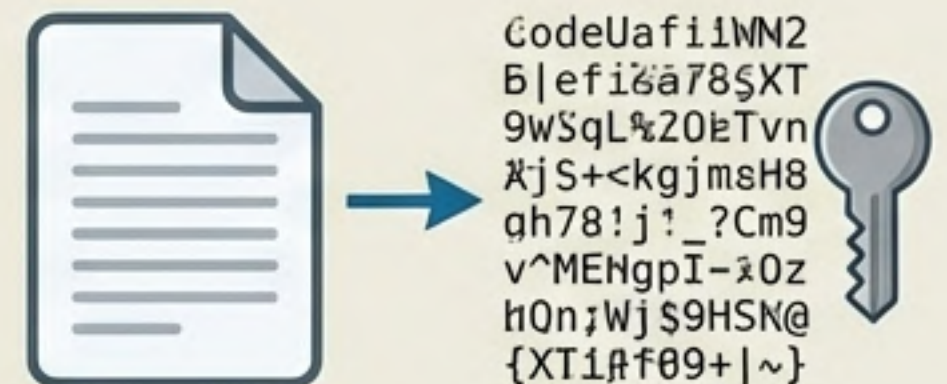
Examples: RPC, SQL, NFS.

Layer 6: Presentation Layer

Role: The 'Translator' of the network. Formats data for the Application layer.

Functions:

- **Translation:** Character code conversion (ASCII to EBCDIC).
- **Compression:** Reducing data size.
- **Encryption:** Ensuring data confidentiality (SSL/TLS lives here).



Encryption

Layer 7: The Application Layer

Function: Closest to the end-user. Provides network services directly to application processes (not the software GUI, but the communication protocols).



File Transfer

Protocols: FTP, TFTP, SFTP.



Electronic Mail

Protocol: SMTP.



Web Browsing

Protocols: HTTP, HTTPS.

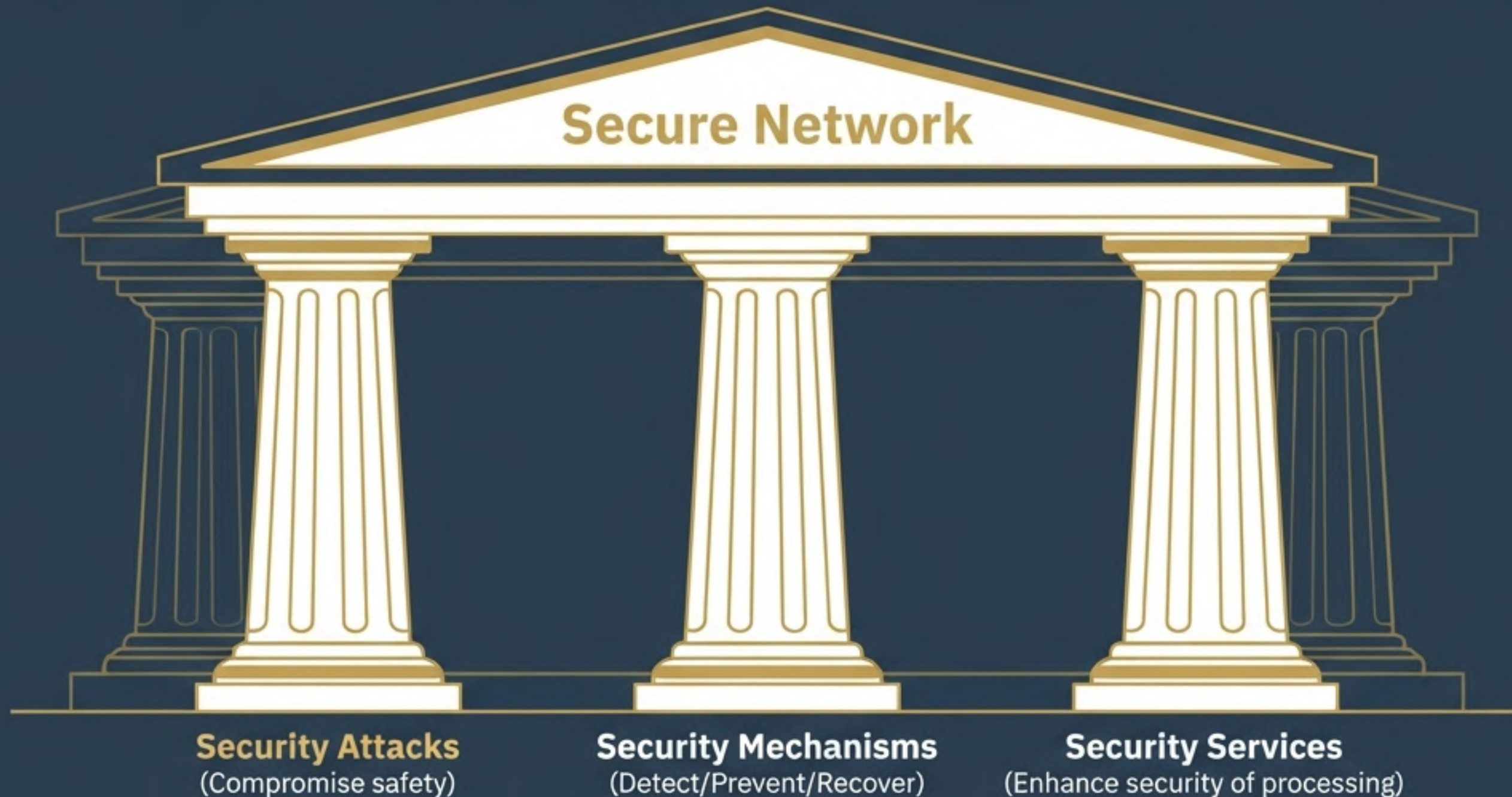


Name Resolution

Protocol: DNS.

OSI Security Architecture (ITU-T X.800)

A systematic framework defining security requirements, useful for managers and vendors to organize security features.



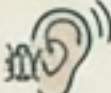
Classification of Security Attacks

Passive Attacks

Nature: Monitoring transmissions *without* affecting them.

Goal: To obtain information. Hard to detect.

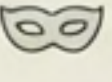
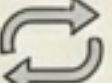
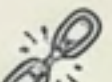
Types:

-  **Eavesdropping:** Release of message contents.
-  **Traffic Analysis:** Observing patterns.

Active Attacks

Nature: Modification of data stream or creation of a false stream.

Types:

1.  **Masquerade:** Pretending to be a different entity.
2.  **Replay:** Capture and re-transmission of messages.
3.  **Modification:** Altering legitimate messages.
4.  **Denial of Service (DoS):** Disrupting service availability.

Security Mechanisms (X.800)

Specific Security Mechanisms

(Implemented in specific layers)

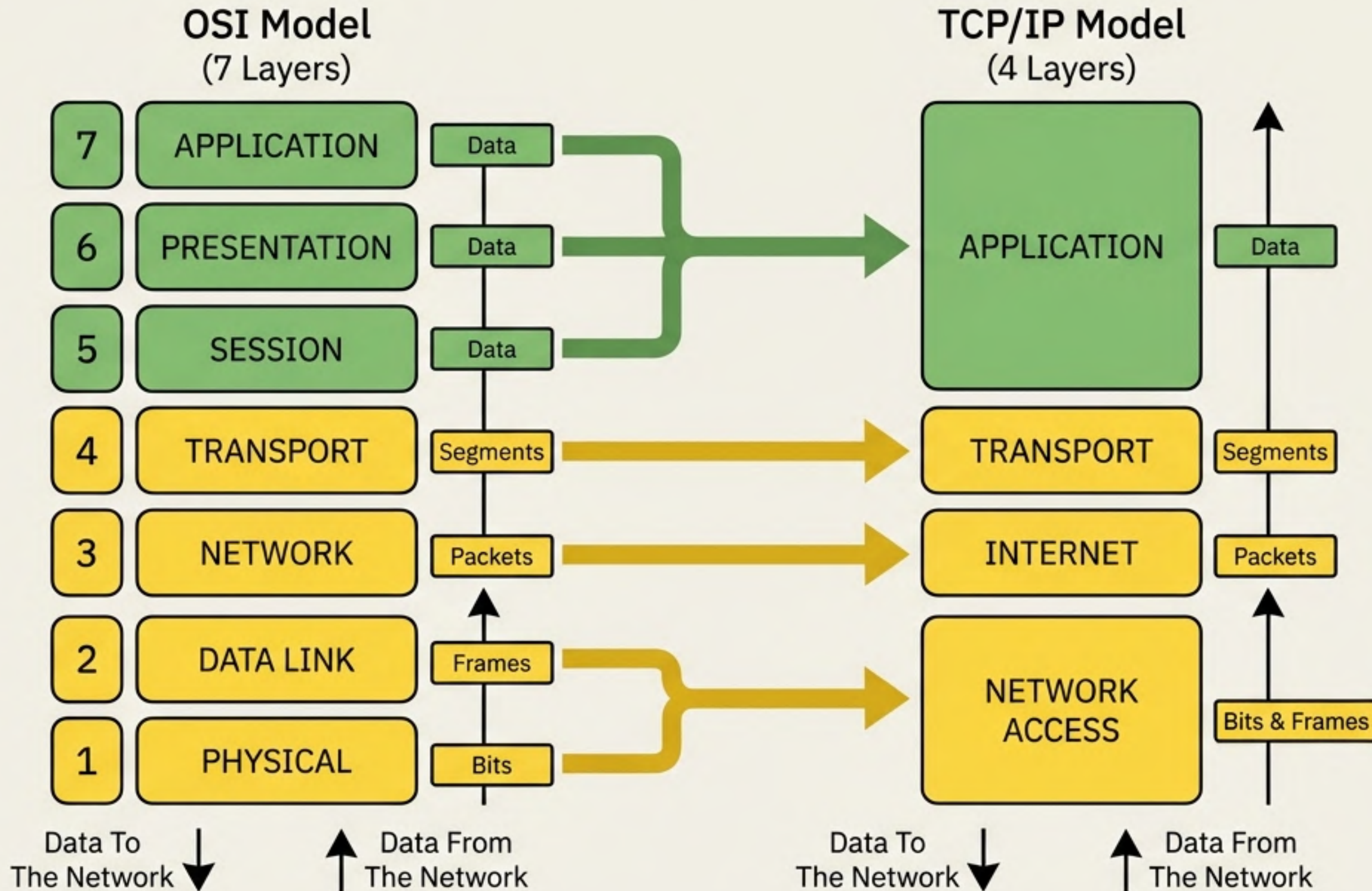
- **Encipherment:** Cryptography/Transformation of data.
- **Digital Signature:** Proof of source and integrity.
- **Access Control:** Enforcing rights to resources.
- **Data Integrity:** Assuring data is unaltered.
- **Authentication Exchange:** Verifying identity.

Pervasive Security Mechanisms

(Not specific to a protocol layer)

- **Trusted Functionality:** Perceived correctness of security criteria.
- **Security Labels:** Marking resources with security attributes.
- **Event Detection:** Identifying security-relevant events.
- **Security Audit Trails:** Independent review of system records.

Comparative Analysis: OSI vs. TCP/IP Models



Summary & Key Takeaways

- ✓ **Architecture:** The OSI model divides network communication into 7 logical layers to ensure interoperability.
- ✓ **Data Flow:** Data undergoes encapsulation (down the stack) and decapsulation (up the stack).
- ✓ **Security:** The ITU-T X.800 Security Architecture layers security services (Confidentiality, Integrity, Availability) over the functional model.
- ✓ **Implementation:** TCP/IP is the practical 4-layer standard of the Internet, while OSI remains the theoretical reference.
- ✓ **Foundation:** Security begins at Layer 1. Without physical security (environmental controls, access restrictions), software controls are vulnerable.

References

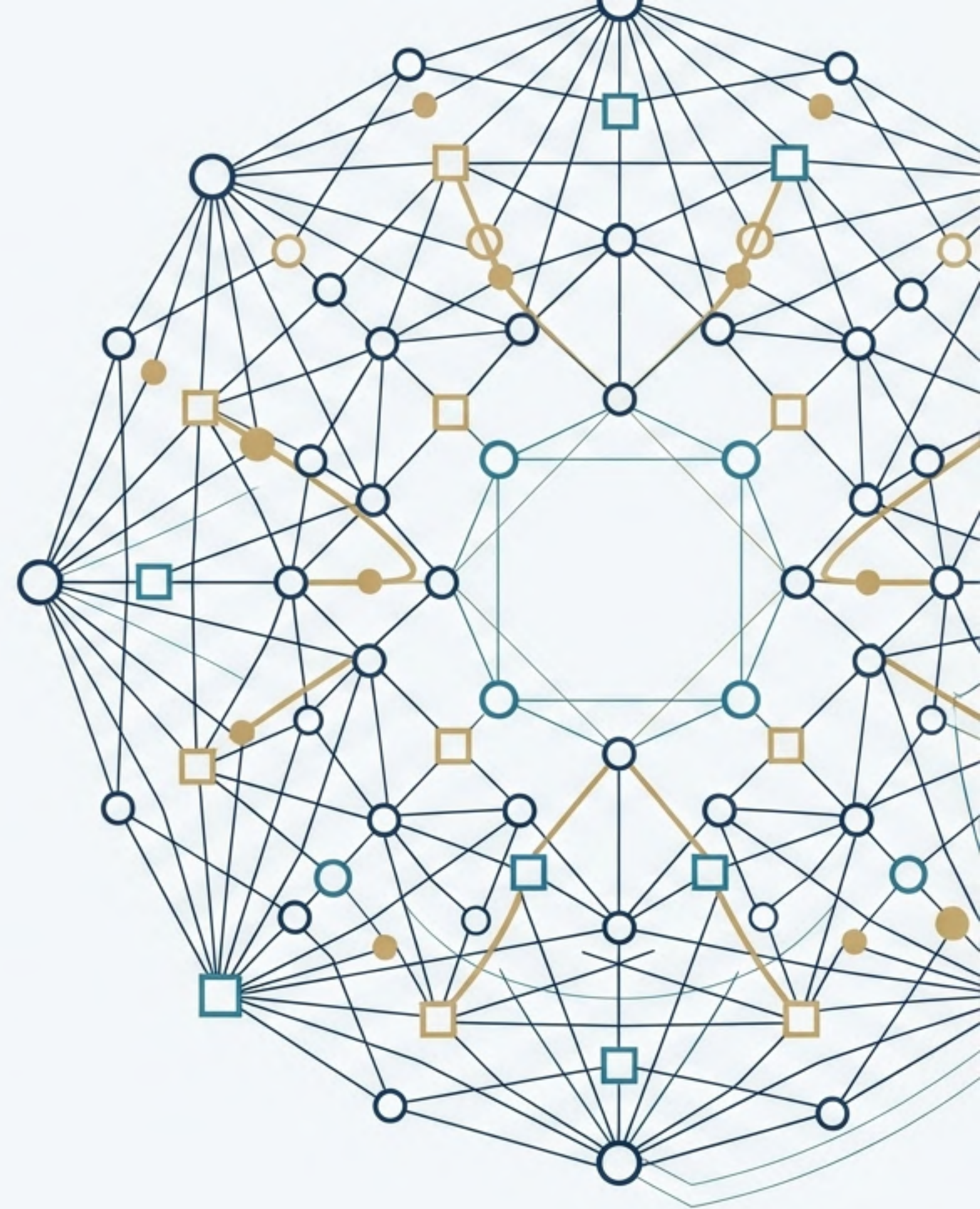
- Stallings, W. Cryptography and Network Security: Principles and Practice.
- Stallings, W. Network Security Essentials.
- ITU-T Recommendation X.800, Security Architecture for OSI.
- IBM Innovation Centre for Education (ICE) Training Materials.

Network Architectures: TCP/IP Model and Comparison with OSI

Protocol Stacks, Layered Models,
and Implementation Realities

Georgia

By: Dr. Richa Kumari



The Imperative for Standard Network Models

Network models provide a set of standards for equipment vendors to ensure “interoperability” and remove proprietary barriers.

An ‘Open System’ is developed based on standardized protocols and interfaces, allowing effective communication with other compliant systems.

1. OSI (Open Systems Interconnection)

A theoretical “yardstick” model developed by ISO (circa 1981) comprising seven functional layers.

2. TCP/IP (Transmission Control Protocol/Internet Protocol)

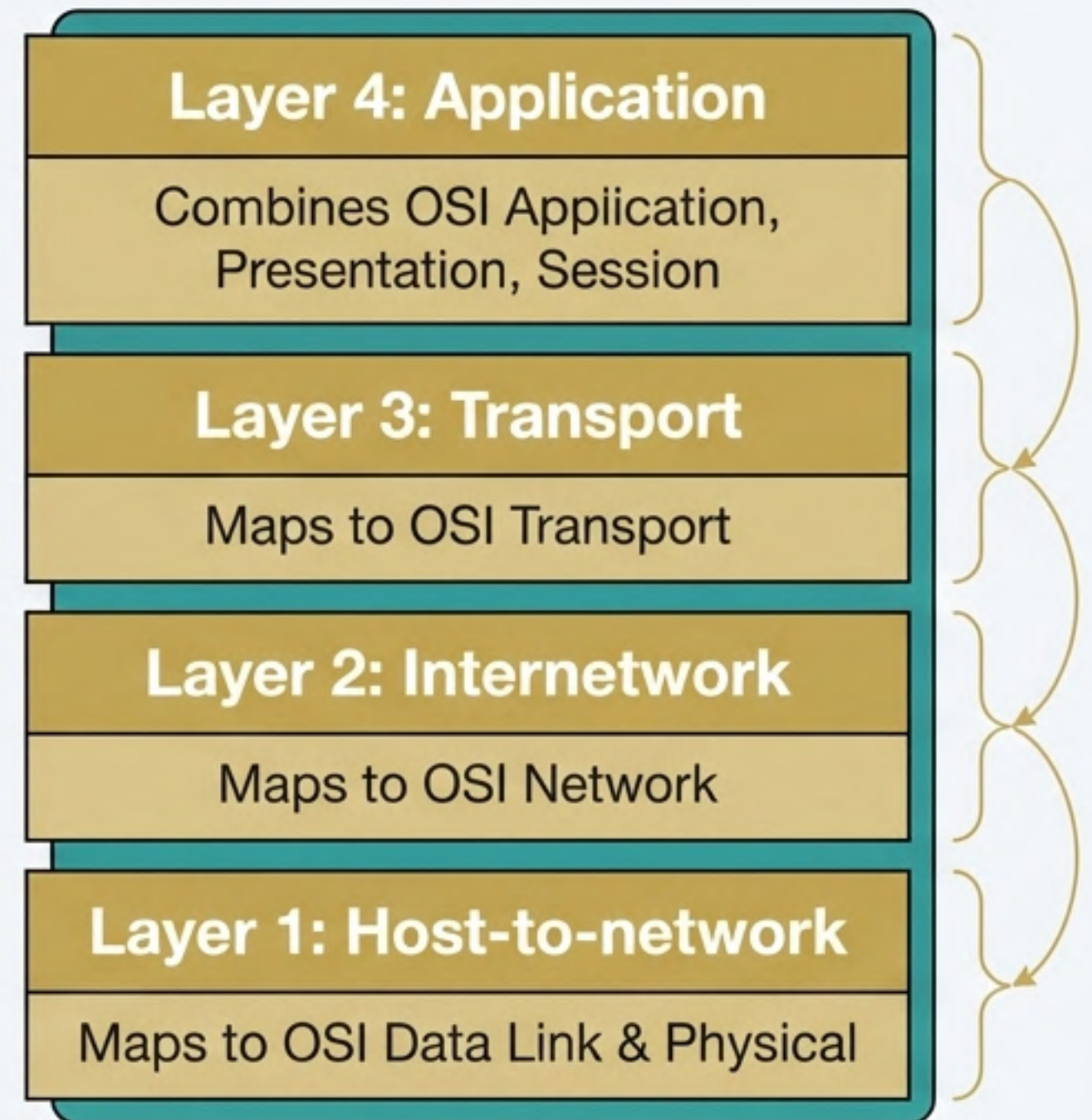
The practical “implementation” model that structures the modern Internet.

The OSI Reference Model: A Theoretical Framework

7	Application	Interface to user; file transfers, email (FTP, SMTP).
6	Presentation	Data translation, encryption, compression (SSL, JPEG).
5	Session	Dialog control, synchronization, session management (NetBIOS, RPC).
4	Transport	End-to-end delivery, segmentation, flow control (TCP, UDP).
3	Network	Routing, logical addressing (IP), path selection.
2	Data Link	Node-to-node delivery, framing, error control (Switch, Bridge).
1	Physical	Transmission of bits over media; electrical specs.

The TCP/IP Model (The DoD Model)

- Also known as the Department of Defense (DoD) model.
- Designed as a robust, flexible architecture for client/server environments.
- Key Feature: Maintains network integrity even if connections fail (designed for survivability).



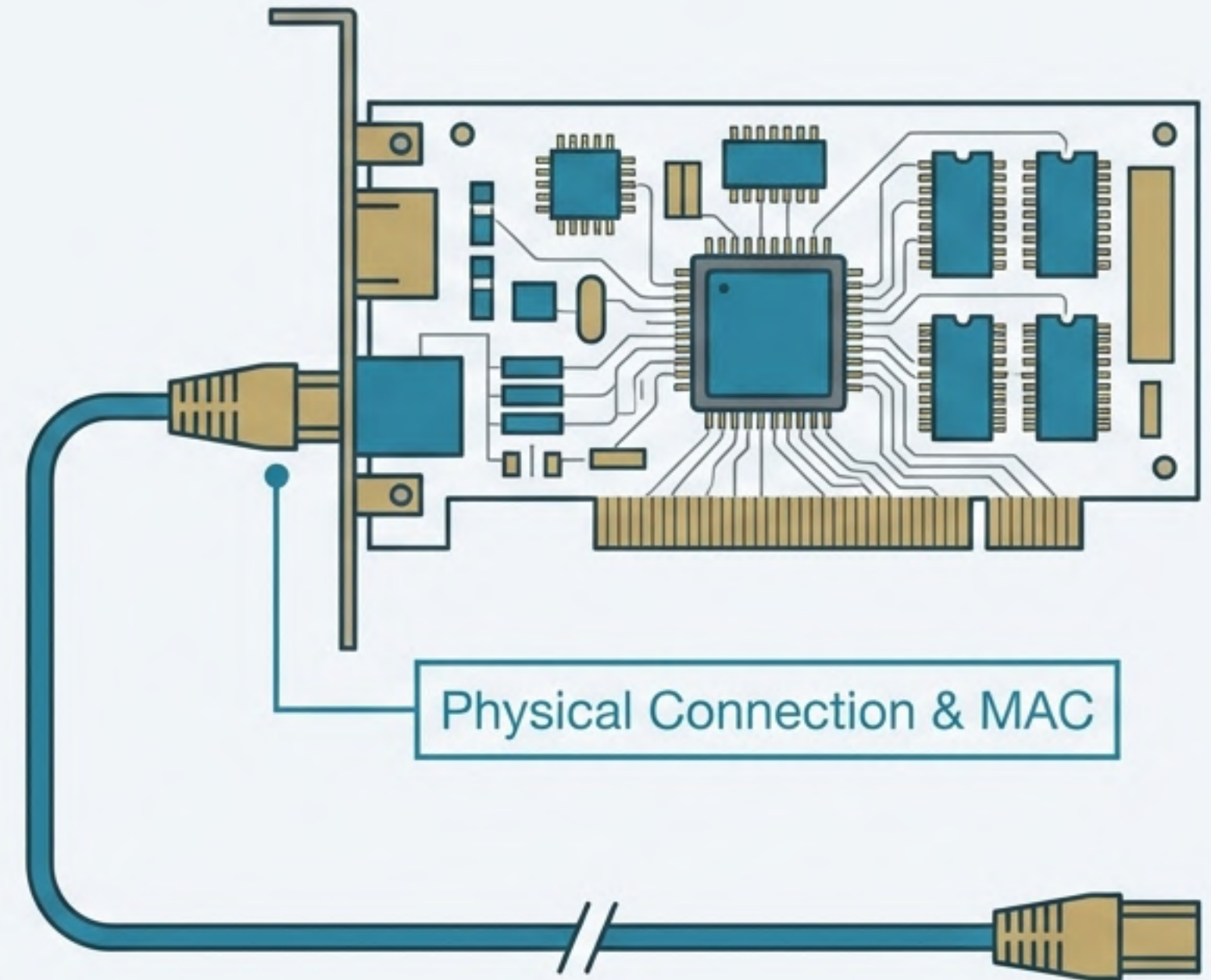
Layer 1: Host-to-Network Layer

Function: The lowest layer of the stack. Defines the protocol used to connect to the host so packets can be sent.

Characteristics:

- Varies from host to host and network to network.
- Includes functions of the Physical Layer (signal encoding/decoding) and Media Access Control (MAC).

Key Technologies: Ethernet, Fast Ethernet, Token Ring, FDDI.



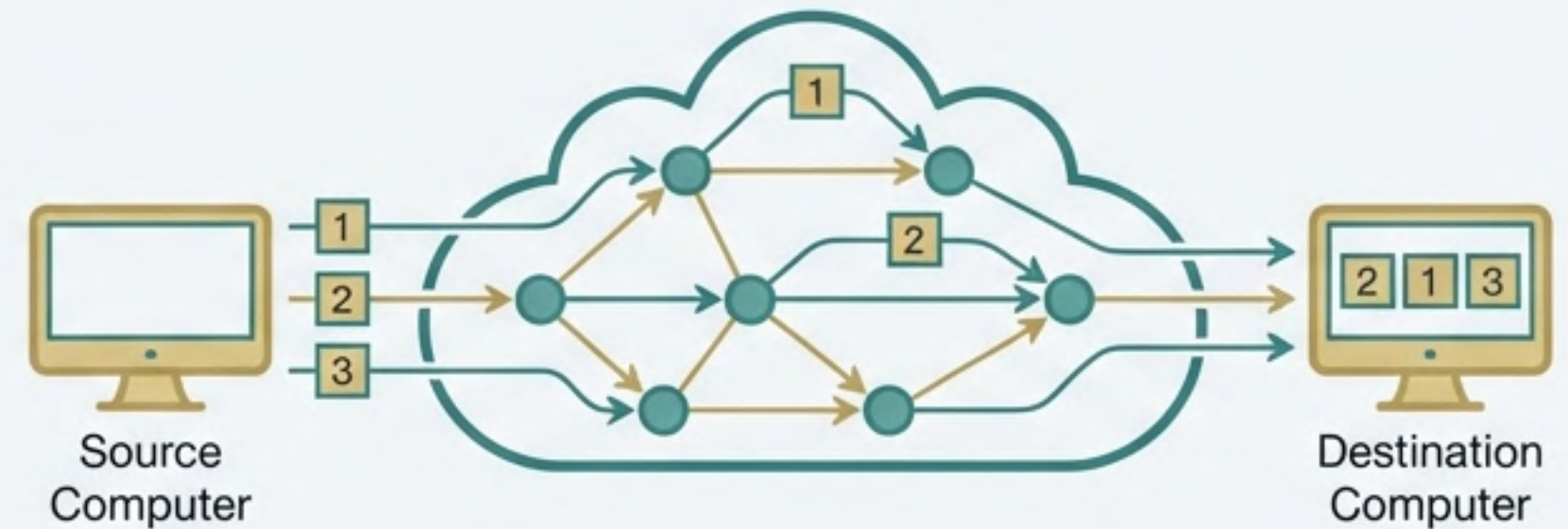
Layer 2: The Internet Layer

Primary Function: Selection of packet switching networks and 'connectionless' internetwork delivery.

Core Protocol: Internet Protocol (IP).

Key Behaviors:

- Performs routing and congestion avoidance.
- Provides 'best effort' service (unreliable, no guaranteed delivery).
- Packets travel independently; arrival order may differ from sending order.

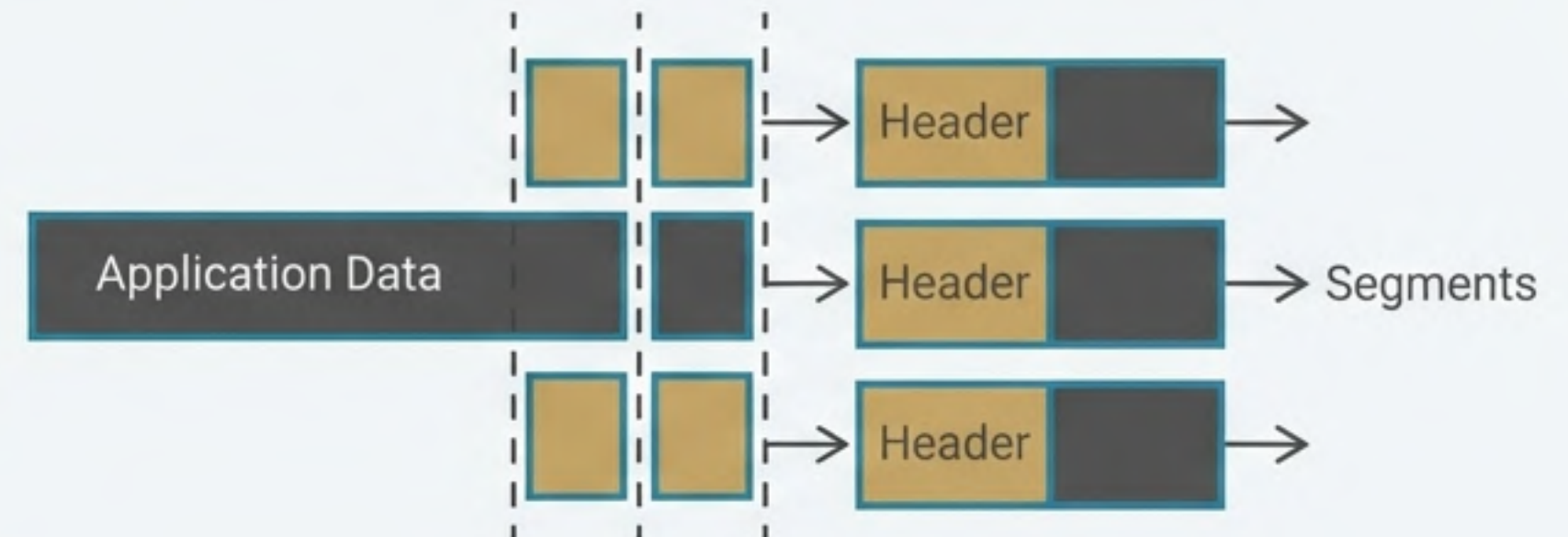


Layer 3: The Transport Layer

Primary Function: Decisions on parallel vs. single path transmission. Breaks messages into segments for efficient network handling.

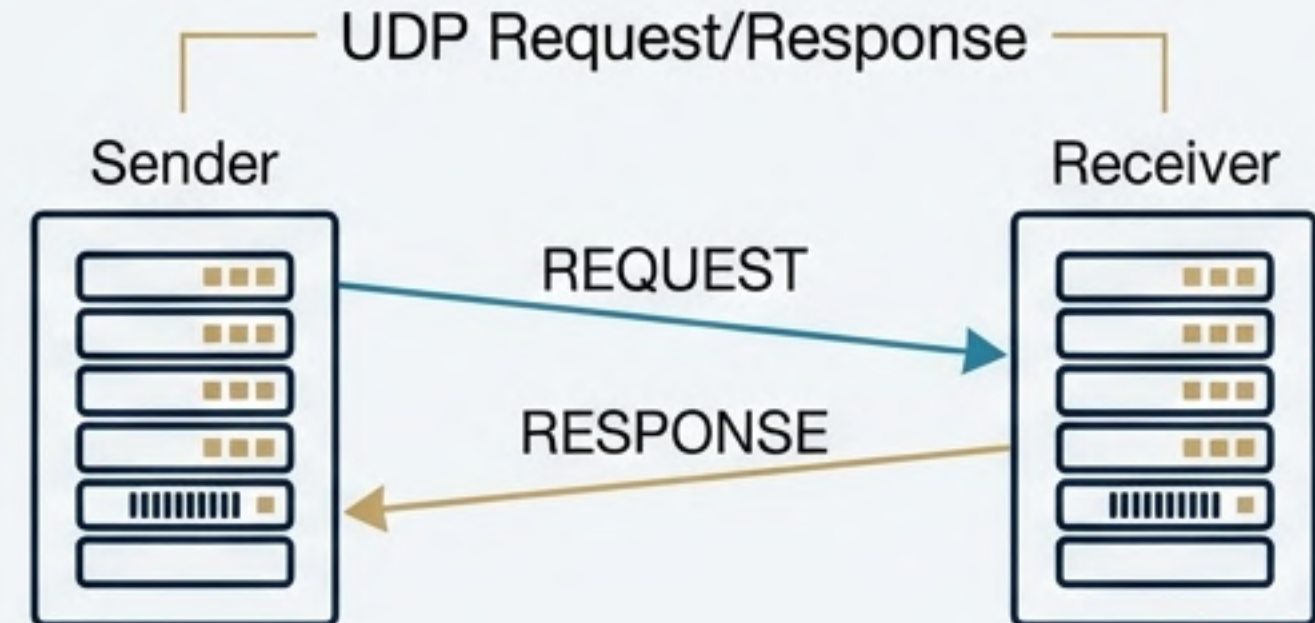
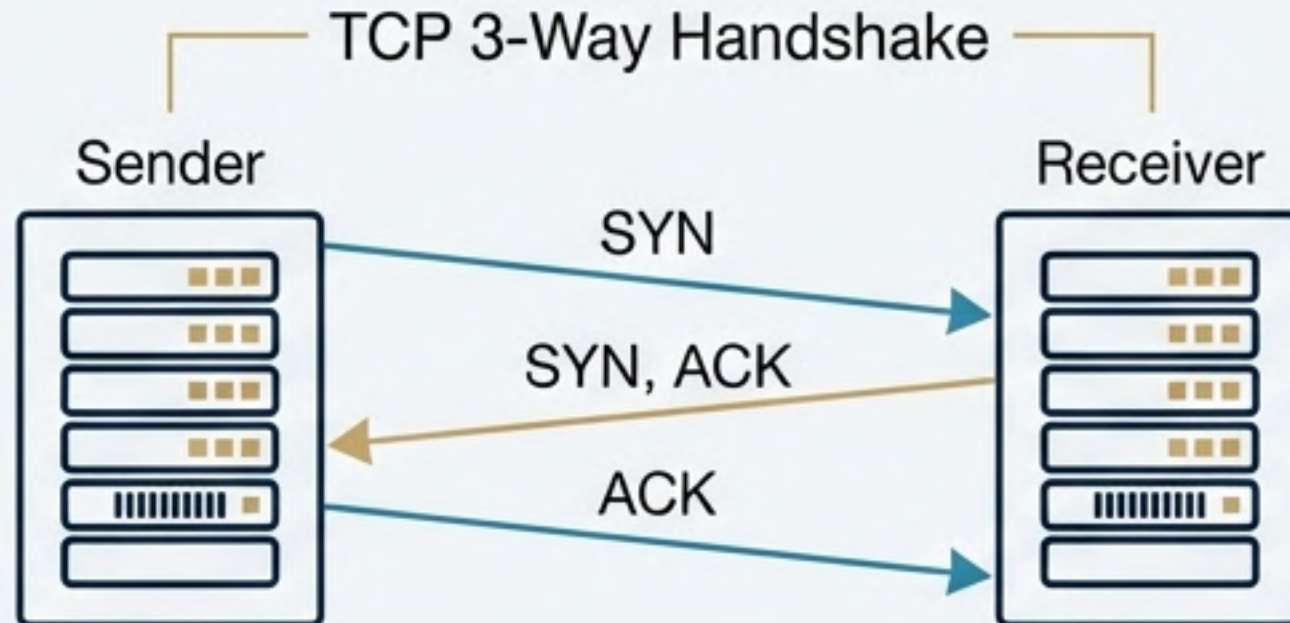
Key Responsibilities:

- Multiplexing & Segmentation: Splitting data into manageable units.
- Sequencing: Arranging packets to be sent/received in correct order.
- Header Addition: Adds control information to data blocks.
- Interface: Allows applications to read/write to the network.



Transport Protocols: TCP vs. UDP

TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
- Reliable, Connection-Oriented - Byte-stream handling with error/flow control - Sequencing ensures order - Header Size: 20 bytes	- Unreliable, Connectionless - One-shot request-reply service - No sequencing or flow control overhead - Header Size: 8 bytes



Layer 4: The Application Layer

Function: Interfaces with the user and defines TCP/IP/IP specifications for various communication services. Encompasses OSI Application, Presentation, and Session layers.

TELNET



Two-way communication protocol for remote command-line connections.

FTP



File Transfer Protocol for moving data between systems.

SMTP



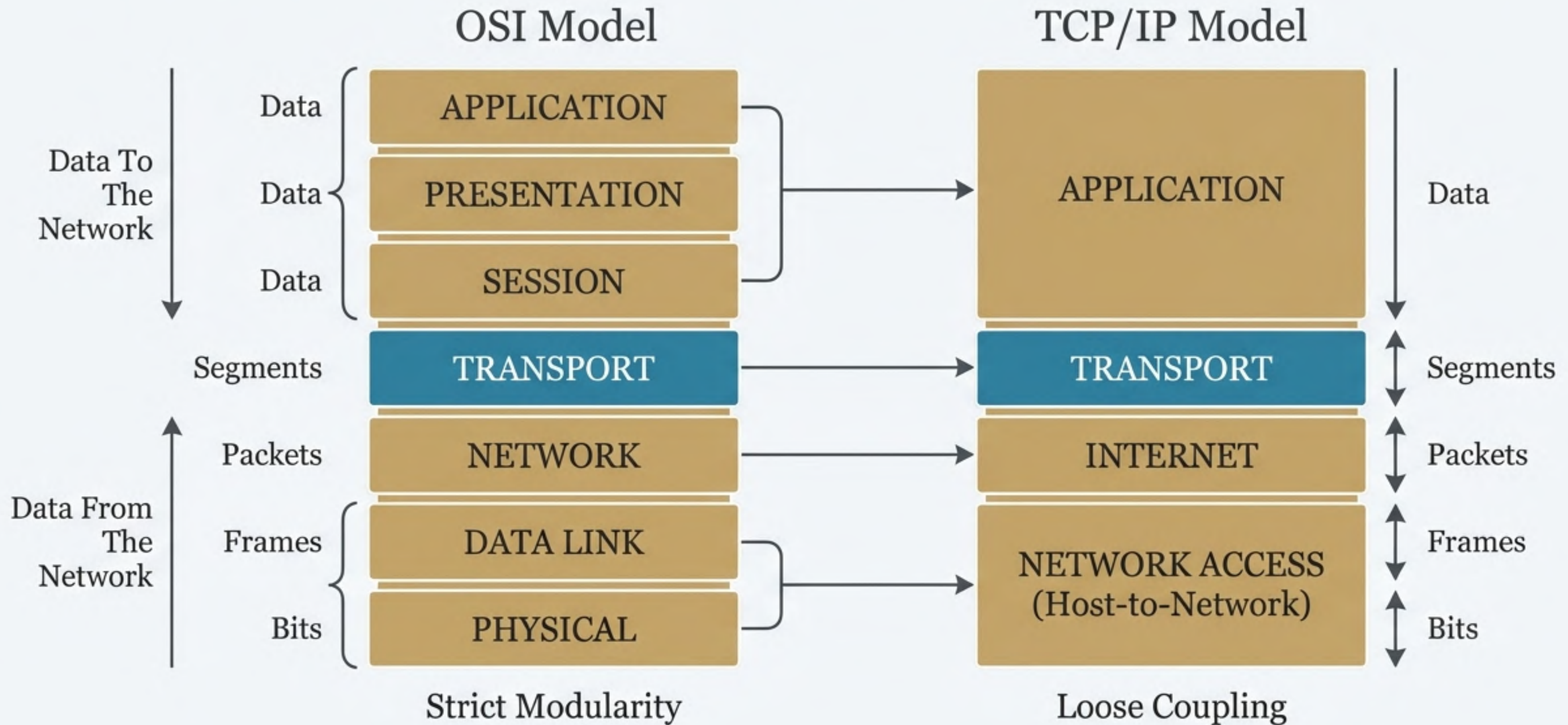
Simple Mail Transfer Protocol for email transmission.

DNS

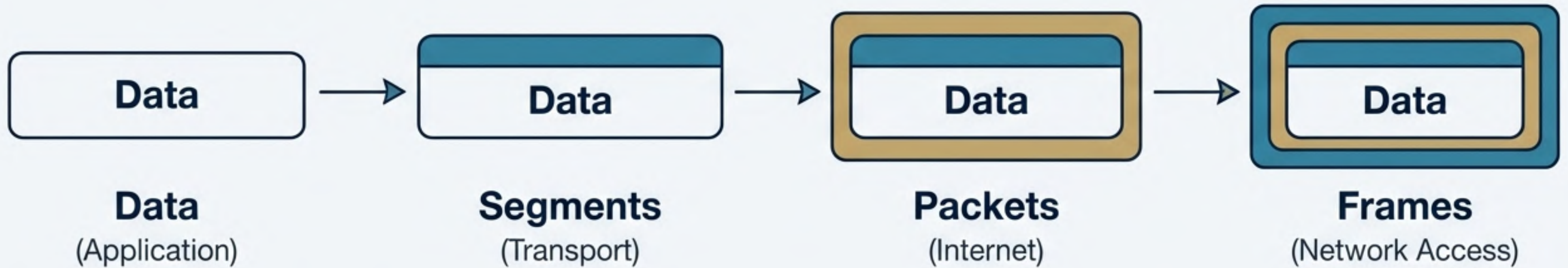


Domain Name System: Resolves IP addresses into textual names.

Comparative Analysis: OSI vs. TCP/IP



Data Transmission and Encapsulation



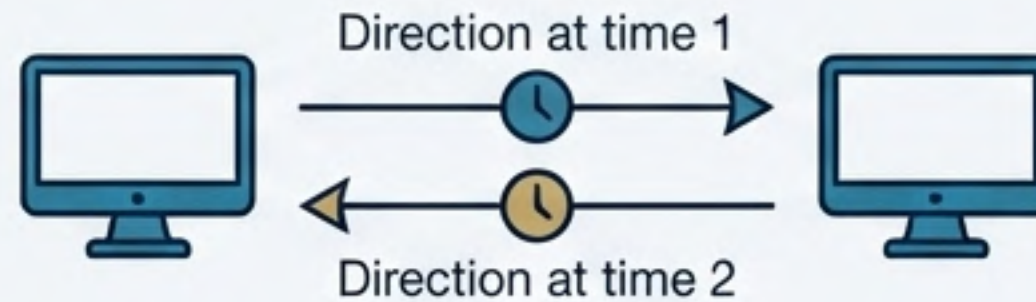
Transmission Modes

Simplex



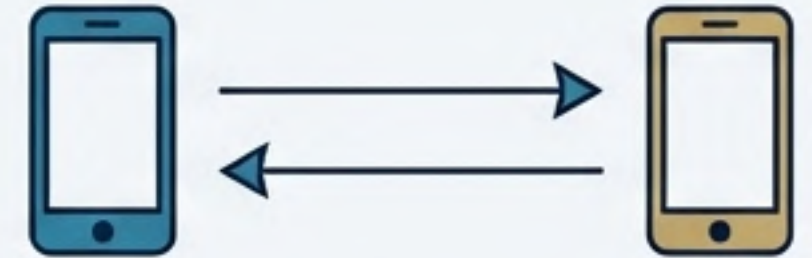
One-way arrow
(Example: Keyboard to CPU)

Half Duplex



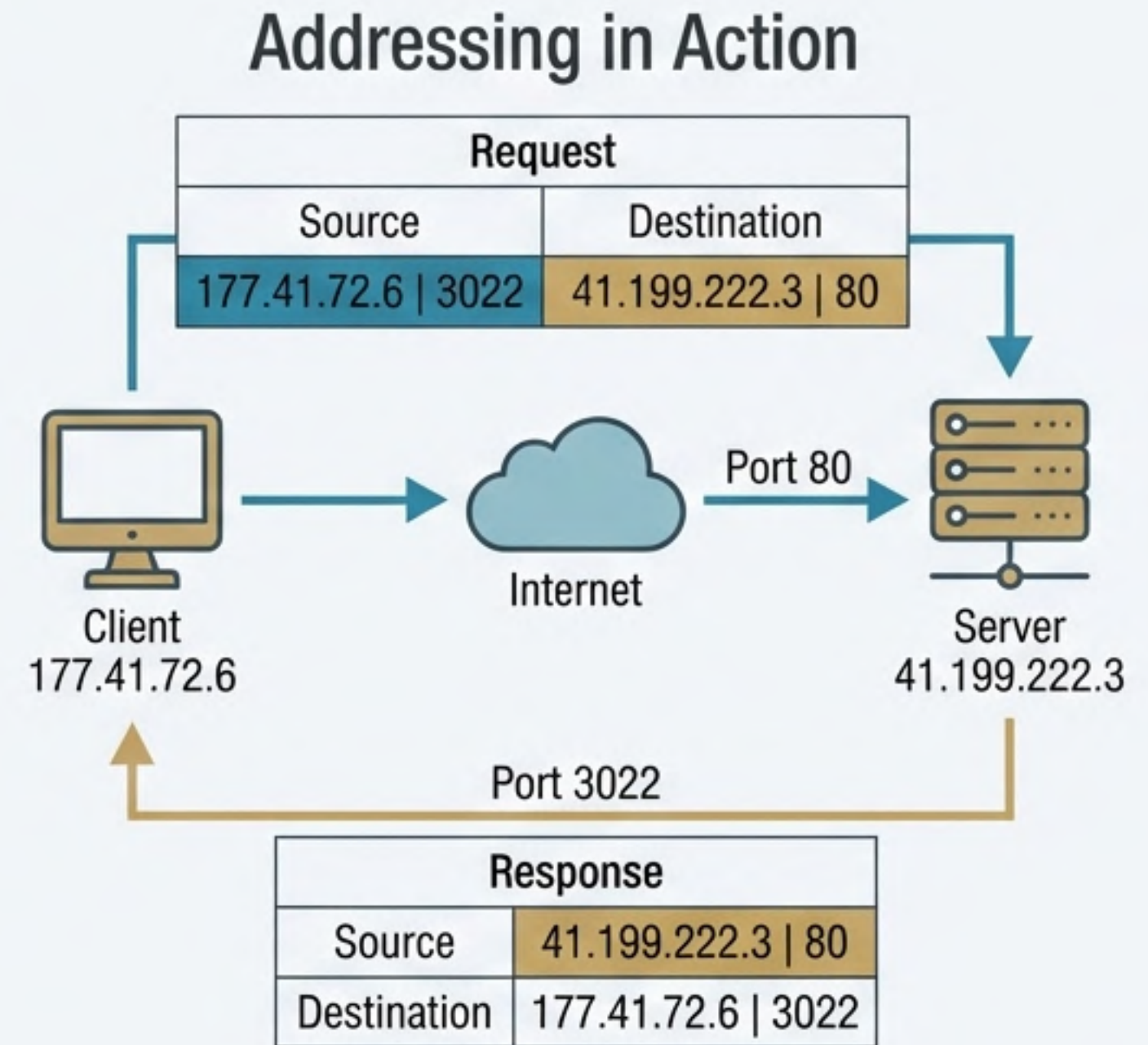
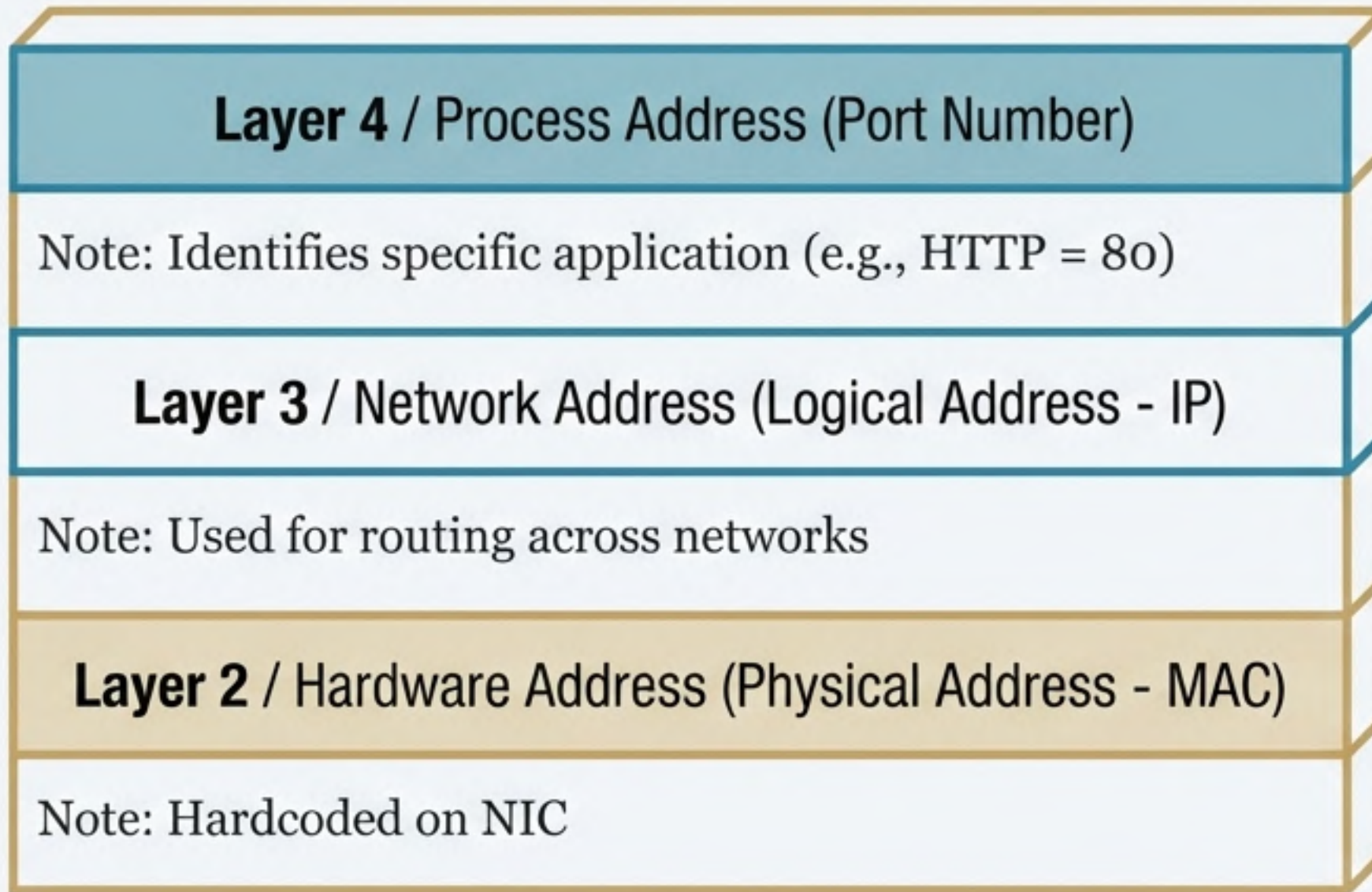
Two-way arrow but one active at a time
(Example: Walkie-Talkies)

Full Duplex



Two-way simultaneous arrows
(Example: Mobile Phones)

Addressing Across the Stack



Knowledge Checkpoint



Q: Which protocol is considered connection-oriented?

A: TCP (Provides reliable connectivity and data transmission).



Q: Order Layers 2, 5, 7, 4, 3 of the OSI model.

A: Data link (2), Network (3), Transport (4), Session (5), Application (7).

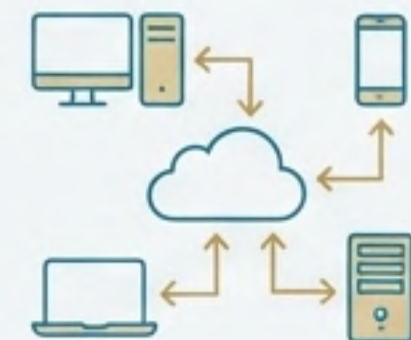
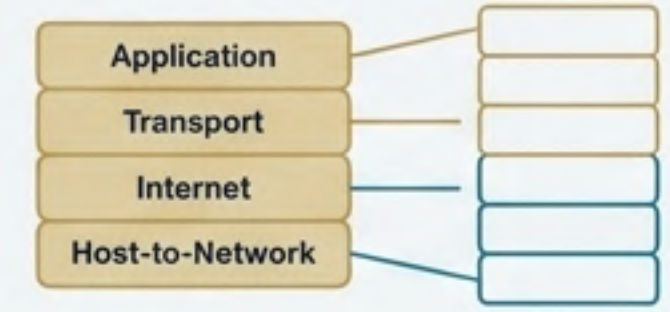


Q: What defines an “Open System”?

A: Systems built with international standards to communicate easily with other systems.

Key Takeaways

- **Architecture:** TCP/IP is a 4-layer model (Application, Transport, Internet, Host-to-Network) that streamlines the theoretical 7-layer OSI model.
- **Reliability:** TCP ensures reliable, connection-oriented delivery (3-way handshake), while UDP offers fast, connectionless transmission.
- **Addressing:** Effective communication requires a combination of MAC addresses (Physical), IP addresses (Logical), and Port numbers (Process).
- **Flexibility:** The TCP/IP suite supports a scalable, robust client/server architecture that is independent of specific underlying hardware.

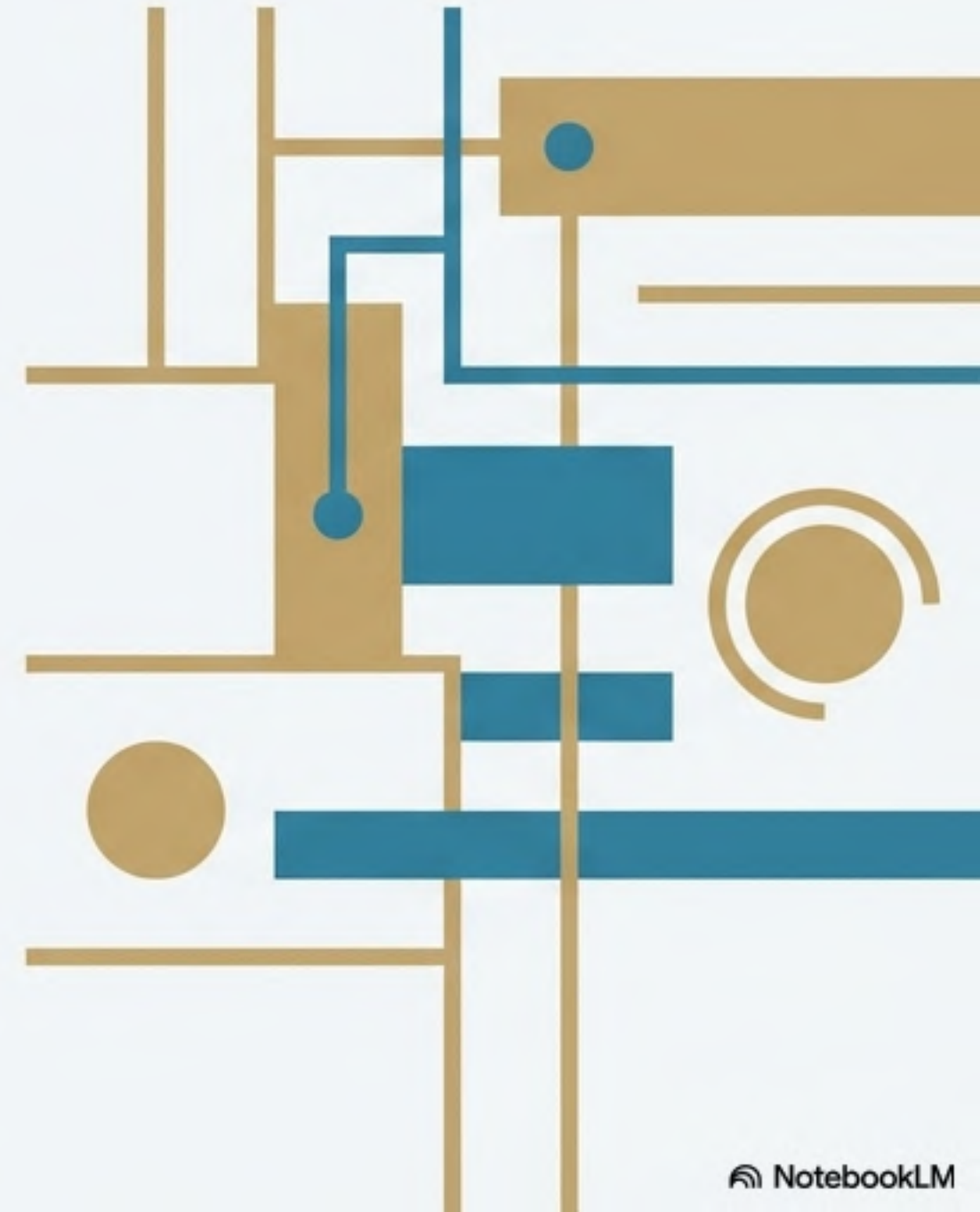


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Forouzan, B. A. "Data Communications and Networking".



Network Devices and Components: Connectivity to Defense Architecture

A comprehensive analysis of network hardware and security controls

By: Dr. Richa Kumari

Network Fundamentals & Core Terminology

Computer Network

A collection of computing devices connected to communicate and share resources. Connections can be physical (wired/cabled) or wireless (radio waves).

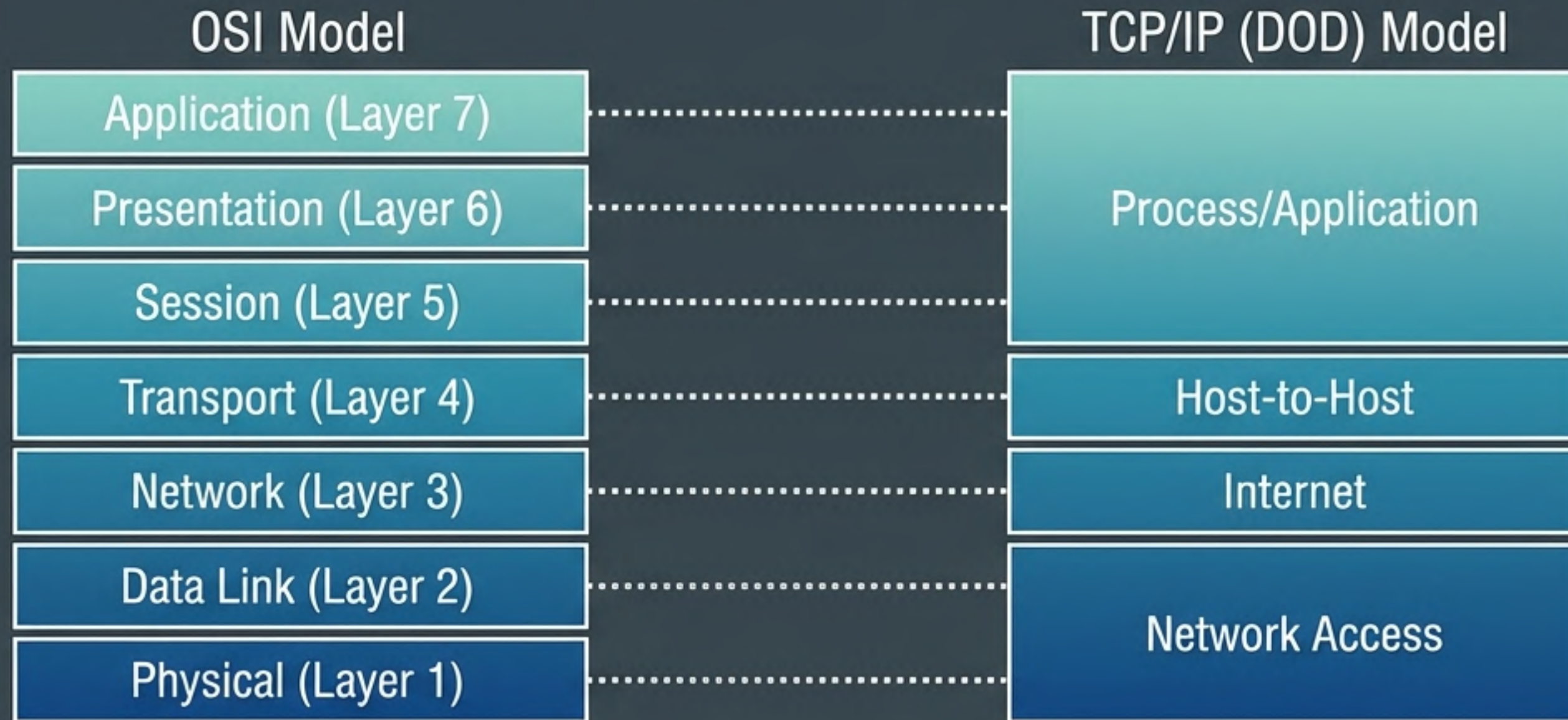


Node (Host): Any device on a network.



Bandwidth: Data transfer rate (speed of movement).

The Architectural Blueprint: OSI vs. TCP/IP Models



Key Insight: The OSI security architecture provides a systematic framework for defining security attacks, mechanisms, and services.

Layer 1: The Physical Layer (Signal Transmission)

Function: Responsible for sending bits from one computer to another; deals with physical connections, synchronization, and transmission modes.

Key Constraints:
Operates purely on the physical layer. No logic.
No intelligence.

1. The Repeater

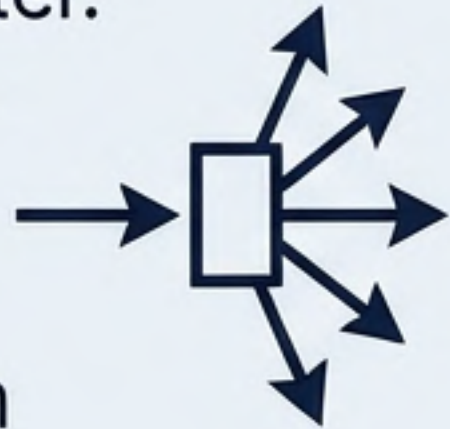


- Regenerates signals to extend transmission distance.

2. The Hub



- Multi-port repeater. Broadcasts data to *all* ports regardless of destination (High Traffic).



Layer 2: The Data Link Layer (Frames & MAC)

Function: Reliable node-to-node delivery, error control, and synchronizing information into 'Frames'.

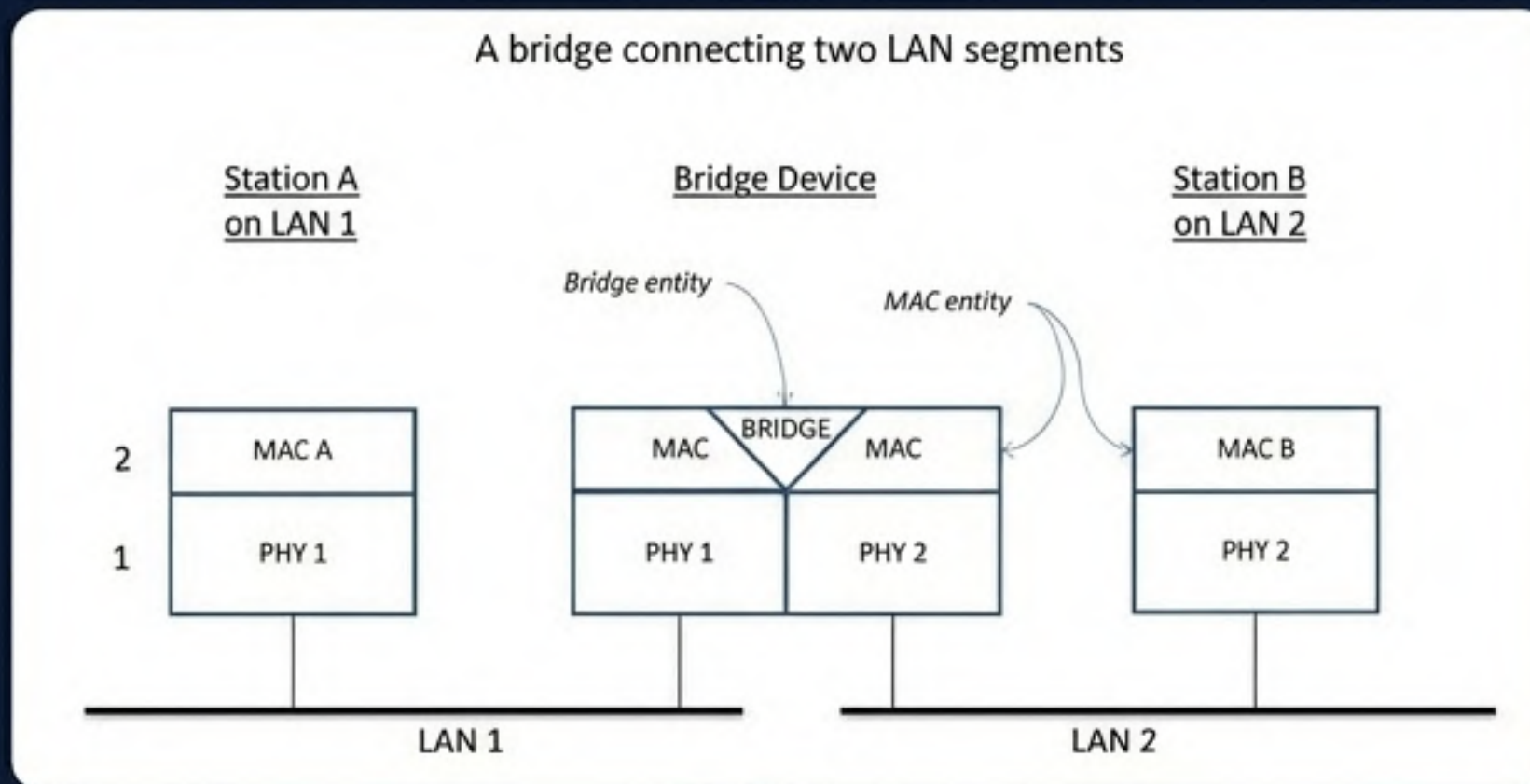
Device 1: The Bridge

Connects two LAN segments using MAC entity logic.

Device 2: The Switch

Intelligence: Maintains a MAC address table to send frames *only* to the specific destination port.

Capability: Breaks up collision domains; supports VLANs.

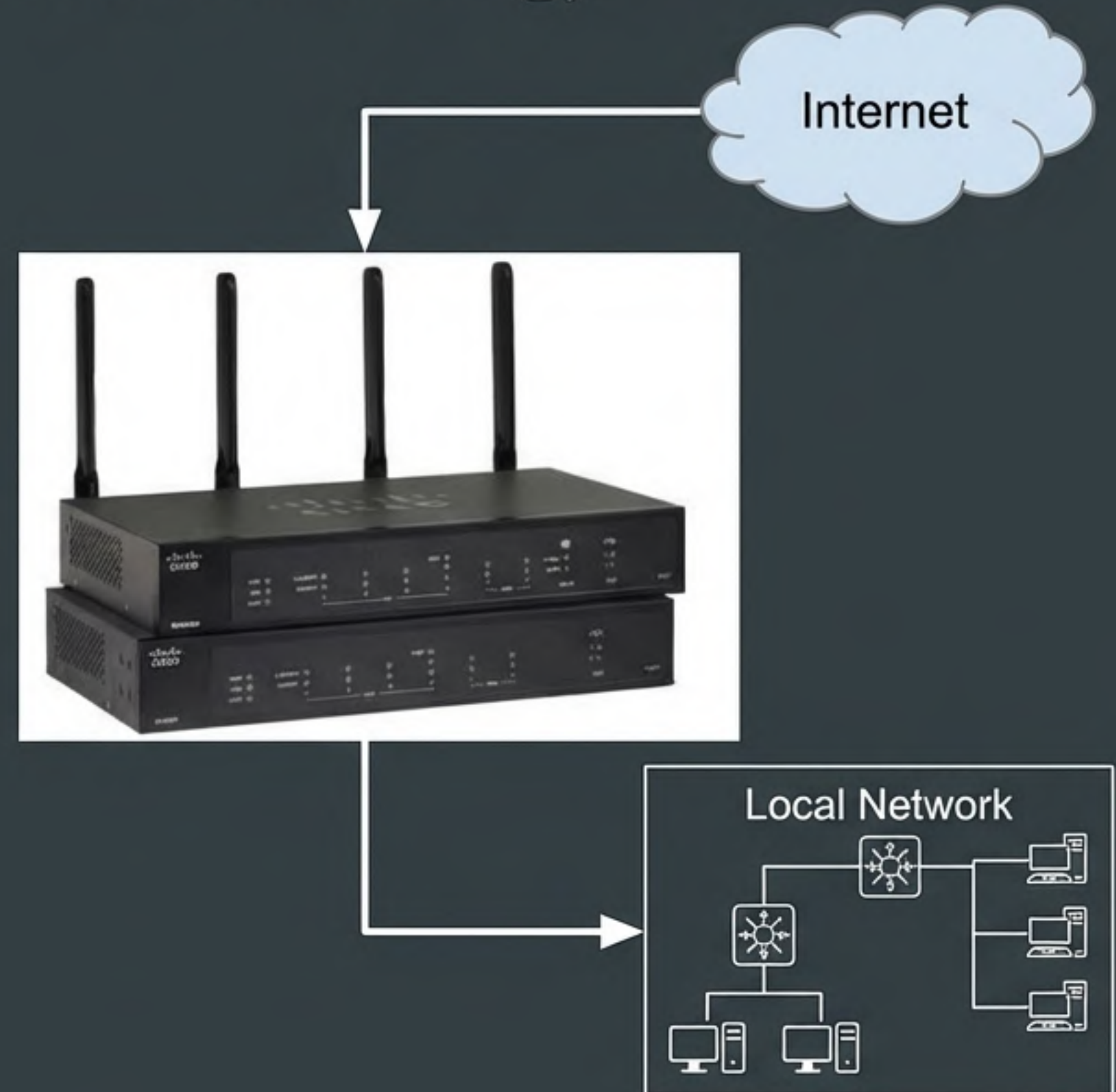


Layer 3: The Network Layer (Packets & Routing)

Function: Translates Logical Network Address (IP) to Physical Address (MAC). Divides outgoing messages into packets.

The Router:

- Facilitates **Network-to-Network communication.**
- Routes packets based on IP address.
- Protocols: IP, ICMP.
- Robust against failure.



Network Interface Components (The Endpoint)

The bridge between the computing device and the medium.



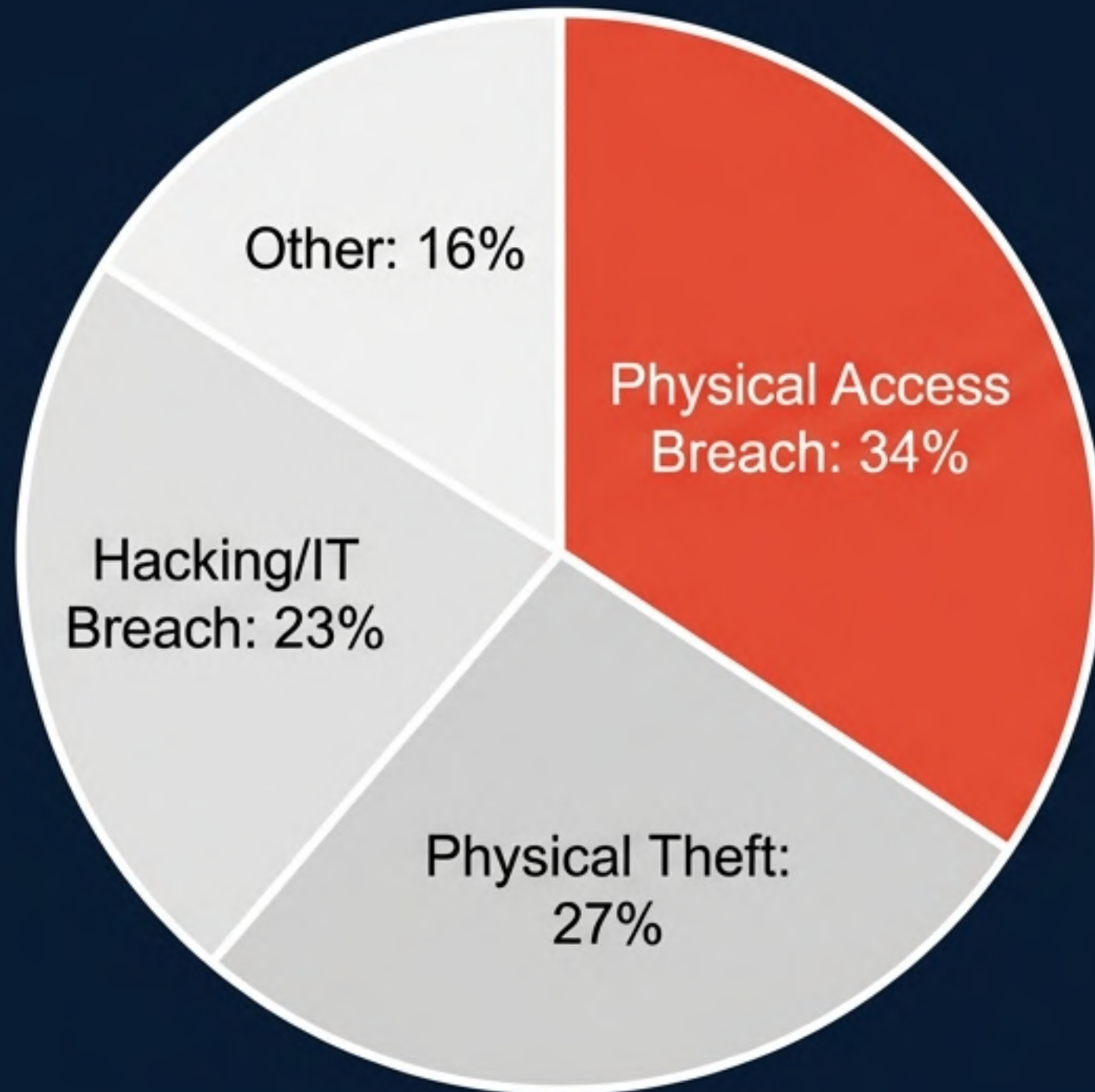
Ethernet Cards (PCI/ISA)



Wireless NICs (PCI, M.2, USB)

****NIC****: Operates at Layer 2 (Data Link). Essential for physical or wireless connectivity.

Transition to Security: The "Defense in Depth" Concept



The First Rule: "Physically Safeguard Your Systems and Networks."

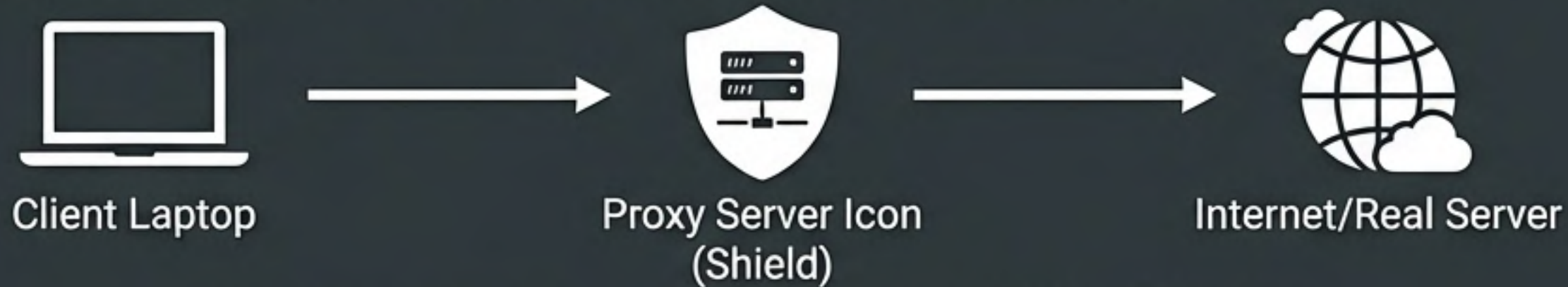
Why? Physical breaches account for **34%** of attacks. Standard firewalls cannot stop a physical intruder.

Defense in Depth: Using multiple overlapping layers so that the failure of one does not leave the system vulnerable.

The Sentinel: Proxy Servers & DMZ

Proxy Server

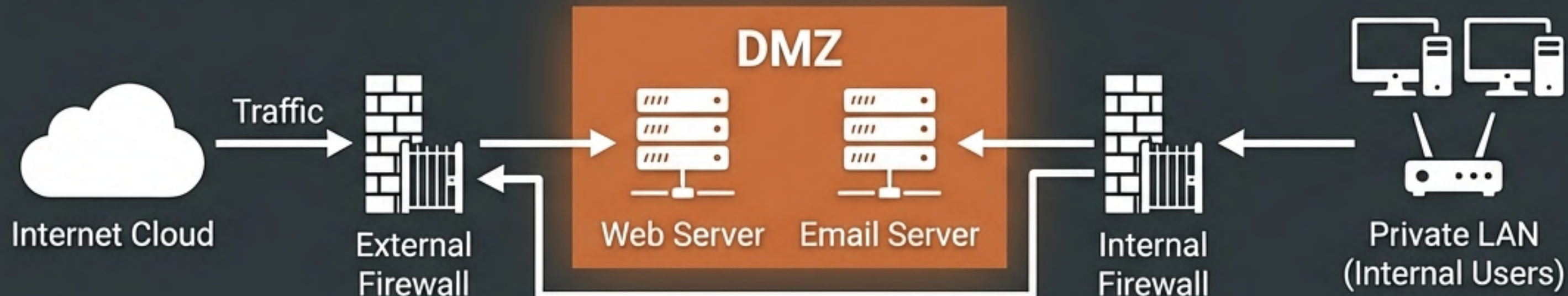
A sentinel virtually located between a client and the actual server.



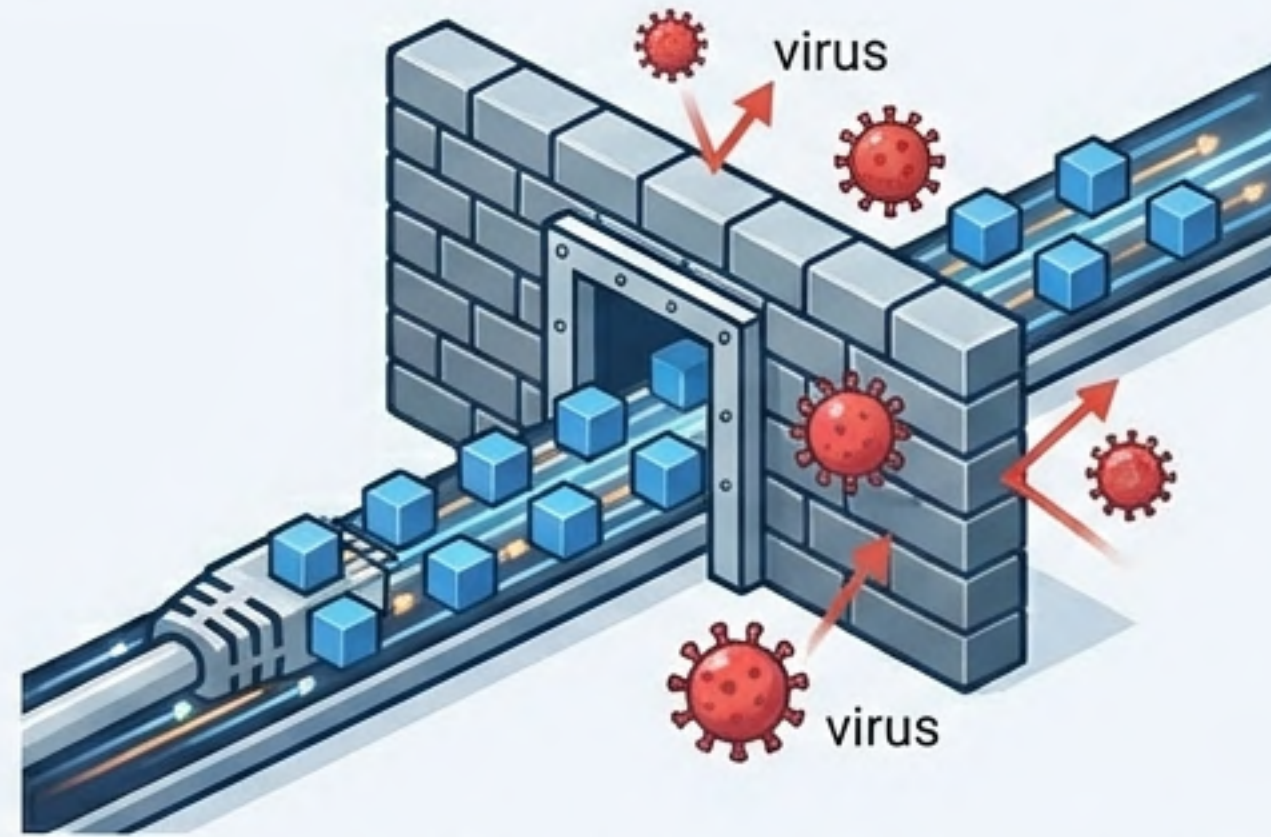
Intercepts requests, filters content, hides identity.

Demilitarized Zone (DMZ)

A buffer zone between the private LAN and the public Internet.



The Choke Point: Firewalls



Definition: Defines a single choke point that keeps unauthorized users out of the protected network.

Traffic Filtering

Prohibits vulnerable services from entering/leaving.

Anti-Spoofing

Protects against IP spoofing and routing attacks.

Management

Consolidates security capabilities on a single system.

"A completely secure and robust network can be designed with proper implementation and configuration of network security elements."

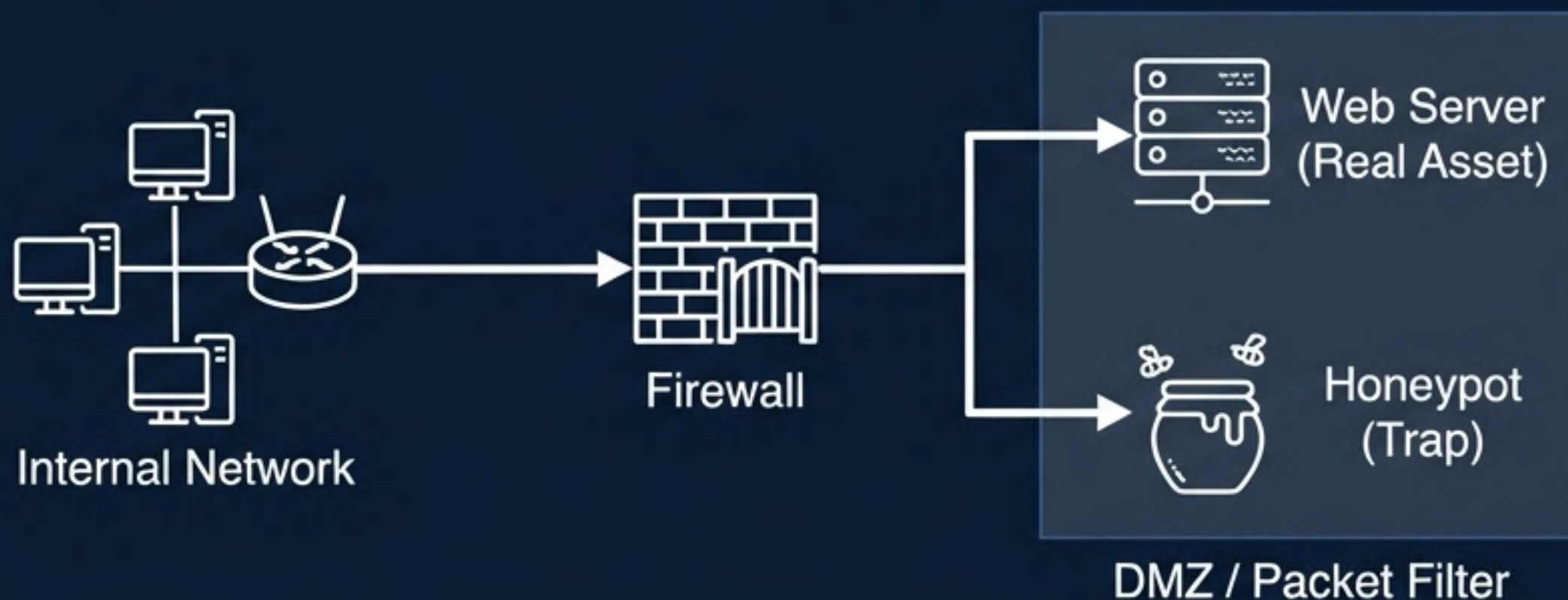
Detection & Deception: IDS & Honeypots

Intrusion Detection System (IDS)

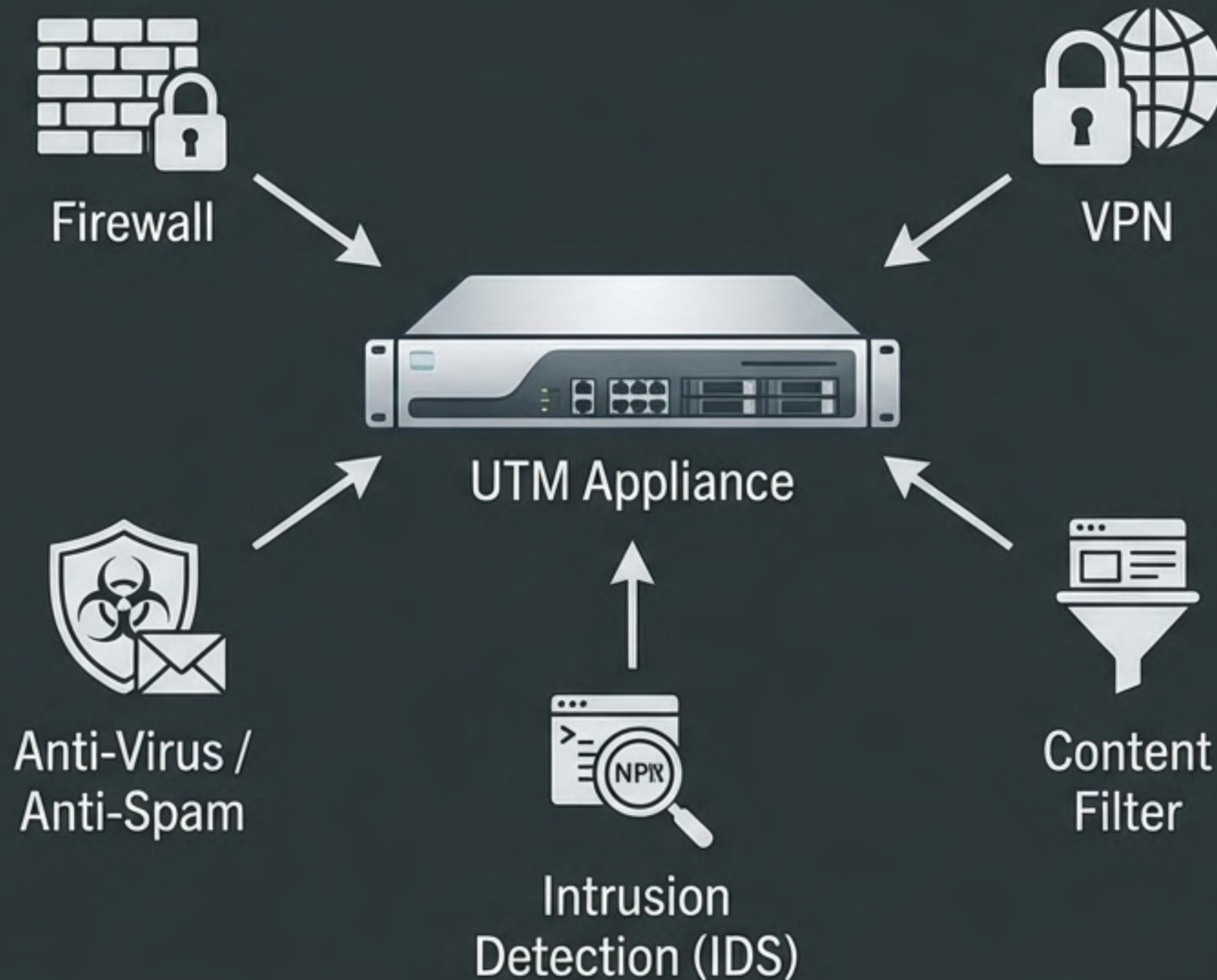
- Inspects inbound/outbound traffic for suspicious patterns indicating a breach.

Honeypots

- A resource explicitly set up to trap attackers. No production value = Any traffic is an attack.



Unified Threat Management (UTM)



- ****Concept****: A comprehensive solution managing network security through a centralized console.
- ****Pros****: Reduced complexity, simplicity, and enhances security.
- ****Cons****: Single point of failure.

Network Access Control (NAC)

****Definition**:** Policy enforcement at the edge. Restricts end-user connection based on security policy.

****Core Functions**:**

1. ****Authentication**:** Verify user identity.
2. ****Health Checks**:** Inspect OS/Platform before admitting.
3. ****Enforcement**:** Determine where a device may connect.

Common Solutions

Cisco NAC

ForeScout CounterACT

Aruba ClearPass



Summary: The Connectivity-Defense Matrix

Device	Layer/Category	Primary Function
Hub	Physical (L1)	Broadcasts data (No intelligence)
Switch	Data Link (L2)	Forwards frames via MAC (Intelligent)
Router	Network (L3)	Routes packets via IP (Network-to-Network)
Firewall	Security Control	Traffic filtering choke point
IDS/Honeypot	Security Analysis	Detection and trapping of threats
NAC	Access Control	Policy enforcement at the edge

Authentic References

[STAL08] Stallings, W., and Brown, L. *Computer Security*. Englewood Cliffs, NJ, 2008. (TAA Award Winner).

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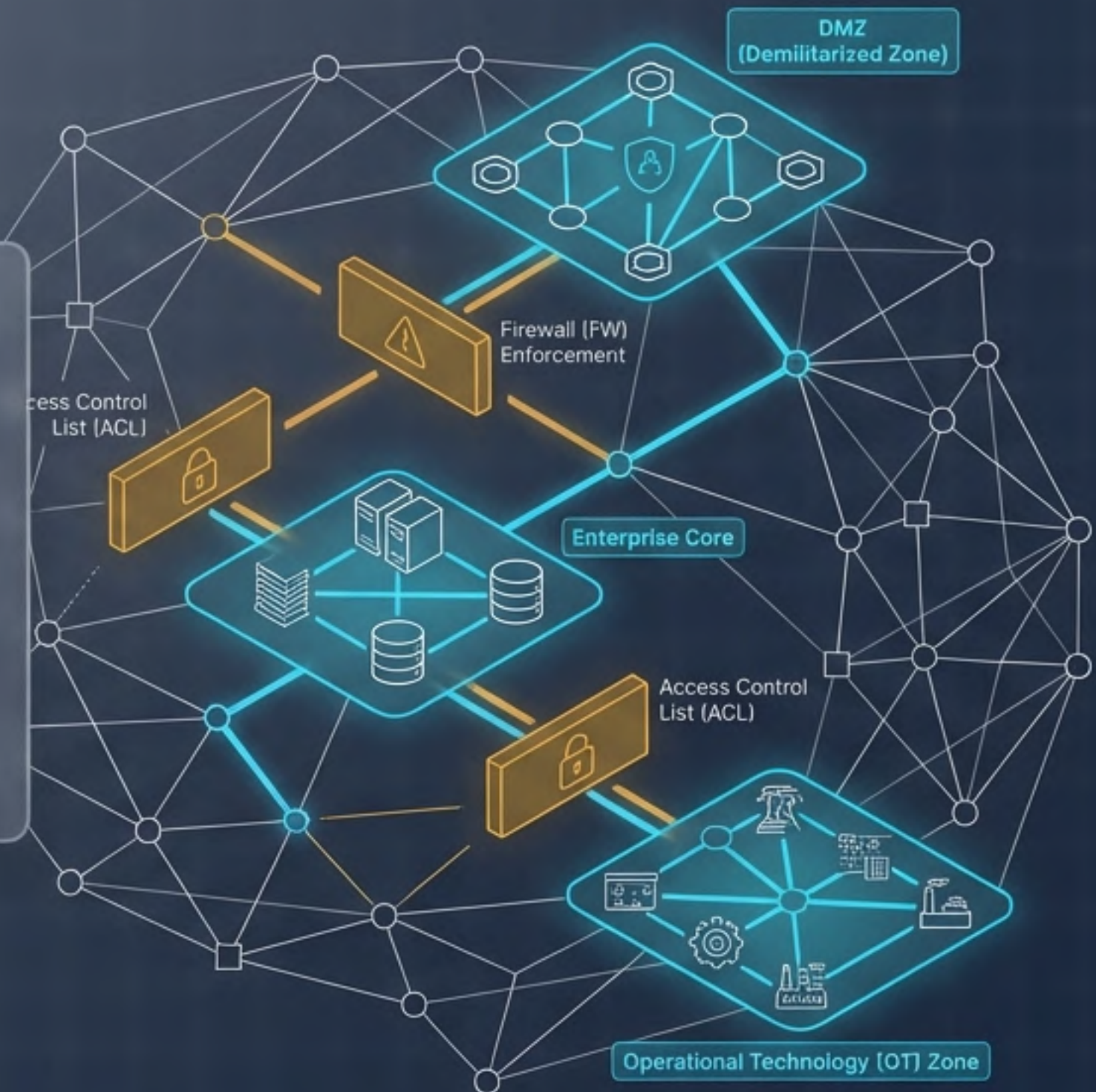
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Network Segmentation and Security Zones

Architectural Strategies for Defense-in-Depth

By: Dr. Richa Kumari



The Vulnerability of Flat Networks

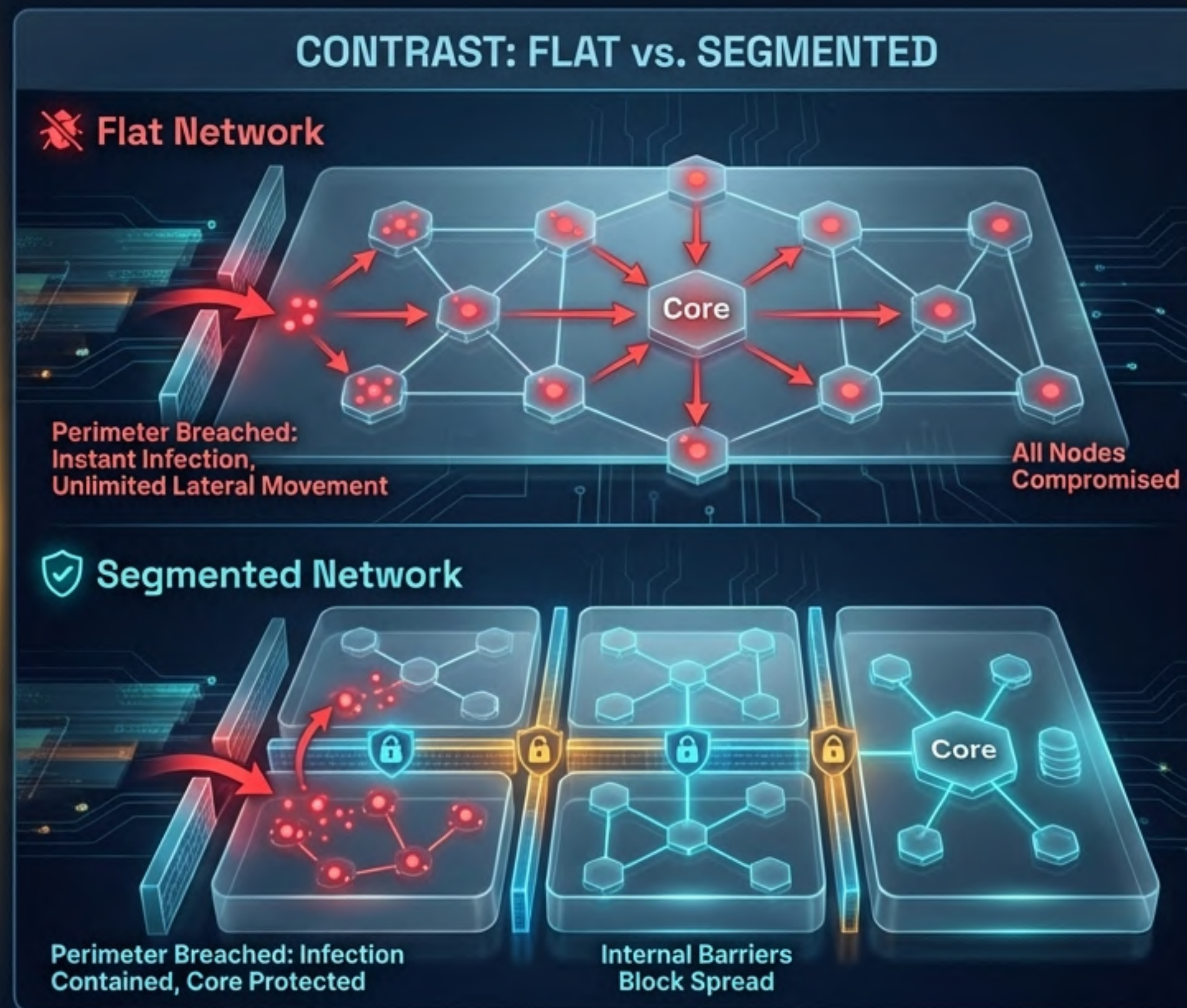
****Core Concept**:** In a 'flat' network, a **single breach** grants unrestricted access to internal resources (**Lateral Movement**).

The Solution: Defense-in-Depth

- Layers multiple security controls.
- Prevents single points of failure.

Key Definitions

- **Network Segmentation:** Dividing a network into smaller sub-networks.
- **Security Zones:** Logical areas with uniform trust levels.



Concept 1: Network Segmentation

Traffic Control

Regulates data flow between segments.



Compliance

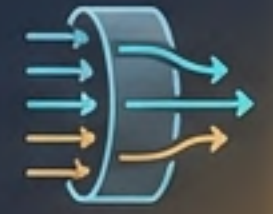
Isolates sensitive data (e.g., PCI-DSS).



Definition*: Dividing a large network into smaller, manageable sub-networks (subnets) to limit communication to only what is necessary (Least Privilege).

Performance

Reduces broadcast traffic and congestion.



Breach Containment

Prevents lateral movement.



Approaches to Segmentation: Physical vs. Logical

Physical Segmentation

- Uses distinct hardware (routers, switches, air gaps).
- **Pros:** Complete isolation, high security.
- **Cons:** High cost, complexity.
- **Use Case:** Critical infrastructure, payment servers.



Critical Systems
(e.g., Payment Servers)

General IT
Infrastructure

Logical Segmentation (VLANs)

- Uses Virtual LANs within shared infrastructure.
- **Pros:** Cost-effective, flexible.
- **Cons:** Misconfiguration risks.
- **Use Case:** Separating HR, Finance, and IT.



Advanced Strategy: Micro-Segmentation

Definition: Granular traffic control at the workload or application level (virtualized/cloud environments).

The Shift:

- **North-South:** Traditional perimeter entry/exit.
- **East-West:** Server-to-server communication (Target of Micro-segmentation).

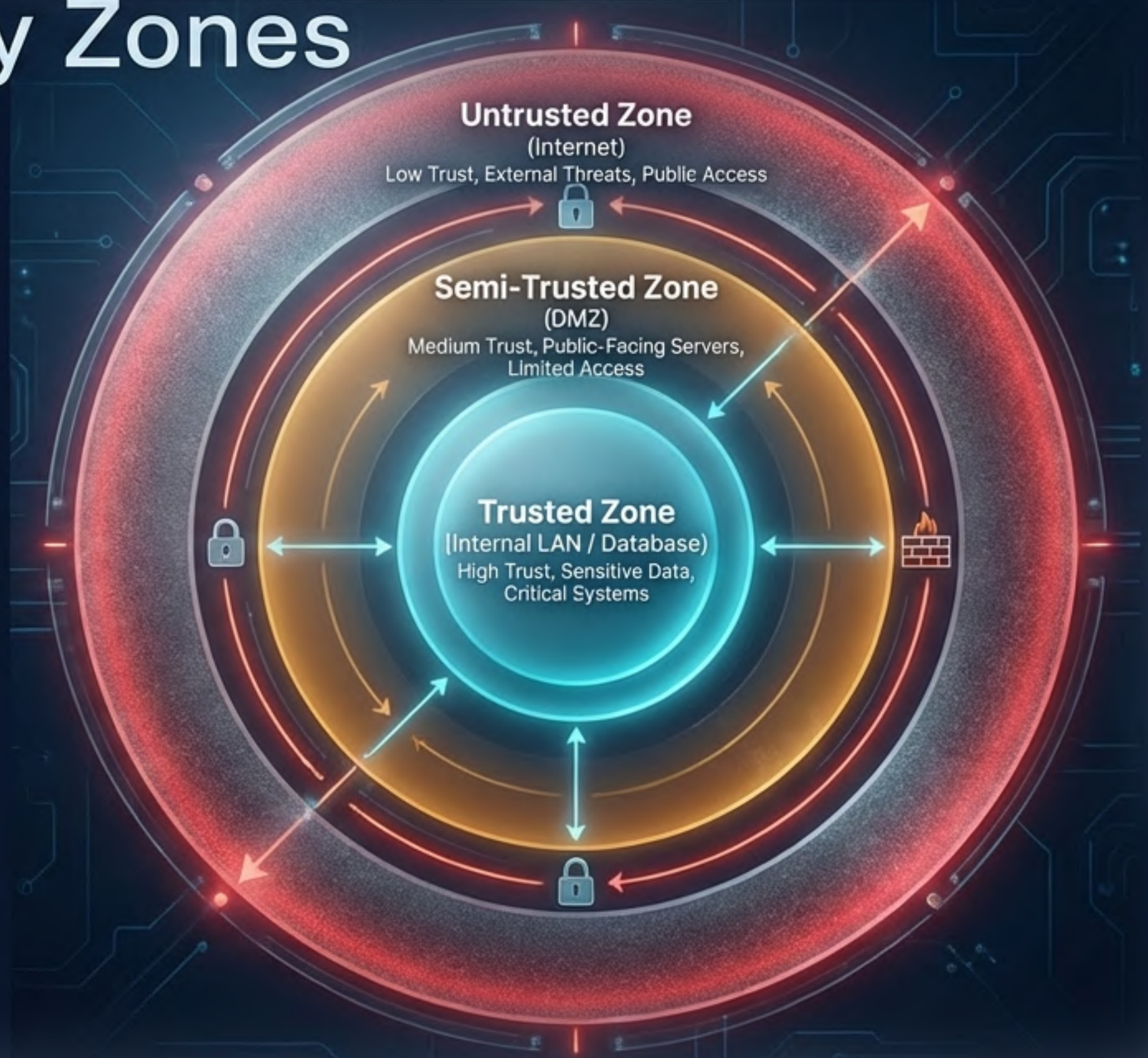
Strategic Value:

- Limits access at the workload level.
- Enabler for Zero Trust ("Never trust, always verify").



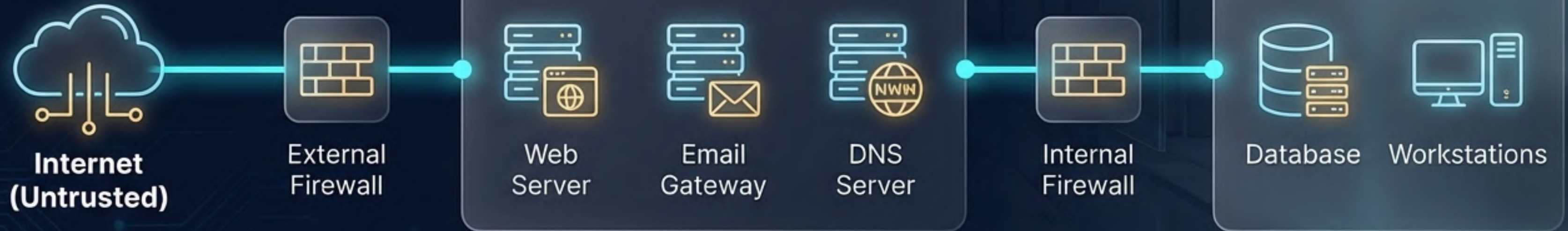
Concept 2: Security Zones

****Definition**:** A logically defined area where systems share uniform trust levels and security controls. Zones create boundaries to contain breaches.



The Demilitarized Zone (DMZ)

Buffer Zone: The DMZ isolates public-facing services, preventing direct access from the Internet to the Internal Network.



Enforcing Zones: The Role of Firewalls

Firewalls act as **gatekeepers** or choke points between zones.

Default Policy: “Deny all traffic unless explicitly allowed.”

Types of Inspection:

1. **Packet Filtering:** Inspects IP addresses and Ports (Layer 3/4).
2. **Stateful Inspection:** Tracks the context of active connections.
3. **Application Proxy:** Deep inspection of Layer 7 data.



Techniques for Implementation



VLANs (Virtual LANs)

Logically groups devices at Layer 2. Reduces broadcast storms and segregates sensitive traffic.



ACLs (Access Control Lists)

Rule sets on routers/switches. Controls traffic based on IPs, ports, and protocols.



Network Access Control (NAC)

Authenticates and scans devices *before* placing them in a segment. Isolates guest devices.

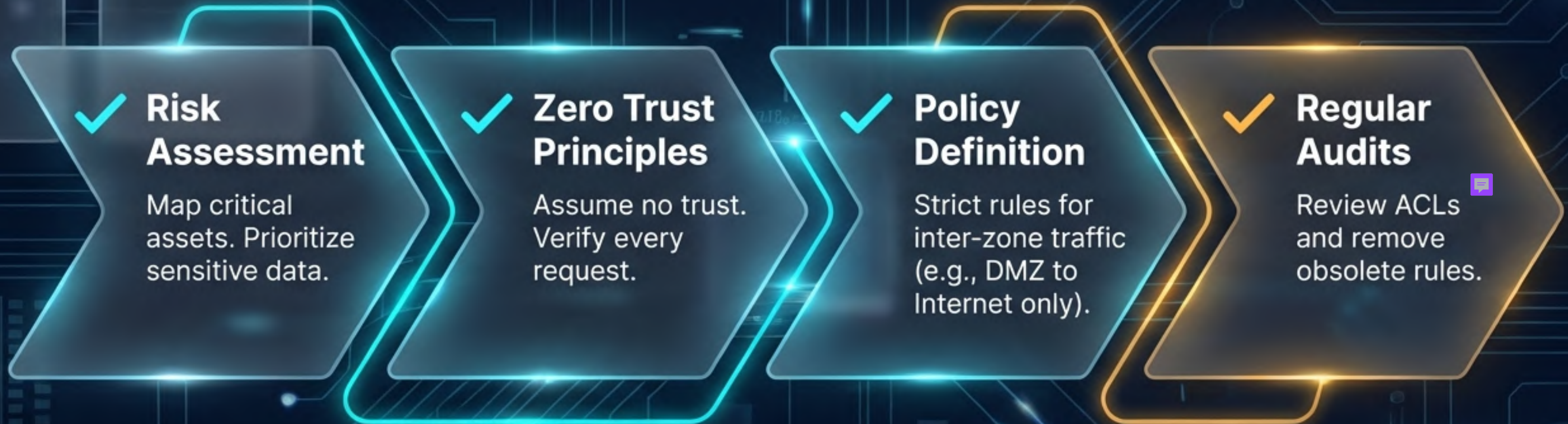
Comparison: Security Zones vs. Network Segmentation

Aspect	Security Zones	Network Segmentation
Focus	 Grouping resources by trust level and policy.	 Dividing network for performance and security.
Implementation	 Firewalls, DMZs.	 VLANs, Subnets.
Scope	 Defines trust boundaries.	 Controls traffic flow & broadcast domains.

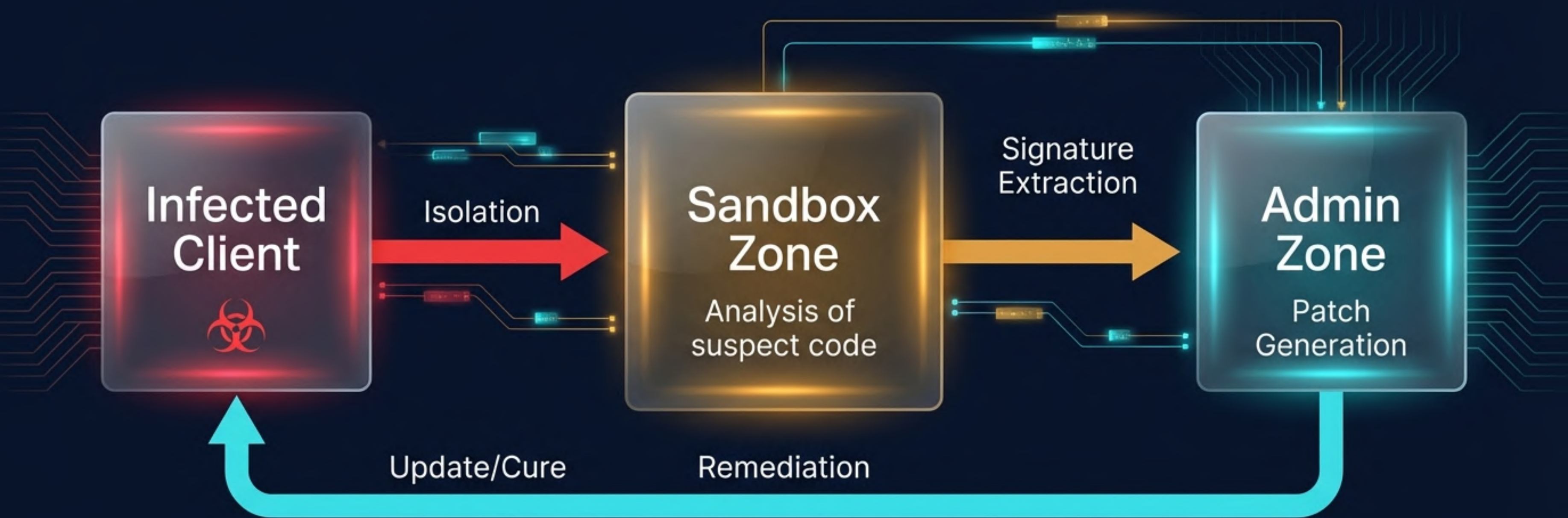
Strategic Importance of Architecture



Best Practices for Implementation



Architecture in Action: The Digital Immune System



Segmentation prevents the virus from spreading while analysis occurs.

Summary

Segmentation divides the network to improve performance and stop lateral movement.

Security Zones establish trust boundaries (Untrusted, DMZ, Trusted) for targeted policies.

Firewalls are the enforcement mechanism policing traffic between these zones.

Conclusion: Single perimeter defense is obsolete. Architectural depth is the modern standard.

References

- Stallings, W. (2017). *Network Security Essentials: Applications and Standards* (6th ed.). Pearson Education.
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